Riemann Surface

Lecture Notes

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Contents

Covering Spaces

CHAPTER 1

1.1. Preliminaries

DEFINITION 1.1. *Let X be a Hausdorff topological space. X is called a 2-dimensional topological manifold if for every $x \in X$, there exists an open neighborhood U of x such that U is homeomorphic to an open subset of $\mathbb{R}^2$.*

DEFINITION 1.2. *Let X be a 2-dimensional topological manifold. A chart of X is a homeomorphism $\phi : U \to V$ where U is an open subset of X and V is an open subset of $\mathbb{C}$. A complex atlas of X is a collection of charts $\{\phi_j : U_j \to V_j\}_{j\in I}$ such that $\{U_j\}_{j\in I}$ is an open cover of X and for every $j, k \in I$ with $U_j \cap U_k \neq \emptyset$, the composition $\phi_k \circ \phi_j^{-1} : \phi_j(U_j \cap U_k) \to \phi_k(U_j \cap U_k)$ is biholomorphic. Two complex atlases $\{\phi_j\}_{j\in I}$ and $\{\psi_\alpha\}_{\alpha\in A}$ are said to be equivalent if their union is also a complex atlas. A complex structure on X is an equivalence class of complex atlases. A Riemann surface is a* ***connected 2-dimensional topological manifold*** *equipped with a* ***complex structure****.*

DEFINITION 1.3. *Let X be a Riemann surface. A function $f : X \to \mathbb{C}$ is holomorphic if $f \circ \phi_j^{-1} : V_j \to \mathbb{C}$ is holomorphic for every chart $\phi_j : U_j \to V_j$ of X.*

Remark. On Riemann surfaces, we often write (U, z) locally instead of $\phi : U \to V$ to denote a chart, where z is the local coordinate on U.

DEFINITION 1.4. *Let X be a Riemann surface. For a open subset $Y \subset X$, the restriction of the complex structure on X to Y gives a complex structure on Y. Let $\mathcal{O}(Y)$ denote the $\mathbb{C}$-algebra of holomorphic functions on Y.*

THEOREM 1.5 (Riemann's removable singularity theorem). *Let X be a Riemann surface and $U \subset X$ be an open subset. If $a \in U$ and $f \in \mathcal{O}(U \setminus \{a\})$ is bounded in a neighborhood of a, then f can be extended uniquely to $\tilde{f} \in \mathcal{O}(U)$.*

Proof. Let ϕ be a chart at a, then since f is bounded in a neighborhood of a, we have

$$\lim_{z\to\phi(a)} (z - \phi(a)) f \circ \phi^{-1}(z) = 0.$$

Apply Riemann's removable singularity theorem in complex analysis, we can extend $f \circ \phi^{-1}$ to a holomorphic function g on $\phi(U)$, and then define $\tilde{f} = g \circ \phi$. The uniqueness of $\tilde{f}$ follows from the identity theorem for holomorphic functions. ∎

DEFINITION 1.6. *A meromorphic function on $Y \subset X$ if there exists $Y' \subset Y$ such that*

(i) *$f \in \mathcal{O}(Y')$;*
(ii) *$Y \setminus Y'$ is a discrete subset of Y;*
(iii) *for every $a \in Y \setminus Y'$,*

$$\lim_{x\to a}|f(x)| = \infty.$$

The set of meromorphic functions on Y is denoted by $\mathcal{M}(Y)$.

DEFINITION 1.7. *Let X and Y be Riemann surfaces. A holomorphic map from X to Y is a continuous map $f : X \to Y$ such that for every chart $\phi : U \to V$ of X and every chart $\psi : U' \to V'$ of Y with $f(U) \subset U'$, the composition $\psi \circ f \circ \phi^{-1} : V \to V'$ is holomorphic. A biholomorphism is a holomorphic map that has a holomorphic inverse.*

Remark. A holomorphic map $f : X \to Y$ induces a pullback map $f^* : \mathcal{O}(Y) \to \mathcal{O}(X)$ defined by $f^*(g) = g \circ f$ for every $g \in \mathcal{O}(Y)$.

THEOREM 1.8 (Identity theorem). *Let X be a Riemann surface and $f, g \in \mathcal{O}(X)$. If f and g agree on a subset of $A \subset X$ that has an accumulation point, then $f \equiv g$ on X.*

Proof. Let

$$G = \{x \in X : \exists \text{ a neighborhood } W \text{ of } x \text{ such that } f|_W = g|_W\}.$$

Then G is open. Let $b \in$ `overline` G, then $f(b) = g(b)$ by the continuity of f and g. Choose connected neighborhoods U of b and V of $f(b)$ such that there exist charts $\phi : U \to \phi(U)$ and $\psi : V \to \psi(V)$. Then $\psi \circ f \circ \phi^{-1}$ and $\psi \circ g \circ \phi^{-1}$ are holomorphic functions on $\phi(U)$ that agree on a subset of $\phi(U)$ that has an accumulation point. By the identity theorem for holomorphic functions, we have $\psi \circ f \circ \phi^{-1} = \psi \circ g \circ \phi^{-1}$ on $\phi(U)$, which implies $f|_U = g|_U$. Hence $b \in G$, which shows that G is closed.

By our assumption, A has an accumulation point a, by a similar argument as above, we can show that $a \in G$. Since G is nonempty, open and closed, we have $G = X$, which implies $f \equiv g$ on X. ∎

DEFINITION 1.9. *Let $\tilde{C} = \mathbb{C} \cup \{\infty\}$ be the one point compactification of $\mathbb{C}$. The Riemann sphere $\mathbb{P}^1$ is the complex structure on $\tilde{C}$ given by the charts $\{\phi_1 : \mathbb{C} \to \mathbb{C}, z \mapsto z\}$ and $\{\phi_2 : \tilde{C} \setminus \{0\} \to \mathbb{C}, z \mapsto \frac{1}{z}\}$.*

THEOREM 1.10. *Let X be a Riemann surface and $f \in \mathcal{M}(X)$. For each pole x of f, define $f(x) = \infty$. Then f can be viewed as a holomorphic map from X to $\mathbb{P}^1$. Conversely, every holomorphic map from X to $\mathbb{P}^1$ is either identically ∞ or a meromorphic function on X.*

Proof. The extended map $f : X \to \mathbb{P}^1$ is clearly continuous. For every chart $\phi : U \to V$ of X and every chart $\psi : U' \to V'$ of $\mathbb{P}^1$ with $f(U) \subset U'$, the composition $\psi \circ f \circ \phi^{-1} : V \to V'$ is holomorphic since it is either a holomorphic function on V or a holomorphic function on $V \setminus \{\phi(x)\}$ that has a pole at $\phi(x)$, which can be extended to a holomorphic function on V by Riemann's removable singularity theorem. Hence f is a holomorphic map from X to $\mathbb{P}^1$.

Conversely, let $f : X \to \mathbb{P}^1$ be a holomorphic map. If $f \not\equiv \infty$, then the set of poles of f is discrete, impling that f is a meromorphic function on X. ∎

From the above theorem, we can conclude that meromorphic functions also satisfy the identity theorem, hence if $f \not\equiv 0$, then $f^{-1}(0)$ is a discrete subset of X. This implies that $\mathcal{M}(X)$ is in fact a field.

Under the local coordinate z of a chart (U, z) such that $z(p) = 0$ for some pole p, then we can write a meromorphic function f on U as a Laurent series

$$f = \sum_{n=m}^{\infty} a_n z^n.$$

If $m < 0$, then f has a pole of order $-m$ at p. If $m \geq 0$, then f is holomorphic at p. If $m > 0$, then f has a zero of order m at p.

THEOREM 1.11. *Let X, Y be Riemann surfaces and $f : X \to Y$ be a non-constant holomorphic map. Suppose $a \in X$ and $b = f(a)$. Then there exist charts (U, z) at a and (V, w) at b such that $f(U) \subset V$ and*

$$w \circ f \circ z^{-1}(\zeta) = \zeta^k$$

for some positive integer k.

Proof. Pick charts (U, z) at a and (V, w) at b such that $f(U) \subset V$, $z(a) = 0$, and $w(b) = 0$. Then $w \circ f \circ z^{-1}$ is a holomorphic map from a neighborhood of 0 in $\mathbb{C}$ to a neighborhood of 0 in $\mathbb{C}$. Since f is non-constant, we can write $f_1 := w \circ f \circ z^{-1} = z^k g$ for some positive integer k and some holomorphic function g with $g(0) \neq 0$. By shrinking U if necessary, we can assume that $g(\zeta) \neq 0$ for every $\zeta \in z(U)$. Then we can take a holomorphic k-th root of g, denoted by h. Replacing z by $h \cdot z$, we get the desired property

$$w \circ f \circ z^{-1}(\zeta) = (z \cdot h(\zeta))^k \Rightarrow w \circ f \circ (z \cdot h)^{-1}(\zeta) = \zeta^k.$$

∎

LEMMA 1.12. *Non-constant holomorphic maps are open maps.*

LEMMA 1.13 (Maximal Principle). *Suppose X is a Riemann surface and $f : X \to \mathbb{C}$ is a non-constant holomorphic function. Then $|f|$ does not attain its maximum on X.*

Proof. If not, say $R = |f(a)| = \sup_{x \in X} |f(x)|$ for some $a \in X$. Then $f(X) \subset \overline{B_R(0)}$. Since f is an open map, $f(X)$ contains an open neighborhood of $f(a)$, which contradicts the definition of R. ∎

THEOREM 1.14. *Suppose X, Y are Riemann surfaces. Suppose X is compact and $f : X \to Y$ is a holomorphic map. Then Y is compact and f is surjective.*

Proof. Since X is compact, $f(X)$ is compact, in particular closed. Since f is an open map, $f(X)$ is also open. Hence $f(X) = Y$, which implies that Y is compact and f is surjective. ∎

LEMMA 1.15. *Every holomorphic function on a compact Riemann surface is constant.*

Proof. This follows from the previous theorem since $\mathbb{C}$ is not compact. ∎

Example 1.1. Let ω_1, ω_2 be linearly independent complex numbers. Define a lattice $\Gamma = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ and the equivalence relation $\sim$ on $\mathbb{C}$ by $z \sim w$ if and only if $z - w \in \Gamma$. Consider the quotient space $X = \mathbb{C}/\Gamma = \mathbb{C}/\sim$ with the quotient topology, that is, the topology given by

$$\{U \subseteq \mathbb{C}/\Gamma : \pi^{-1}(U) \text{ is open in } \mathbb{C}\},$$

where $\pi : \mathbb{C} \to \mathbb{C}/\Gamma$ is the projection map. Let $V \subset \mathbb{C}$ be an open set such that no two points in V are equivalent under $\sim$. Then $\pi|_V : V \to \pi(V)$ is a homeomorphism, whose inverse $\phi : \pi(V) \to V$ is a chart of X. The collection of charts $\{\phi : \pi(V) \to V\}$ gives a complex structure on X, which makes X a Riemann surface.

1.2. The Fundamental Group

DEFINITION 1.16. *Let X be a topological space and $a, b \in X$. A curve from a to b is a continuous map $u : I \to X$ such that $u(0) = a$ and $u(1) = b$, where $I = [0, 1]$. Two curves u, v from a to b are homotopic, denoted by $u \sim v$, if there exists a continuous map $A : I \times I \to X$ such that*

(i) *$A(t, 0) = u(t)$ for every $t \in I$;*
(ii) *$A(t, 1) = v(t)$ for every $t \in I$;*
(iii) *$A(0, s) = a, A(1, s) = b$ for every $s \in I$.*

THEOREM 1.17. *The homotopy relation $\sim$ is an equivalence relation on the set of curves from a to b.*

Proof. The reflexivity and symmetry of $\sim$ are clear. To show the transitivity, let u, v, w be curves from a to b such that $u \overset{A}{\sim} v$ and $v \overset{B}{\sim} w$. Define a continuous map $C : I \times I \to X$ by

$$C(t, s) = \begin{cases} A(t, 2s) \text{ if } \ 0 \le s \le \frac{1}{2} \\ B(t, 2s - 1) \text{ if } \ \frac{1}{2} < s \le 1 \end{cases}$$

Then C is a homotopy from u to w, which shows that $u \sim w$. ∎

DEFINITION 1.18. *Let u be a curve from a to b and v be a curve from b to c. Define the product of u and v to be the curve $u \cdot v : I \to X$ from a to c given by*

$$(u \cdot v)(t) = \begin{cases} u(2t) \text{ if } \ 0 \le t \le \frac{1}{2} \\ v(2t - 1) \text{ if } \ \frac{1}{2} < t \le 1 \end{cases}$$

DEFINITION 1.19. *Let u be a curve from a to b. The inverse of u is the curve u^{-1} from b to a given by $u^{-1}(t) = u(1 - t)$ for every $t \in I$.*

THEOREM 1.20. *If $u \sim u'$ and $v \sim v'$, then $u \cdot v \sim u' \cdot v'$ and $u^{-1} \sim (u')^{-1}$.*

Proof. Let A be a homotopy from u to u' and B be a homotopy from v to v'. Define a continuous map $C : I \times I \to X$ by

$$C(t, s) = \begin{cases} A(2t, s) \text{ if } \ 0 \le t \le \frac{1}{2} \\ B(2t - 1, s) \text{ if } \ \frac{1}{2} < t \le 1 \end{cases}$$

Then C is a homotopy from $u \cdot v$ to $u' \cdot v'$, which shows that $u \cdot v \sim u' \cdot v'$.

Define a continuous map $D : I \times I \to X$ by $D(t, s) = A(1 - t, s)$. Then D is a homotopy from u^{-1} to $(u')^{-1}$, which shows that $u^{-1} \sim (u')^{-1}$. ∎

DEFINITION 1.21. *A curve u is closed if $u(0) = u(1)$. A curve u is constant if $u(t) = u(0)$ for every $t \in I$. A constant curve at a may be denoted simply by a.*

Theorem 1.22. *The set $\pi_1(X, a)$ of homotopy classes of closed curves at a forms a group under the product operation, called the fundamental group of X at a.*

Proof. The identity element of $\pi_1(X, a)$ is the homotopy class of the constant curve at a. To see this, for every curve u from a to a, construct a homotopy $A : I \times I \to X$ from $u \cdot a$ to u by

$$A(t,s) = \begin{cases} u\left(\frac{2t}{1+s}\right) \text{ if } \ 0 \leq t \leq \frac{1+s}{2} \\ a \text{ if } \ \frac{1+s}{2} < t \leq 1 \end{cases}$$

The case for $a \cdot u$ is similar. For every curve u from a to a, construct a homotopy $B : I \times I \to X$ from $u \cdot u^{-1}$ to a by

$$B(t,s) = \begin{cases} u\left(\frac{2t}{1-s}\right) \text{ if } \ 0 \leq t \leq \frac{1}{2} \\ u\left(\frac{2(1-t)}{1-s}\right) \text{ if } \ \frac{1}{2} < t \leq 1 \end{cases}$$

For curves u, v, w from a to a, construct a homotopy $C : I \times I \to X$ from $(u \cdot v) \cdot w$ to $u \cdot (v \cdot w)$ by

$$C(t,s) = \begin{cases} u\left(\frac{4t}{s+1}\right) \text{ if } \ 0 \leq t \leq \frac{s+1}{4} \\ v(4t - s - 1) \text{ if } \ \frac{s+1}{4} < t \leq \frac{s+2}{4} \\ w\left(\frac{4t-s-2}{2-s}\right) \text{ if } \ \frac{s+2}{4} < t \leq 1 \end{cases}$$

Hence the product operation on $\pi_1(X, a)$ is associative. ∎

Denote the homotopy class of a curve u by $\mathrm{cl}(u)$.

Remark. If X is arcwise connected, then $\pi_1(X, a)$ is independent of the choice of a up to isomorphism. To see this, let $b \in X$ and u be a curve from a to b. Define a map $\Phi : \pi_1(X, a) \to \pi_1(X, b)$ by $\Phi(\mathrm{cl}(v)) = \mathrm{cl}(u^{-1} \cdot v \cdot u)$ for every closed curve v at a, then Φ is a group isomorphism.

Definition 1.23. *An arcwise connected space X is simply connected if $\pi_1(X, a) = 0$.*

Definition 1.24. *Suppose u, v are two close curves whose initial point are not necessarily the same. We say that u and v are free homotopic if there exists a continuous map $A : I \times I \to X$ such that*

(i) $A(t, 0) = u(t)$ *for every* $t \in I$;
(ii) $A(t, 1) = v(t)$ *for every* $t \in I$;
(iii) $A(0, s) = A(1, s)$ *for every* $s \in I$.

Theorem 1.25. *The following are equivalent for an arcwise connected space X:*

(i) *X is simply connected;*
(ii) *for every u, v from a to b, we have $u \sim v$;*
(iii) *for every closed curve u, v at a, b respectively, we have u and v are free homotopic.*

Proof.

- (i) ⇒ (ii): Note that
$$u \sim a \cdot u \sim (v \cdot v^{-1}) \cdot u \sim v \cdot (v^{-1} \cdot u) \sim v \cdot b \sim v.$$

- (ii) $\Rightarrow$ (iii): Let w be a curve from a to b. Then consider the concatenation of $u \sim a$, $a \sim b$ and $b \sim v$, where the middle map is given by w. This gives a free homotopy from u to v.
- (iii) $\Rightarrow$ (i): For every closed curve u at a, we have $\mathrm{cl}(u) = \mathrm{cl}(a)$, which implies that $\pi_1(X, a) = 0$.

∎

DEFINITION 1.26. *Let X, Y be topological spaces and $p : Y \to X$ be a continuous map. p is a local homeomorphism if for every $\in Y$, there exists an open neighborhood V of y such that $p(V)$ is open in X and $p|_V : V \to p(V)$ is a homeomorphism. p is a **covering map** if for every $x \in X$, there exists an open neighborhood U of x such that $p^{-1}(U)$ is a disjoint union of open subsets of Y, each of which is homeomorphic to U via p.*

THEOREM 1.27 (Curve Lifting Property). *Let X, Y be topological spaces and $p : Y \to X$ be a covering map. Let u be a curve with initial point $x_0 \in X$ and $y_0 \in p^{-1}(x_0)$. Then there exists a curve $\tilde{u}$ with initial point y_0 such that $p \circ \tilde{u} = u$.*

$$\begin{tikzcd}
& Y \arrow[d, "p"] \\
{[0,1]} \arrow[ur, dashed, "\tilde{u}"] \arrow[r, "u"'] & X
\end{tikzcd}$$

Proof. Let

$$J = \Big\{ t \in [0,1] : u|_{[0,t]} \text{ can be lifted to a curve } \tilde{u} \text{ with initial point } y_0 \Big\}.$$

$J \neq \emptyset$ since $0 \in J$.

For $t \in J$, choose an open neighborhood $\mathcal{U}$ of $u(t)$ such that $p^{-1}(\mathcal{U}) = \bigsqcup_\alpha V_\alpha$ and $p|_{V_\alpha} : V_\alpha \to \mathcal{U}$ is a homeomorphism for every α. Let $\tilde{u}(t) \in V_{\alpha_0}$ for some α_0. Since u is continuous, there exists $\varepsilon > 0$ such that $[t, t+\varepsilon] \subset u^{-1}(\mathcal{U})$. Let $\phi : \mathcal{U} \to V_{\alpha_0}$ be the inverse of $p|_{V_{\alpha_0}}$, then $\phi \circ u|_{[t,t+\varepsilon]}$ is a lift of $u|_{[t,t+\varepsilon]}$ with initial point $\tilde{u}(t)$. Hence $[t, t+\varepsilon] \subset J$, which shows that J is open.

Suppose $\{t_n\} \subset J$ is a sequence such that $t_n \to t_0$ with $t_n < t_0$ for every n. Choose $u(t_0) \in \mathcal{U}$ as above. Assume that $u([t_n, t_0]) \subset \mathcal{U}$ and let $\tilde{u}(t_n) \in V_{\beta_0}$ for some β_0. Let $\phi = \left(p|_{V_{\beta_0}}\right)^{-1}$, then we can define $\tilde{u}(t) = \phi \circ u(t)$ for every $t \in [t_n, t_0]$, thus $[t_n, t_0] \subset J$. Hence $t_0 \in J$, which shows that J is closed.

Since J is nonempty, open and closed, we have $J = [0,1]$, which shows the existence of $\tilde{u}$. ∎

THEOREM 1.28 (Uniqueness). *Suppose X, Y are Hausdorff topological spaces and $p : Y \to X$ is a local homeomorphism. Suppose Z is connected and $f : Z \to X$ is a continuous map. If $\tilde{f}_1, \tilde{f}_2 : Z \to Y$ are two lifts of f such that $\tilde{f}_1(z_0) = \tilde{f}_2(z_0)$ for some $z_0 \in Z$, then $\tilde{f}_1 = \tilde{f}_2$.*

Proof. Let

$$G = \{ z \in Z \mid \tilde{f}_1(z) = \tilde{f}_2(z) \}.$$

Then G is nonempty since $z_0 \in G$. G is closed since it is the preimage $\left(\tilde{f}_1, \tilde{f}_2\right)^{-1}(\delta)$ of the diagonal δ in $Y \times Y$. To show that G is open, let $z \in G$ and $y = \tilde{f}_1(z) = \tilde{f}_2(z)$. Choose a neighborhood U of y such that $p|_U : U \to p(U)$ is a homeomorphism. Since $\tilde{f}_1, \tilde{f}_2$ are continuous, there exists a neighborhood W of z such that $\tilde{f}_{i(W)} \subseteq U$ for $i = 1, 2$. By definition of lift, $p \circ \tilde{f}_i = f \Rightarrow \tilde{f}_i = (p|_U)^{-1} \circ f$ on W, which implies that $\tilde{f}_1 = \tilde{f}_2$ on W. Hence G is open.

By the connectedness of Z, we have $G = Z$, which shows that $\tilde{f}_1 = \tilde{f}_2$. ∎

THEOREM 1.29 (Homotopy Lifting). *Suppose X, Y are Hausdorff topological spaces and $p : Y \to X$ is a covering map. Suppose u, v are curves from a to b in X and $A : I \times I \to X$ is a homotopy from u to v. Then for every $\hat{a} \in p^{-1}(a)$, there exists a homotopy $\tilde{A} : I \times I \to Y$ from the lift of u with initial point $\hat{a}$ to the lift of v with initial point $\hat{a}$ such that $p \circ \tilde{A} = A$.*

Proof. By the curve lifting property (Theorem 1.27), for each $u_{s(t)} = A(t, s)$ there exists a lift $\tilde{u}_s$ with $\tilde{u}_{s(0)} = \hat{a}$. Define $\tilde{A}(t, s) := \tilde{u}_{s(t)}$, then clearly $p \circ \tilde{A} = A$. It remains to show that $\tilde{A}$ is continuous.

Take a neighborhood $\mathcal{U}$ of $\hat{a}$ such that U is homeomorphic to $p(\mathcal{U})$ via p. Let $\phi = (p|_{\mathcal{U}})^{-1}$. Since A is continuous, there exists $\varepsilon > 0$ such that $A([0, \varepsilon] \times I) \subset p(\mathcal{U})$. By uniqueness of lift, we have $\tilde{u}_{s(t)} = \phi \circ A(t, s)$ for every $(t, s) \in [0, \varepsilon] \times I$, which implies that $\tilde{A}$ is continuous on $[0, \varepsilon] \times I$.

Now suppose that $\tilde{A}$ is discontinuous at some point (t, c), and let t_0 be the infimum of such t. Choose a neighborhood $\mathtt{cal}\ V$ of $\tilde{A}(t_0, c)$ such that $p|_{\mathtt{cal}\ V} : \mathtt{cal}\ V \to p(\mathtt{cal}\ V)$ is a homeomorphism. Again, let $\phi = (p|_{\mathtt{cal}\ V})^{-1}$. Since A is continuous, there exists $\delta > 0$ such that $A\left(B_{\delta(t_0, c)}\right) \subset p(\mathtt{cal}\ V)$. By uniqueness of lift, we have $\tilde{A}(t, s) = \phi \circ A(t, s)$ for every $(t, s) \in B_{\delta(t_0, c)}$ with $t < t_0$, which implies that $\tilde{A}$ is continuous at (t_0, c), a contradiction. Hence $\tilde{A}$ is continuous on $I \times I$.

Intuition: by uniqueness of lifting, the mapping $\tilde{A}$ is locally given by $\phi \circ A$, which is the composition of two continuous maps, hence $\tilde{A}$ is locally continuous, hence continuous ∎

1.3. Deck Transformations

THEOREM 1.30 (Lifting of a continuous map). *Suppose X, Y are Hausdorff topological spaces and $p : Y \to X$ is a covering map. Suppose Z is simply connected and $f : Z \to X$ is a continuous map. Then for every $y_0 \in p^{-1}(f(z_0))$, there exists a unique lift $\tilde{f} : Z \to Y$ of f such that $\tilde{f}(z_0) = y_0$.*

Proof. Pick $z_0 \in Z$, $x_0 = f(z_0)$ and $y_0 \in p^{-1}(x_0)$. For every $z \in Z$, choose a curve u from z_0 to z and $v = f \circ u$. By the curve lifting property (Theorem 1.27), there exists a lift $\tilde{v}$ of v with initial point y_0. Define $\tilde{f}(z) = \tilde{v}(1)$.

To show that $\tilde{f}$ is well-defined, let u' be another curve from z_0 to z and $v' = f \circ u'$. Since Z is simply connected, we have $u \sim u'$, which implies that $v \sim v'$. By the homotopy lifting property, we have $\tilde{v} \sim \tilde{v}'$, which implies that $\tilde{v}(1) = \tilde{v}'(1)$, hence $\tilde{f}(z) = \tilde{v}(1) = \tilde{v}'(1)$.

To show that $\tilde{f}$ is continuous, let $z \in Z$ and $y = \tilde{f}(z)$. Choose a neighborhood $\mathcal{U}$ of y such that $p|_{\mathcal{U}} : \mathcal{U} \to p(\mathcal{U})$ is a homeomorphism and let $\phi = (p|_{\mathcal{U}})^{-1}$. Since f is continuous, there exists a neighborhood W of z such that $f(W) \subset p(\mathcal{U})$. For every $w \in W$, choose a curve u from z to w and let $v = f \circ u$. By the curve lifting property, there exists a lift $\tilde{v}$ of v with initial point $\tilde{f}(z) = y$. By uniqueness of lift, we have $\tilde{v}(1) = \phi \circ f(w)$, which implies that $\tilde{f}(w) = \tilde{v}(1) = \phi \circ f(w)$ for every $w \in W$, hence $\tilde{f}$ is continuous at z. Since z is arbitrary, $\tilde{f}$ is continuous on Z. ∎

Remark. In the proof above, the only properties of p that are used are that it is a local homeomorphism and has the curve lifting property.

Theorem 1.31. *Let X be a manifold, Y be a Hausdorff topological space, then $p : Y \to X$ is a covering map if and only if p is a local homeomorphism and has the curve lifting property.*

Proof. The forward direction is clear. For the backward direction, let $x_0 \in X$ and choose a simply connected neighborhood U of x_0 and consider the injection $f : U \to X$. By the previous theorem, for each $y_j \in p^{-1}(x_0)$, there exists a lift $\tilde{f} : U \to Y$ of f such that $\tilde{f}(x_0) = y_j$. Let $V_j = \tilde{f}(U)$, then $p|_{V_j} : V_j \to U$ is a homeomorphism. It remains to show that $p^{-1}(U) = \bigsqcup_j V_j$.

If $y \in V_i \cap V_j$, then $\tilde{f}_i$ and $\tilde{f}_j$ are two lifts of f such that $\tilde{f}_{i(x_0)} = \tilde{f}_{j(x_0)} = y_i$, which implies that $\tilde{f}_i = \tilde{f}_j$, hence $V_i = V_j$. For $y \in p^{-1}(U)$, let $x = p(y)$ and choose a curve u from x_0 to x, then there exists a lift $\tilde{u}$ of u with initial point y_i, which implies that $y \in \tilde{f}_{i(U)} = V_i$. Hence $p^{-1}(U) = \bigsqcup_j V_j$, which shows that p is a covering map. ∎

Definition 1.32. *Let $p : Y \to X$ be a covering map. A deck transformation of p is a fiber-preserving homeomorphism $f : Y \to Y$, that is, $p \circ f = p$.*

The set of deck transformations of p forms a group under composition, called the deck transformation group of p, denoted by $\mathrm{Deck}\left(\frac{Y}{X}\right)$.

Proposition 1.33. *If $p : Y \to X$ is a covering map, then p induces an injective group homomorphism $p_* : \pi_1(Y, y_0) \to \pi_1(X, x_0)$ for every $y_0 \in Y$ and $x_0 = p(y_0)$.*

Proof. For every closed curve u at y_0, $p \circ u$ is a closed curve at x_0. Define $p_*(\mathrm{cl}(u)) = \mathrm{cl}(p \circ u)$, then p_* is a well-defined group homomorphism. To show that p_* is injective, let u be a closed curve at y_0 such that $p_*(\mathrm{cl}(u)) = \mathrm{cl}(p \circ u) = 0$. By the homotopy lifting property, there exists a homotopy $\tilde{A} : I \times I \to Y$ from u to the constant curve at y_0 such that $p \circ \tilde{A} = A$, where A is a homotopy from $p \circ u$ to the constant curve at x_0. Hence $\mathrm{cl}(u) = 0$, which shows that p_* is injective. ∎

Proposition 1.34. *Let $p : Y \to X$ be a covering map. Let $\mathrm{cl}(u) \in \pi_1(X, x_0)$ and $\tilde{u}(0) = y_0$ for some, then $u \in p_*(\pi_1(Y, y_0))$ if and only if $\tilde{u}(1) = y_0$.*

Proof. If $u \in p_*(\pi_1(Y, y_0))$, then there exists a closed curve v at y_0 such that $p \circ v = u$. By uniqueness of lift, we have $\tilde{u} = v$, which implies that $\tilde{u}(1) = v(1) = y_0$.

Conversely, if $\tilde{u}(1) = y_0$, then $\tilde{u}$ is a closed curve at y_0 such that $p \circ \tilde{u} = u$, which implies that $u \in p_*(\pi_1(Y, y_0))$. ∎

LEMMA 1.35. *For two loops u_1, u_2 at x_0, $u_1 \cdot u_2^{-1} \in p_*(\pi_1(Y, y_0))$ if and only if the lifts of u_1 and u_2 with initial point y_0 have the same terminal point.*

Proof. This follows from the previous proposition since $u_1 \cdot u_2^{-1} \in p_*(\pi_1(Y, y_0))$ if and only if the lift of $u_1 \cdot u_2^{-1}$ with initial point y_0 is a closed curve at y_0, which is the case if and only if the lifts of u_1 and u_2 with initial point y_0 have the same terminal point. ∎

Let $p : Y \to X$ be a covering map. A curve $\mathrm{cl}(u) \in \pi_1(X, x_0)$ acts on the fibers of p in the following way: for every $y \in Y$, exists a curve v from y_0 to y, then v is the unique lift of $p \circ v$ with initial point y_0. Consider the lift v' of $p \circ v \cdot u$ with initial point y_0, then $v'(1)$ is the image of y under the action of $\mathrm{cl}(u)$. This action is well-defined if Y is simply-connected, hence we have a group homomorphism $\Phi : \pi_1(X, x_0) \to \mathrm{Deck}\left(\frac{Y}{X}\right)$.

PROPOSITION 1.36. *Two covering maps $p, p' : Y \to X$ are said to be isomorphic if there exists a homeomorphism $f : Y \to Y$ such that $p' \circ f = p$. Two covering maps $p, p' : Y \to X$ are isomorphic if and only if $p_*(\pi_1(Y, y_0)) = p'_*(\pi_1(Y, y_{0'}))$ where $p(y_0) = p'(y_{0'})$.*

Proof. "$\Rightarrow$": Since $p = p' \circ f$, for $v \in \pi_1(Y, y_0)$, we have
$$p_*(\mathrm{cl}(v)) = \mathrm{cl}(p \circ v) = \mathrm{cl}(p' \circ f \circ v) = p'_*(\mathrm{cl}(f \circ v)).$$
Hence $p_*(\pi_1(Y, y_0)) \subseteq p'_*(\pi_1(Y, y_{0'}))$. Since f is a homeomorphism, by symmetry, we have $p_*(\pi_1(Y, y_0)) = p'_*(\pi_1(Y, y_{0'}))$.

"$\leq$": Similar to the proof of Theorem 1.30, for each $y \in Y$, choose a curve v from y_0 to y and let $u = p \circ v$. By the curve lifting property (Theorem 1.27), there exists a lift $\tilde{u}$ of u with initial point $y_{0'}$. Define $\tilde{p}(y) = \tilde{u}(1)$. The well-definedness follows from the assumption that $p_*(\pi_1(Y, y_0)) = p'_*(\pi_1(Y, y_{0'}))$. The continuity of f follows from the same argument as in Theorem 1.30.

$$\begin{tikzcd} & Y \arrow[d, "p'"] \\ Y \arrow[r, "p"'] \arrow[ur, dashed, "\exists\tilde{p}"] & X \end{tikzcd}$$

By symmetry, there exists a continuous map $\tilde{p}' : Y \to Y$ such that $p \circ \tilde{p}' = p'$. By uniqueness of lift, we have $\tilde{p}' \circ \tilde{p} = \tilde{p} \circ \tilde{p}' = \mathrm{id}_Y$, which shows that p and p' are isomorphic. ∎

DEFINITION 1.37. *A covering map $p : Y \to X$ is Galois if the deck transformation group* Deck $\left(\frac{Y}{X}\right)$ *acts transitively on the fibers of p, i.e. for every $x \in X$ and $y_1, y_2 \in p^{-1}(x)$, there exists a deck transformation f such that $f(y_1) = y_2$.*

PROPOSITION 1.38. *Given a covering map $p : Y \to X$, its deck transformation group is isomorphic to*
$$\mathrm{Deck}\left(\frac{Y}{X}\right) \cong \frac{N(\pi_1(Y, y_0))}{p_*}(\pi_1(Y, y_0)),$$

where $N(\pi_1(Y, y_0))$ is the normalizer of $p_*(\pi_1(Y, y_0))$ in $\pi_1(Y, y_0)$. Moreover, p is Galois if and only if $p_*(\pi_1(Y, y_0))$ is a normal subgroup of $\pi_1(X, x_0)$, in which case we have $\text{Deck}\left(\frac{Y}{X}\right) \cong \pi_1 \frac{X, x_0}{p_*}(\pi_1(Y, y_0))$.

Proof. For each path $u \in N(\pi_1(Y, y_0))$, define a deck transformation $f_u : Y \to Y$ by $f_{u(y)} = \tilde{u}(1)$, where $\tilde{u}$ is the lift of $p \circ u$ with initial point y. The well-definedness of f_u follows from the definition of normalizer. The map $\Phi : N(\pi_1(Y, y_0)) \to \text{Deck}\left(\frac{Y}{X}\right)$ given by $\Phi(u) = f_u$ is a group homomorphism since $f_{u \cdot v} = f_u \circ f_v$ for every $u, v \in N(\pi_1(Y, y_0))$. The kernel of Φ is exactly $p_*(\pi_1(Y, y_0))$, hence Φ induces an injective group homomorphism $\tilde{\Phi} : \frac{N(\pi_1(Y, y_0))}{p_*}(\pi_1(Y, y_0)) \to \text{Deck}\left(\frac{Y}{X}\right)$. To show that $\tilde{\Phi}$ is surjective, let f be a deck transformation of p. For every $\in Y$, choose a curve v from y_0 to y, then there exists a lift $\tilde{v}$ of $p \circ v$ with initial point y_0. Let $u = f \circ \tilde{v}$, then $p \circ u = p \circ f \circ \tilde{v} = p \circ \tilde{v} = p \circ v$, which implies that u is a lift of $p \circ v$ with initial point y_0. By uniqueness of lift, we have $u = \tilde{v}$, which implies that $f(y) = u(1) = \tilde{v}(1) = y$. Hence f is the identity map on Y, which shows that $\tilde{\Phi}$ is surjective. ∎

1.4. The Universal Covering

Recall that if $p : Y \to X$ is a covering map and $p_*(\pi_1(Y, y_0)) \lhd \pi_1(X, x_0)$, then p is Galois and the action of $\pi_1(X, x_0)$ on the fibers of p is well-defined.

Definition 1.39. *$p : Y \to X$ is a universal covering of X if for all covering map $p' : Y' \to X$, there exists a continuous map $f : Y \to Y'$ such that $p' \circ f = p$.*

Proposition 1.40. *If Y is simply connected, then a covering map $p : Y \to X$ is universal, Galois and* $\text{Deck}\left(\frac{Y}{X}\right) \cong \pi_1(X, x_0)$.

Proof. Existence of lifting follows from Theorem 1.30. Since $p_*(\pi_1(Y, y_0)) = 0$, p is Galois and $\text{Deck}\left(\frac{Y}{X}\right) \cong \pi_1(X, x_0)$. ∎

Theorem 1.41. *If X is a connected manifold and $H \leq \pi_1(X, x_0)$ is a subgroup, then there exists a covering map $p : Y \to X$ such that $p_*(\pi_1(Y, y_0)) = H$ for some $y_0 \in Y$ with $p(y_0) = x_0$.*

Proof. Let
$$Y = \{(x, [[u]]) \mid [[u]] \text{ is equiv class of curves } u : x_0 \mapsto x \text{ under } u \sim v \text{ if } u \cdot v^{-1} \in H\}).$$
Define a manifold structure on Y as follows: for each $y = (x, [[u]]) \in Y$ and an open ball U_x around x, let
$$[U_x, y] := \{(z, [[u \cdot v]]) \mid z \in U_x, v : x \mapsto z \ \text{ is contained in } U_x\}.$$
We first show that the collection $\{[U_x, y] \mid \in Y\}$ forms a basis of topology on Y. Let $y_1 = (x_1, [[u_1]]), y_2 = (x_2, [[u_2]]) \in Y$ and say $y_0 \in [U_{x_1}, y_1] \cap [U_{x_2}, y_2]$. We can write $y_0 = (z_0, [[u_1 \cdot v_1]])$ for some curve v_1 from x_1 to z_0 contained in U_{x_1} and $y_0 = (z_0, [[u_2 \cdot v_2]])$ for some curve v_2 from x_2 to z_0 contained in U_{x_2}. By definition, $(u_1 \cdot v_1) \sim (u_2 \cdot v_2)$ in H. Consider a smaller open ball W around z_0 such that $W \subseteq U_{x_1} \cap U_{x_2}$, then for every $y \in (W, y_0)$, there exists a curve w from z_0 to z contained in W such that $y = (z, [[u_1 \cdot v_1 \cdot w]]) = (z, [[u_2 \cdot v_2 \cdot w]])$, which implies that $y \in [U_{x_1}, y_1] \cap [U_{x_2}, y_2]$. Hence $\{[U_x, y] \mid \in Y\}$ forms a basis of topology on Y.

Now we show that Y is Hausdorff. It suffices to prove that for $y_1 = (x, [[u_1]]), y_2 = (x, [[u_2]]) \in Y$ with $[[u_1]] \neq [[u_2]]$, there exist U_x such that $[U_x, y_1] \cap [U_x, y_2] = \emptyset$. If not, then exists $(z, r) \in [U_x, y_1] \cap [U_x, y_2]$. By definition, $r = [[u_1 \cdot v_1]]$ for some curve v_1 from x to z contained in U_x and $r = [[u_2 \cdot v_2]]$ for some curve v_2 from x to z contained in U_x. Since v_1, v_2 are both contained in U_x, we have $v_1 \sim v_2$, which implies that $u_1 \cdot v_1 \sim u_2 \cdot v_2$, hence $[[u_1]] = [[u_2]]$, a contradiction. Hence Y is Hausdorff.

Define $p : Y \to X$ by $p(x, [[u]]) = x$. Suppose u is a curve from x_0 to x. Let u_s be the curve from x_0 to $u(s)$ given by $u_{s(t)} = u(st)$, then

$$\hat{u} : [0,1] \to Y, \quad t \mapsto (u(t), [[v \cdot u_t]])$$

is a lift of u with initial point $(x_0, [[x_0]])$. Hence p has the curve lifting property. Since p is a local homeomorphism, p is a covering map. Finally, by definition, we have $p_*(\pi_1(Y, (x_0, [[x_0]])) = H$. ∎

From the construction above, we can see that the universal covering of X is given by taking $H = 0$. Since universal covering is unique up to isomorphism, we know that $p : Y \to X$ is the universal covering of X if and only if Y is simply connected.

Remark. Given a universal covering $p : \tilde{X} \to X$ and a covering $p' : Y \to X$, the lifting $f : \tilde{X} \to Y$ such that $p' \circ f = p$ is in fact also a covering map.

Remark. A covering $p : Y \to X$ is Galois iff $\pi_1(Y, y_0) \lhd \pi_1(X, x_0)$, in this case $\mathrm{Deck}\left(\frac{Y}{X}\right) \cong \mathrm{Deck}\, \frac{\frac{\tilde{X}}{X}}{\pi_1}(Y, y_0)$, where $\tilde{X}$ is the universal covering of X. Hence the Galois coverings of X are in one-to-one correspondence with the normal subgroups of $\pi_1(X, x_0)$.

1.5. Covering for Riemann Surfaces

DEFINITION 1.42. *A topological space X is locally compact if it is Hausdorff and every point has a compact neighborhood. A continuous map $f : X \to Y$ is proper if X, Y are locally compact and $f^{-1}(K)$ is compact for every compact subset K of Y.*

PROPOSITION 1.43. *A proper map is closed.*

Proof. Recall that in a locally compact space, a subset is closed iff its intersection with every compact subset is compact. Let $f : X \to Y$ be a proper map and A be a closed subset of X. For every compact subset K of Y, we have

$$f(A) \cap K = f(A \cap f^{-1}(K)),$$

which is compact since $A \cap f^{-1}(K)$ is compact and f is continuous. Hence $f(A)$ is closed. ∎

PROPOSITION 1.44. *If $p : Y \to X$ is proper and discrete, then $\forall x \in X$, $p^{-1}(x)$ is finite.*

Proof. Since p is discrete, for every $x \in X$, $p^{-1}(x)$ is discrete and compact, hence $p^{-1}(x)$ is finite. ∎

Theorem 1.45. *If $p: Y \to X$ is proper and a local homeomorphism, then p is a covering map.*

Proof. Let $x \in X$ and $p^{-1}(x) = \{y_1, ..., y_n\}$. Since p is a local homeomorphism, for each y_j, there exists a neighborhood W_j of y_j such that $p|_{W_j} : W_j \to p(W_j)$ is a homeomorphism. Since Y is Hausdorff, we can choose W_j such that $\{W_j\}$ are pairwise disjoint. Let $V = W_1 \cup ... \cup W_n$ and $\mathcal{U} = \bigcap_{j=1}^n p(W_j)$, then $p^{-1}(\mathcal{U}) \subseteq V$. Set $V_j = W_j \cap p^{-1}(\mathcal{U})$, then $p|_{V_j} : V_j \to \mathcal{U}$ is a homeomorphism for every j and $p^{-1}(\mathcal{U}) = \bigsqcup_{j=1}^n V_j$, which shows that p is a covering map. ∎

We now assume that X, Y are Riemann surfaces and $p: Y \to X$ is a non-constant holomorphic map.

By the identity theorem $p^{-1}(x)$ is discrete for every $x \in X$.

Definition 1.46. *A point $y \in Y$ is a branch point of p if there does not exist a neighborhood U of y such that $p|_U$ is injective. A non-constant holomorphic map $p: Y \to X$ is unbranched if it has no branch point.*

Proposition 1.47. *A non-constant holomorphic map $p: Y \to X$ is unbranched iff it is a local homeomorphism.*

Proof. "$\Rightarrow$": For $y \in Y$, exist a neighborhood U of y such that $p|_U$ is injective. Since p is holomorphic, by the open mapping theorem, $p(U)$ is open in X, which implies that $p|_U : U \to p(U)$ is a homeomorphism. "$\leq$": By definition, p has no branch point. ∎

Lemma 1.48. *A proper unbranched holomorphic map $p: Y \to X$ is a covering map.*

Proof. Since p is proper and a local homeomorphism, p is a covering map. ∎

In general, let A be the set of branch points of p, then A is closed and discrete. Let $B = p(A)$ be the set of critical values of p.

Proposition 1.49. *If p is proper, then $p: Y \setminus p^{-1}(B) \to X \setminus B$ is an unbranched holomorphic covering map.*

Proof. Since p is proper, $p^{-1}(B)$ is compact, hence $Y \setminus p^{-1}(B)$ is open. Since B is discrete, $X \setminus B$ is open. By definition of branch point, $p: Y \setminus p^{-1}(B) \to X \setminus B$ is a local homeomorphism. Since p is proper, $p: Y \setminus p^{-1}(B) \to X \setminus B$ is a covering map. ∎

Note that the fibers of a covering map have the same cardinality, hence we say $p: Y \to X$ is an n-sheeted covering if $|p^{-1}(x)| = n$ for every $x \in X$.

Theorem 1.50. *Let $\mathbb{D} = \{z \in \mathbb{C} \mid |z| < 1\}$ be the unit disk. If $p: X \to \mathbb{D}$ is a proper non-constant holomorphic map, which is unbranched on $\mathbb{D}^* := \mathbb{D} \setminus \{0\}$, then p is isomrophic to the map $\mathbb{D} \to \mathbb{D}, z \mapsto z^k$ for some $k \in \mathbb{N}$.*

Proof. Let $X^* = p^{-1}(\mathbb{D}^*)$, then $p: X^* \to \mathbb{D}^*$ is an unbranched holomorphic covering map. Let $H = \{z \in \mathbb{C} \mid \text{Re}\,(z) < 0\}$ be the left half-plane, then the map $\exp: H \to \mathbb{D}^*$ is a universal covering of $\mathbb{D}^*$. Hence there exists a lift $f: H \to X^*$ of exp such that

$p \circ f = \exp$. Moreover, X^* is isomorphic to $\frac{H}{G}$, where G is a subgroup of Deck $\left(\frac{H}{\mathbb{D}^*}\right) \cong \pi_1(\mathbb{D}^*) \cong \mathbb{Z}$.

If $G = 0$, then $X^* \cong H$, which implies that p has an infinite fiber, contradicting the fact that p is proper. Hence $G = k\mathbb{Z}$ for some $k \in \mathbb{N}$. Let p_k denote the map $p_k : \mathbb{D}^* \to \mathbb{D}^*, z \mapsto z^k$. Note that $p_*(\pi_1(X^*, x_0)) = p_{k*}(\pi_1(\mathbb{D}^*, z_0))$, hence p and p_k are isomorphic as covering maps. Since p and p_k are locally biholomorphic, the isomorphism between p and p_k is in fact a local biholomorphism, hence an isomorphism of holomorphic covering maps. It now remains to show that p can be extended to a holomorphic map on X.

If $p^{-1}(0) = \{b_1, ..., b_n\}$, then exists disjoint neighborhoods V_i of b_i and $B_{r(0)}$ of 0 such that $p^{-1}\left(B_{r(0)}\right) \subseteq \bigcup_{i=1}^n V_i$. However, $p^{-1}\left(B^*_{r(0)}\right) \cong p_k^{-1}\left(B^*_{r(0)}\right) = B_{\sqrt[k]{r}}(0)^*$, which is connected, hence $p^{-1}\left(B_{r(0)}\right)$ is connected, which implies that $n = 1$. Thus $p^{-1}(0) = \{b_1\}$, which implies that $p|_{X^*}$ can be extended to a holomorphic map on X by defining $p(b_1) = 0$. Hence p is isomorphic to the map $\mathbb{D} \to \mathbb{D}, z \mapsto z^k$ for some $k \in \mathbb{N}$. ∎

DEFINITION 1.51. *Let X be a Riemann surface and $a \in X$. Define the stalk of holomorphic functions at a to be the set of equivalence classes of $f \in \mathcal{O}(U)$ where U is a neighborhood of a, under the equivalence relation $f \sim g$ if there exists a neighborhood W of a such that $W \subseteq U \cap V$ and $f|_W = g|_W$. The stalk of holomorphic functions at a is denoted by $\mathcal{O}_a$. Let $\rho_a : \mathcal{O}(U) \to \mathcal{O}_a$ be the natural map given by $\rho_{a(f)} = [f]$.*

DEFINITION 1.52. *Let*

$$|\mathcal{O}| := \bigsqcup_{a \in X} \mathcal{O}_a$$

be the disjoint union of stalks of holomorphic functions on X. Define a basis of topology on $|\mathcal{O}|$ as follows: for each $U \subseteq X$ and $f \in \mathcal{O}(U)$, let

$$[U, f] := \{\rho_{x(f)} \mid x \in U\} \subseteq |\mathcal{O}|.$$

Let $p : |\mathcal{O}| \to X$ be the natural map given by $p\left(\rho_{x(f)}\right) = x$.

PROPOSITION 1.53. *The collection* `cal` *B of all sets $[U, f]$ forms a basis of topology on $|\mathcal{O}|$.*

Proof. If $\phi \in [U, f] \cap [V, g]$, $p(\phi) = x \in U \cap V$ and $\phi = \rho_{x(f)} = \rho_{x(g)}$, then there exists a neighborhood W of x such that $W \subseteq U \cap V$ and $f|_W = g|_W$, which implies that $[W, f] = [W, g] \subseteq [U, f] \cap [V, g]$. Hence `cal` B forms a basis of topology on $|\mathcal{O}|$.

Let $\phi_1, \phi_2 \in |\mathcal{O}|$ be distinct germs of holomorphic functions at x_1, x_2 respectively. If $x_1 \neq x_2$, then just take disjoint neighborhoods of x_1 and x_2. If $x_1 = x_2 = x$, let $\rho_{x(f_1)} = \phi_1$ and $\rho_{x(f_2)} = \phi_2$ with $f_1, f_2 \in \mathcal{O}(U)$ for some neighborhood U of x. Then $[U, f_1|_U]$ and $[U, f_2|_U]$ are disjoint neighborhoods of ϕ_1 and ϕ_2 respectively. Hence $|\mathcal{O}|$ is Hausdorff. ∎

PROPOSITION 1.54. *$p : |\mathcal{O}| \to X$ is a local homeomorphism and hence inherits a complex structure from X such that p is holomorphic.*

Proof. For $\phi = \rho_{x(f)} \in |\mathcal{O}|$, let U be a neighborhood of x such that $f \in \mathcal{O}(U)$, then $p|_{[U,f]} : [U, f] \to U$ is a homeomorphism with inverse given by $x' \mapsto \rho_{x'}(f)$, which shows that p is a local homeomorphism. Hence we can define a complex structure on $|\mathcal{O}|$ by taking the pullback of the complex structure on X via p, which makes p holomorphic. ∎

Let X be a Riemann surface and $a \in X$, $\phi \in \mathcal{O}_a$. We want to find a maximal analytic continuation of ϕ.

Let Y be the connected component of $|\mathcal{O}|$ containing ϕ, then Y is a Riemann surface and $p : Y \to X$ is locally biholomorphic. Define the function

$$f : Y \to \mathbb{C}, \quad \psi \mapsto \psi(p(\psi)).$$

Then f is well-defined and holomorphic. Moreover, $p_*\left(\rho_{\psi(f)}\right) = \psi$.

1.6. Analytic Continuation

Let X be a Riemann surface and $a \in X$, $\phi \in \mathcal{O}_{X,a}$. Let Y be the connected component of $|\mathcal{O}|$ containing ϕ, then Y is a Riemann surface and $p : Y \to X$ is locally biholomorphic and induces an isomorphism $p^* : \mathcal{O}_{X,x} \to \mathcal{O}_{Y,y}$ for every $y \in Y$ with $p(y) = x$. Define the function

$$f : Y \to \mathbb{C}, \quad \psi \mapsto \psi(p(\psi)).$$

Then f is well-defined and holomorphic. Moreover, $p_*\left(\rho_{\psi(f)}\right) = \psi$. Hence f is a maximal analytic continuation of ϕ.

In this section, we give a more concrete construction of analytic continuation of ϕ along curves.

Let $v : I \to Y$ be a curve with $v(0) = \phi$. Let $u = p \circ v$, then u is a curve in X with $u(0) = a$. For each $t \in I$, let ϕ_t be the germ of holomorphic functions at $u(t)$ given by $\phi_t = p_*\left(\rho_{v(t)}(f)\right)$, then $\phi_0 = \phi$.

For $t_0 \in I$, exists $V_0 \subset Y$ such that $v(t_0) \in V_0$ and $U_0 = p(V_0)$ is a neighborhood of $u(t_0)$ such that $p|_{V_0} : V_0 \to U_0$ is a homeomorphism, and let $q := \left(p|_{V_0}\right)^{-1} : U_0 \to V_0$. Let $g_0 = q^*\left(f|_{V_0}\right) \in \mathcal{O}(U_0)$, then $p_*\left(\rho_{\eta(f)}\right) = \rho_{p(\eta)}(g_0)$ for every $\eta \in V_0$. Since v is continuous, there exists a neighborhood $T_0 \subset I$ of t_0 such that $v(T_0) \subset V_0$, i.e. $u(T_0) \subset U_0$. For all $t \in T_0$, we have $\phi_t = p_*\left(\rho_{v(t)}(f)\right) = \rho_{p(v(t))}(g_0) = \rho_{u(t)}(g_0)$. Since I is compact, exists $0 = t_0 < t_1 < ... < t_n = 1$ and for each i, there exists V_i such that $v(t_i) \in V_i$ and $U_i = p(V_i)$ is a neighborhood of $u(t_i)$ such that $p|_{V_i} : V_i \to U_i$ is a homeomorphism, and let $g_i = \left(p|_{V_i}\right)^*\left(f|_{V_i}\right) \in \mathcal{O}(U_i)$, then for each $t \in [t_{i-1}, t_i]$, we have $\phi_t = p_*\left(\rho_{v(t)}(f)\right) = \rho_{p(v(t))}(g_i) = \rho_{u(t)}(g_i)$. Hence ϕ_1 is the analytic continuation of ϕ along the curve u.

DEFINITION 1.55. *Suppose X is a Riemann surface and $a \in X$, $\phi \in \mathcal{O}_a$. A quadrupel (Y, p, f, b) is called a analytic continuation of ϕ if*

(i) *Y is a Riemann surface and $p : Y \to X$ is an unbranched holomorphic covering map;*
(ii) *f is a holomorphic on Y;*
(iii) *b is a point of Y such that $p(b) = a$ and*

$$p_*\left(\rho_{b(f)}\right) = \phi.$$

An analytic continuation (Y, p, f, b) of ϕ is maximal if for any analytic continuation (Z, q, g, c) of ϕ, there exists a holomorphic map $F : Z \to Y$ such that $p \circ F = q$, $F(c) = b$ and $F^*(f) = g$.

Remark. The construction (Y, p, f, ϕ) from the sheaf of holomorphic functions gives a maximal analytic continuation of ϕ. For another analytic continuation (Z, q, g, c) of

ϕ, define $F : Z \to Y$ as follows: for $z \in Z$, let $q(z) = x$. Since Y consists of all germs obtained by analytic continuation of ϕ, there exists $\eta \in Y$ such that $q_*\left(\rho_{z(g)}\right) = \eta$. Define $F(z) = \eta$, then $p \circ F(z) = p(\eta) = q(z)$ and $F^*(f) = g$ by definition of f.

THEOREM 1.56. *Let X be a Riemann surface and A be a discrete closed subset of X. Let $X' = X \setminus A$. Let $\pi' : Y' \to X'$ be a proper unbranched holomorphic covering map, then exists a Riemann surface Y and a proper holomorphic map $\pi : Y \to X$ such that $\pi' \circ \phi = \pi$ on $Y \setminus \pi^{-1}(A)$, where ϕ is a biholomorphism from $Y \setminus \pi^{-1}(A)$ to Y'.*

Proof. Let $A = \{a_i \mid i \in \Lambda\}$ and each a_i has a neighborhood U_i such that $U_i \cap U_j = \emptyset$ for $i \neq j$. Let $U_i^* = U_i \setminus \{a_i\}$. Since $\pi' : Y' \to X'$ is proper, $\pi'^{-1}(U_i^*)$ is a disjoint union of finitely many connected components V_{ij}, each of which is biholomorphic to U_i^* via π'. Let U_i be isomorphic to the unit disk $\mathbb{D}$ via z_i

$$\begin{tikzcd} V_{ij}^* \arrow[r, "\phi_{ij}^*"] \arrow[d, "\pi'"'] & \mathbb{D}_i^* \arrow[d, "z \mapsto z^{k_{ij}}"] \\ U_i^* \arrow[r, "z_i"'] & \mathbb{D}_i^* \end{tikzcd}$$

Choose P_{ij} for each $i \in \Lambda$ and $j = 1, ...n_i$, where n_i is the number of connected components of $\pi'^{-1}(U_i^*)$. The topology near P_{ij} is given by the neighborhood basis $\{P_{ij\}} \cup \left(\pi'^{-1}(W_{it}) \cap V_{ij}^*\right)$ where $W_{it}, t \in I_i$ is a neighborhood basis of a_i. The complex charts near P_{ij} is given by the mappings $\phi_{ij} : V_{ij} \to D_i$. ∎

1.7. *Riemann Surfaces of Algebraic Functions*

Algebraic functions are functions that satisfy a polynomial equation with coefficients in $\mathcal{M}(X)$, the field of meromorphic functions on X. Most algebraic functions are multivalued, hence we want to find a Riemann surface on which the algebraic function is single-valued.

PROPOSITION 1.57. *If $c_1, ..., c_n \in \mathcal{O}(B_R(0))$ are holomorphic functions on the disk $B_R(0)$, and $w_0 \in \mathbb{C}$ is a simple zero of the polynomial*

$$T^n + c_1(0)T^{n-1} + ... + c_{n(0)} \in \mathbb{C}[T],$$

then there exists $0 < r \leq R$ and a holomorphic function $\phi \in \mathcal{O}\left(B_{r(0)}\right)$ such that $\phi(0) = w_0$ and

$$\phi^n + c_1\phi^{n-1} + ... + c_n = 0.$$

Proof. For $z \in B_R(0)$, $w \in \mathbb{C}$, let

$$F(z, w) = w^n + c_1(z)w^{n-1} + ... + c_{n(z)}.$$

By assumption, there exists $\varepsilon > 0$ such that $F(0, w)$ has a unique zero in $B_\varepsilon(w_0)$. Since $F(0, w) \neq 0$ on $\partial B_\varepsilon(w_0)$, by continuity, there exists $0 < r \leq R$ such that $F(z, w) \neq 0$ on $\partial B_\varepsilon(w_0)$ for every $z \in B_{r(0)}$. By the argument principle, for every $z_0 \in B_{r(0)}$,

$$n(z_0) := \frac{1}{2\pi i}\int_{\partial B_\varepsilon(w_0)} \frac{\frac{\partial F}{\partial} w(z_0, w)}{F(z_0, w)} dw$$

is the number of zeros of $F(z_0, w)$ in $B_\varepsilon(w_0)$ counting multiplicity. Since $n(z_0)$ is an integer-valued continuous function on $B_{r(0)}$, $n(z_0)$ is constant on $B_{r(0)}$. Since $n(0) = 1$, we have $n(z_0) = 1$ for every $z_0 \in B_{r(0)}$. Recall that

$$\frac{1}{2\pi i}\int_{\partial B_\varepsilon(w_0)} \frac{w\frac{\partial F}{\partial} z(z_0, w)}{F(z_0, w)} dw$$

is equal to the sum of zeros of $F(z_0, w)$ in $B_\varepsilon(w_0)$. Define $\phi(z_0)$ to be the above integral, then ϕ is holomorphic on $B_{r(0)}$ and $F(z_0, \phi(z_0)) = 0$ for every $z_0 \in B_{r(0)}$. ∎

LEMMA 1.58. *Let X be a Riemann surface and $a \in X$. If*

$$\overline{p}(T) = T^n + c_1 T^{n-1} + \ldots + c_n \in \mathcal{O}_{a[T]}$$

and

$$p(T) := T^n + c_1(a) T^{n-1} + \ldots + c_{n(a)} \in \mathbb{C}[T]$$

has n distinct zeros $w_1, \ldots, w_n$, then there exists $\phi_1, \ldots, \phi_n \in \mathcal{O}_a$ such that $\phi_{j(a)} = w_j$ and

$$\overline{p}(T) = \prod_{j=1}^{n}(T - \phi_j).$$

Proof. Apply the above proposition to each w_j iteratively, we can find $\phi_1, \ldots, \phi_n$ such that $\phi_{j(a)} = w_j$ and $\overline{p}(T) = (T-\phi_1)\ldots(T-\phi_n)$ in $\mathcal{O}_{a[T]}$. ∎

Let X, Y be Riemann surfaces and $\pi : Y \to X$ be an n-sheeted holomorphic covering map. Let $f \in \mathcal{M}(Y)$. Fix a $x \in X$ and a neighborhood U of x such that $\pi^{-1}(U)$ is a disjoint union of n connected components $V_1, \ldots, V_n$, each of which is biholomorphic to U via π. Let $\tau_j : U \to V_j$ be the inverse of $\pi|_{V_j}$, and let $f_i := \tau_i^* f = f \circ \tau_i \in \mathcal{M}(U)$. Consider

$$\prod_{j=1}^{n}(T - f_j) = T^n + c_1 T^{n-1} + \ldots + c_n \in \mathcal{M}(U)[T]$$

with $c_i = (-1)^i s_{i(f_1, \ldots, f_n)} \in \mathcal{M}(U)$ where s_i is the i-th elementary symmetric polynomial. Let x run through X to get $c_i \in \mathcal{M}(X)$. Then f satisfies the polynomial equation

$$f^n + c_1 f^{n-1} + \ldots + c_n = 0$$

in the sense that for every $y \in Y$, $x = \pi(y)$ and j such that $y \in V_j$, we have

$$f(y)^n + c_1(x) f(y)^{n-1} + \ldots + c_{n(x)} = f_{j(x)}^n + c_1(x) f_{j(x)}^{n-1} + \ldots + c_{n(x)} = 0.$$

More generally, if $\pi : Y \to X$ is now an n-sheeted branched holomorphic covering map. Let $A \subset X$ be the set of critical values of π and $B = \pi^{-1}(A)$. For $x \in X \setminus A$, we can define f_j, c_i as above. If c_i can be extended to holomorphic functions on X for every i, then f can also be extended to a holomorphic function on Y: for $a \in A$, let $\pi^{-1}(a) = \{b_1, \ldots, b_m\}$ and let $\pi^{-1}(U) = V$, $U \cap A = \{a\}$. For all $y \in V \setminus \{b_1, \ldots, b_m\}$, we have

$$f(y)^n + c_1(\pi(y)) f(y)^{n-1} + \ldots + c_{n(\pi(y))} = 0,$$

which implies that $f(y)$ is bounded near b_i for every i iff c_i is bounded near a for every i.

Similarly, if c_i can be extended to meromorphic functions on X for every i, then f can also be extended to a meromorphic function on Y.

PROPOSITION 1.59. *Let X, Y be Riemann surfaces and $\pi : Y \to X$ be an n-sheeted branched holomorphic covering map. Let $f, c_1, ..., c_n$ be constructed as above.*

(i) *If $f \in \mathcal{M}(Y)$, then $f_n + (\pi^* c_1) f^{n-1} + ... + (\pi^* c_n) = 0$ in $\mathcal{M}(Y)$.*

(ii) *Let $L := \mathcal{M}(Y)$, $K := \pi^* \mathcal{M}(X)$, then $\frac{L}{K}$ is an algebraic extension of degree at most n.*

(iii) *If exists $x \in X$ with $\pi^{-1}(x) = \{y_1, ..., y_n\}$ and some $f \in \mathcal{M}(Y)$ such that $f(y_i)$ are distinct for $i = 1, ..., n$, then $[L : K] = n$.*

Part (i) is clear by construction. For part (ii), note that $[K(f) : K] \leq n$ since f satisfies a polynomial equation of degree n with coefficients in K. Let $f_0 \in \mathcal{M}(Y)$ be such that $n_0 = [K(f_0) : K]$ is maximal. For each $f \in \mathcal{M}(Y)$, since char $K = 0$*, by the primitive element theorem, there exists $g \in \mathcal{M}(Y)$ such that $K(f, f_0) = K(g)$, then*

$$[K(g) : K] = [K(f, f_0) : K] \geq [K(f_0) : K] = n_0 \geq [K(g) : K],$$

which implies that $n_0 = [K(g) : K]$ and hence $f \in K(g) = K(f_0)$, thus $L = K(f_0)$ and $[L : K] = n_0 \leq n$.

For part (iii), if $[K(f) : K] = m \leq n$, then f satisfies a polynomial equation of degree m, say $f^m + \pi^* g_1 f^{m-1} + ... + \pi^* g_m = 0$, then for all $x \in X$, $y \in \pi^{-1}(x)$, $f(y)$ is a root of

$$T^m + g_1(x) T^{m-1} + ... + g_{m(x)} = 0,$$

which implies that f takes at most m distinct values on $\pi^{-1}(x)$. By assumption, f takes n distinct values on $\pi^{-1}(x)$, hence $m = n$ and $[L : K] = n$.

THEOREM 1.60. *Let X be a Riemann surface and*

$$p(T) = T^n + c_1 T^{n-1} + ... + c_n \in \mathcal{M}(X)[T]$$

be a polynomial irreducible over $\mathcal{M}(X)$, then there exists a Riemann surface Y, a n-sheeted branched holomorphic covering map $\pi : Y \to X$ and $F \in \mathcal{M}(Y)$ such that $\pi^* p(F) = 0$ in $\mathcal{M}(Y)$. Moreover, (Y, π, F) is unique up to isomorphism in the sense that if (Z, q, G) is another triple with the same property, then there exists a biholomorphism $f : Y \to Z$ such that $q \circ f = \pi$ and $f^* G = F$.

$$\begin{tikzcd} & Z \arrow[d, "q"] \\ Y \arrow[ur, "f"] \arrow[r, "\pi"'] & X \end{tikzcd}$$

Proof. Since char $\mathcal{M}(X) = 0$ and p is irreducible over $\mathcal{M}(X)$, the discriminant $\delta \in \mathcal{M}(X)$ of p is nonzero, hence the set zeros of δ is discrete and closed. Let A be the union of the set of zeros of δ and the set of poles of c_i for $i = 1, ..., n$, then A is a discrete closed subset of X. Let $X' = X \setminus A$. By definition, $c_i \in \mathcal{O}(X')$ and for all $x \in X'$, $p(T) \in \mathcal{O}(X')[T]$ has n distinct zeros, say $w_1, ..., w_n$. By Proposition 1.59, there exists a neighborhood U_x of x and $\phi_1, ..., \phi_n \in \mathcal{O}(U_x)$ such that $\phi_{j(x)} = w_j$ and $p(T) = \prod_{j=1}^{n} (T - \phi_j)$ in $\mathcal{O}(U_x)[T]$.

Let

$$Y' = \{\phi \in |\mathcal{O}| : \phi \in \mathcal{O}_x, x \in X', p(\phi) = 0\}$$

and $\pi' : Y' \to X'$ be the natural map given by $\pi'(\phi) = x$. π' is n to 1 hence proper, also $\pi'^{-1}(U_x) = \bigcup_{j=1}^{n} [U_x, \phi_j]$ is a disjoint union of n connected components (each root w_j are distinct hence ϕ_j are distinct) each of which is biholomorphic to U_x via π'. Hence $\pi' : Y' \to X'$ is a proper unbranched holomorphic covering map. Let $f : Y' \to \mathbb{C}$ be given by $f(\phi) = \phi(\pi'(\phi))$. Note that since $\pi'_*\left(\rho_{\phi(f)}\right) = \phi$ for every $\phi \in Y'$, we have

$$\pi'^* p(f) = p\left(\pi'_*\left(\rho_{\phi(f)}\right)\right) = p(\phi) = 0$$

in $\mathcal{M}(Y')$.

By the above theorem, we can extend $\pi' : Y' \to X'$ to a proper holomorphic map $\pi : Y \to X$, and f can also be extended to a meromorphic function on Y, denoted by F, such that $\pi^* p(F) = 0$ in $\mathcal{M}(Y)$.

If Y is not connected, let $Y_1, ..., Y_k$ be the connected components of Y with $k \geq 2$ and $\pi : Y_i \to X$ be n_i-sheeted coverings. From each $F|_{Y_i}$, we may construct a degree n_i polynomial P_i as in Proposition 1.59, then $p(T) = \prod_{i=1}^{k} P_{i(T)}$ in $\mathcal{M}(X)[T]$, contradicting the irreducibility of p. Hence Y is connected.

For the uniqueness, let (Z, q, G) be another triple with the same property. Let $B = \{\text{poles of } G\} \cup \{\text{the branched points of } \tau\} \subset Z$ and $A' = \tau(B)$. Set $X'' = X' \setminus A'$, $Y'' = \tau^{-1}(X'')$ and $Z'' = q^{-1}(X'')$. Let σ map G to F by taking the germ of G at $z \in Z''$ to the germ of F at $\tau(z)$. Then σ is a well-defined holomorphic map from Z'' to Y'' such that $\pi \circ \sigma = q$ and $\sigma^* F = G$. Since Z'' is dense in Z, σ can be extended to a holomorphic map from Z to Y, which is still denoted by σ. Since $\pi \circ \sigma = q$ and both π, q are proper, σ is proper. Since $\pi \circ \sigma = q$, σ is injective, hence $\sigma : Z \to Y$ is a biholomorphism with inverse given by the same construction by sending the germ of F at $y \in Y''$ to the germ of G at $\sigma^{-1}(y)$. ∎

Remark. In fact, Deck $\left(\frac{Y}{X}\right) \cong \mathrm{Gal}\left(\frac{L}{K}\right)$ where $L = \mathcal{M}(Y)$ and $K = \pi^* \mathcal{M}(X)$, and the isomorphism is given by sending $\sigma \in$ Deck $\left(\frac{Y}{X}\right)$ to the automorphism of L over K given by $f \mapsto \sigma^* f$. Hence the Galois group of the algebraic extension $\frac{L}{K}$ can be interpreted as the group of deck transformations of the branched holomorphic covering map $\pi : Y \to X$.

1.8. Differential Forms

A differential 1-form on $B_{r(0)}$ can be written as $\omega = f, dx + g, dy$ where $f, g \in \mathcal{E}\left(B_{r(0)}\right)$ are smooth functions. The collection of all differential 1-forms on $B_{r(0)}$ is denoted by $\mathcal{E}^{(1)}\left(B_{r(0)}\right)$. ω is closed iff ω is closed iff $\frac{\partial F}{\partial} y = \partial \frac{g}{\partial} x$. ω is exact iff there exists $F \in \mathcal{E}\left(B_{r(0)}\right)$ such that $\omega = dF = \frac{\partial F}{\partial} x, dx + \frac{\partial F}{\partial} y, dy$. exact implies closed, but in general closed does not imply exact.

Let X be a Riemann surface, a differential 1-form on X is a collection of differential 1-forms on each chart of X such that they are compatible on the overlaps, i.e. on each char $U \subset X$, $\omega|_U = f, dx + g, dy$ where $f, g \in \mathcal{E}(U)$ are smooth functions. Given a covering map $p : Y \to X$ and a differential 1-form ω on X, we can define the pullback $p^* \omega$ of ω on Y by pulling back the local expressions of ω. Precisely, on a local chart U with $p^{-1}(U) = \bigcup V_i$ where V_i are disjoint and $p|_{V_i} : V_i \to U$ is a homeomorphism, if $\omega|_U = f, dx + g, dy$, then $(p^* \omega)|_{V_i} = (f \circ p), d(x \circ p) + (g \circ p), d(y \circ p)$. Given a closed differential 1-form ω on X, we would like to know whether there exists a holomorphic function F on Y such that $dF = p^*\omega$.

THEOREM 1.61. *Suppose X is a Riemann surface and ω is a closed differential 1-form on X. Then there exists a covering map $p : Y \to X$ and a holomorphic function F on Y such that $dF = p^*\omega$.*

Proof. For $U \subset X$, let
$$\mathcal{F}(U) = \{f \in \mathcal{E}(U) \mid df = \omega|_U\},$$
and let $\mathcal{F}$ be the sheaf of $\mathcal{E}$-modules on X given by $U \mapsto \mathcal{F}(U)$. Recall that $|\mathcal{F}| = \bigsqcup_{x\in X} \mathcal{F}_x$ is the disjoint union of stalks of $\mathcal{F}$ on X.

Consider the projection map $p : |\mathcal{F}| \to X$ given by $p\big(\rho_{x(f)}\big) = x$ where $\rho_{x(f)}$ is the germ of f at x. Locally, if U is a neighborhood of x isomorphic to the unit disk $\mathbb{D}$, so that there exists a local primitive $f \in \mathcal{F}(U)$, then $p^{-1}(U) = \bigcup_{c\in\mathbb{C}}[U, f + c]$. Pick Y to be a connected component of $|\mathcal{F}|$, then $p : Y \to X$ is a local homeomorphism and hence inherits a complex structure from X such that p is holomorphic. Define $F : Y \to \mathbb{C}$ by $F(\phi) = \phi(p(\phi))$, then F is well-defined and holomorphic, and $p_*\big(\rho_{\phi(F)}\big) = \phi$ for every $\phi \in Y$. Hence $dF = p^*\omega$ on Y. ∎

LEMMA 1.62. *Suppose X is a Riemann surface and $\pi : \tilde{X} \to X$ is the universal covering map, then for every closed differential 1-form ω on X, there exists a holomorphic function F on $\tilde{X}$ such that $dF = \pi^*\omega$.*

Proof. Let $p : Y \to X$ and F be given by the above theorem, Since π is the universal covering map, there exists a holomorphic map $\tau : \tilde{X} \to Y$ such that $p \circ \tau = \pi$, hence τ^*F is a holomorphic function on $\tilde{X}$ such that $d(\tau^*F) = \tau^*dF = \tau^*p^*\omega = \pi^*\omega$. ∎

THEOREM 1.63. *Suppose X is a Riemann surface and $\pi : \tilde{X} \to X$ is the universal covering map. For every closed differential 1-form ω on X, exists a primitive $F \in \mathcal{E}\big(\tilde{X}\big)$ of $\pi^*\omega$. If $c : I \to X$ is a curve and $\tilde{c} : I \to \tilde{X}$ is a lift of c, then*
$$\int_c \omega = F(\tilde{c}(1)) - F(\tilde{c}(0))$$
is well-defined, i.e. independent of the choice of the primitive F and the lift $\tilde{c}$.

Proof. For another primitive $F' \in \mathcal{E}\big(\tilde{X}\big)$ of $\pi^*\omega$, we have $d(F - F') = 0$, hence $F - F'$ is a constant function on $\tilde{X}$.

Let $\tilde{c}_1, \tilde{c}_2 : I \to \tilde{X}$ be two lifts of c. Since $\pi : \tilde{X} \to X$ is Galois, there exists a deck transformation $\sigma : \tilde{X} \to \tilde{X}$ such that $\sigma \circ \tilde{c}_1 = \tilde{c}_2$, hence
$$d(\sigma^*F) = \sigma^*dF = \sigma^*\pi^*\omega = \pi^*\omega = dF,$$
which implies that $\sigma^*F - F$ is a constant function on $\tilde{X}$. Thus
$$\begin{aligned} F(\tilde{c}_2(1)) - F(\tilde{c}_2(0)) &= F(\sigma \circ \tilde{c}_1(1)) - F(\sigma \circ \tilde{c}_1(0)) \\ &= \sigma^*F(\tilde{c}_1(1)) - \sigma^*F(\tilde{c}_1(0)) = F(\tilde{c}_1(1)) - F(\tilde{c}_1(0)). \end{aligned}$$
∎

PROPOSITION 1.64. *Let X be a simply connected Riemann surface, then every closed differential 1-form on X is exact.*

Proof. The universal covering map of X is the identity map, hence every closed differential 1-form on X has a primitive on X. ∎

PROPOSITION 1.65. *Let X be a Riemann surface and ω be a closed differential 1-form on X. Let $a, b \in X$ and u, v be two homotopic curves from a to b, then $\int_u \omega = \int_v \omega$.*

Proof. Since u, v are homotopic, their liftings $\tilde{u}, \tilde{v}$ with the same initial point are also homotopic, hence $\tilde{u}(1) = \tilde{v}(1)$, which implies that $\int_u \omega = F(\tilde{u}(1)) - F(\tilde{u}(0)) = F(\tilde{v}(1)) - F(\tilde{v}(0)) = \int_v \omega$. ∎

PROPOSITION 1.66. *Let X be a Riemann surface and ω be a closed differential 1-form on X. Let u, v be two closed curves that are free homotopic, then $\int_u \omega = \int_v \omega$.*

Proof. Say u, v are based on a, b respectively, and let w be a cuve from a to b such that $w \cdot v \cdot w^{-1} \sim u$, then

$$\int_u \omega = \int_{w \cdot v \cdot w^{-1}} \omega = \int_w \omega + \int_v \omega + \int_{w^{-1}} \omega = \int_v \omega.$$

∎

THEOREM 1.67. *Let X be a Riemann surface and $\pi : \tilde{X} \to X$ be the universal covering map.*

(i) *Let $\omega \in \mathcal{E}^{(1)}(X)$ be a closed differential 1-form on X with primitive $dF = \pi^*\omega$, then for all $\sigma \in \mathrm{Deck}\left(\frac{\tilde{X}}{X}\right)$, define $\sigma F = F \circ \sigma^{-1}$, then σF is also a primitive of $\pi^*\omega$ and $F - \sigma F$ is a constant function on $\tilde{X}$. The constant equals*

$$F - \sigma F = \int_\sigma \omega$$

where $\int_\sigma \omega$ is understood by integrating along the curve associated to σ by the isomorphism $\mathrm{Deck}\left(\frac{\tilde{X}}{X}\right) \cong \pi_1(X)$.

(ii) *If $F \in \mathcal{E}(\tilde{X})$ such that for all $\sigma \in \mathrm{Deck}\left(\frac{\tilde{X}}{X}\right)$, exists a constant c_σ such that $F - \sigma F = c_\sigma$, then dF is the pullback of a closed differential 1-form on X.*

Proof.

(1) Let $F - \sigma F = c$. Let z_0, y_0 be two points in $\tilde{X}$ such that $\sigma(y_0) = z_0$ and v be a curve from y_0 to z_0, then $u = \pi \circ v$ is associated to σ by the isomorphism $\mathrm{Deck}\left(\frac{\tilde{X}}{X}\right) \cong \pi_1(X)$. Hence

$$\int_\sigma \omega = \int_u \omega = F(v(1)) - F(v(0)) = F(z_0) - F(y_0) = F(z_0) - \sigma F(z_0) = c.$$

(2) For every $\sigma \in \mathrm{Deck}\left(\frac{\tilde{X}}{X}\right)$,

$$\sigma^* dF = d(\sigma F) = d(F - c_\sigma) = dF,$$

hence dF is invariant under the action of $\mathrm{Deck}\left(\frac{\tilde{X}}{X}\right)$, which implies that dF is the pullback of a closed differential 1-form on X.

∎

1.9. Linear Differential Equations

THEOREM 1.68 (Local solution). *If $A \in M(n \times n, \mathcal{O}(B_R(0)))$ is a matrix of holomorphic functions on the disk $B_R(0)$, then for every $w_0 \in \mathbb{C}^n$, there exists a unique holomorphic function $w \in M\big(n \times 1, \mathcal{O}\big(B_{r(0)}\big)\big)$ such that $w'(z) = A(z)w(z)$ for every $z \in B_R(0)$ and $w(0) = w_0$.*

Proof. Expand into Taylor series for each entry of A, so that we can write

$$A(z) = \sum_{nu=0}^{\infty} A_n u z^n u$$

where $A_n u = \left(a_{ijnu}\right) \in M(n \times n, \mathbb{C})$ for every nu. Suppose the solution has the form

$$w(z) = \sum_{nu=0}^{\infty} c_n u z^n u$$

where $c_n u = (c_{inu}) \in \mathbb{C}^n$ for every nu. If this series converges, then it should satisfy the equation $w'(z) = A(z)w(z)$, which is equivalent to

$$\sum_{k=1}^{\infty} k c_k z^{k-1} = \left(\sum_{nu=0}^{\infty} A_n u z^n u\right)\left(\sum_{nu=0}^{\infty} c_n u z^n u\right) = \sum_{k=0}^{\infty}\left(\sum_{nu=0}^{k} A_n u c_{k-nu}\right) z^k,$$

hence

$$(k+1)c_{k+1} = \sum_{nu=0}^{k} A_n u c_{k-nu}$$

for every $k \geq 0$. Since $c_0 = w_0$ is given, we can determine $c_n u$ for every nu inductively, uniqueness also follows from this. It now remains to show that the series converges.

For arbitrary $0 < r < R$, the series

$$\sum_{nu=0}^{\infty} |a_{ijnu}|\ r^n u$$

converges for every i, j, hence there exists $N > 0$ such that $|a_{ijnu}|\ r^n u \leq \frac{N}{r}$, i.e. $|a_{ijnu}| \leq \frac{N}{r^{nu+1}}$ for every nu. Define

$$b_{ij}(z) = \frac{N}{r}\left(1 - \frac{z}{r}\right)^{-1} = \sum_{nu=0}^{\infty} \frac{N}{r^{nu+1}} z^n u,$$

which is holomorphic in $|z| < r$. Write $w_0 = (w_{i0}) \in \mathbb{C}^n$ and define $K = \max\limits_{1 \leq i \leq n} |w_{i0}|$. Define $\psi(z) = K\big(1 - \frac{z}{r}\big)^{-N}$, which is holomorphic in $|z| < r$, then

$$\psi'(z) = -NK\left(1 - \frac{z}{r}\right)^{-N-1} \cdot \left(-\frac{1}{r}\right) = \frac{N}{r}K\left(1 - \frac{z}{r}\right)^{-N-1} = b_{ij}(z)\psi(z)$$

and satisfies $\psi(0) = K$. Let $v(z) = (\psi(z), ..., \psi(z)) \in \mathbb{C}^n$, then $v'(z) = b_{ij}(z)v(z)$ and $v(0) = (K, ..., K)$. Write

$$\psi(z) = \sum_{nu=0}^{\infty} \psi_n u z^n u.$$

Note that $|a_{ijnu}| \leq \frac{N}{r^{nu+1}} = b_{ijnu}$. We show by induction that $|c_{inu}| \leq \psi_n u$ for every i, nu. For $nu = 0$, $|c_{i0}| = |w_{i0}| \leq K = \psi_0$. Assume that $|c_{inu}| \leq \psi_n u$ for every i and $nu \leq k$, then

$$
\begin{aligned}
(k+1)|c_{i,k+1}| &= |\left(\sum_{nu=0} | \sum_{j=1}^{n} a_{ijnu} c_{j,k-nu}\right) \\
&\leq \sum_{nu=0}^{k} \sum_{j=1}^{n} |a_{ijnu}|\ |c_{j,k-nu}| \\
&\leq \sum_{nu=0}^{k} \sum_{j=1}^{n} b_{ijnu} \psi_{k-nu} \\
&= (k+1)\psi_{k+1},
\end{aligned}
$$

which implies that $|c_{i,k+1}| \leq \psi_{k+1}$. Hence the series $\sum_{nu=0}^{\infty} c_n u z^n u$ converges for every $|z| < r$, which completes the proof. ∎

Let $\omega(X)$ be the space of holomorphic 1-forms on a Riemann surface X, i.e. on every chart U, ω can be written as $\omega = f, dz$ where $f \in \mathcal{O}(U)$.

Theorem 1.69. *Let X be a simply connected Riemann surface and $A \in M(n \times n, \omega(X))$, $x_0 \in X$, then for all $c \in \mathbb{C}^n$, there exists a unique $w \in \mathcal{O}(X)^n$ is a solution of the equation $dw = Aw$ with initial condition $w(x_0) = c$.*

Proof. Note that locally we can write $A = F, dz$ where $F \in M(n \times n, \mathcal{O}(U_0))$ for some chart $U_0 \cong B_1(0)$. Hence the equation $dw = Aw$ is equivalent to $d\frac{w}{d}z = w' = Fw$, which has a unique local solution by the above theorem. Let $\alpha : [0,1] \to X$ be a curve with $\alpha(0) = x_0$. If $\alpha([0,t_1]) \subset U_0$, then let $x_1 = \alpha(t_1)$ and $c_1 = f(\alpha(t_1))$, then by the above theorem again, exists a local solution $f_1 \in \mathcal{O}(U_1)^n$ with $U_1 \cong B_1(0)$ a neighborhood of x_1 such that $f_1(x_1) = c_1$. By the uniqueness of the local solution, f and f_1 coincide on $U_0 \cap U_1$, hence we can glue them together to get a solution on $U_0 \cup U_1$. Since $[0,1]$ is compact, we can cover $\alpha([0,1])$ by finitely many charts $U_0, ..., U_k$ and glue the local solutions together to get a solution along the curve α. The maximal analytic continuation (Y, p, F, y_0) of the local solution f is a covering map, and since X is simply connected, $Y \cong X$ and thus F is a global solution on X. ∎

Lemma 1.70. *Let $p : \tilde{X} \to X$ be the universal covering, $A \in M(n \times n, \omega(X))$ and $c \in \mathbb{C}^n$, then there exists a unique $w \in \mathcal{O}(\tilde{X})^n$ is a solution of the equation $dw = (p^*A)w$ with initial condition $w(\tilde{x}_0) = c$ where $\tilde{x}_0 \in \tilde{X}$ is a lift of x_0.*

Proof. Since $\tilde{X}$ is simply connected, the equation $dw = (p^*A)w$ has a unique solution on $\tilde{X}$ with the given initial condition by the above theorem. ∎

Definition 1.71. *Suppose X is a Riemann surface and $A \in M(n \times n, \omega(X))$. On the universal covering $p : \tilde{X} \to X$, define*

$$L_A := \{w \in \mathcal{O}(\tilde{X})^n : dw = (p^*A)w\},$$

then L_A is an n-dimensional vector space over $\mathbb{C}$ and $w_1, ..., w_n \in L_A$ are linearly independent iff $w_1(x_0), ..., w_{n(x_0)}$ are linearly independent for some $x_0 \in \tilde{X}$. We call L_A the solution space of the equation $dw = Aw$.

For a basis $w_1, ..., w_n$ of L_A, define a fundamental system of solutions to be

$$\Phi := (w_1, ..., w_n) \in \mathrm{GL}\ (n, \mathcal{O}(\tilde{X})),$$

then Φ satisfies $d\Phi = (p^*A)\Phi$.

Let $G = \mathrm{Deck}\left(\frac{\tilde{X}}{X}\right)$, then for any $\sigma \in G$, define $\sigma\Phi := \Phi \circ \sigma^{-1}$, then

$$d(\sigma\Phi) = d(\Phi \circ \sigma^{-1}) = (p^*A)(\Phi \circ \sigma^{-1}) = (p^*A)\sigma\Phi,$$

hence $\sigma\Phi$ is also a fundamental system of solutions. Since Φ is a fundamental system of solutions, there exists a unique $T_\sigma \in \mathrm{GL}(n, \mathbb{C})$ such that $\sigma\Phi = \Phi T_\sigma$. The matrices T_σ are called the factors of automorphy of Φ.

Consider the special case where $X = B_R(0)^*$ with $0 < R \leq \infty$ and $p : \tilde{X} \to X$ be the universal covering. Note that $\exp : \{\mathrm{Re}\ z < \log R\} \to X$ is also a universal covering, hence there exists a single-valued branch of log on $\tilde{X}$ such that $\exp \circ \log = p$. Pick a generator $\sigma \in \mathrm{Deck}\left(\frac{\tilde{X}}{X}\right)$ such that $\sigma \circ \log = \log + 2\pi i$, then the behavior of Φ under deck transformations is determined by the factor of automorphy T_σ. Recall that there exists $S \in \mathrm{GL}(n, \mathbb{C})$ such that $S^{-1}T_\sigma S$ is in Jordan normal form, then we can consider $\Psi = \Phi S$, so that

$$\sigma\Psi = \sigma(\Phi S) = \sigma\Phi S = \Phi T_\sigma S = \Phi S(S^{-1}T_\sigma S) = \Psi(S^{-1}T_\sigma S).$$

DEFINITION 1.72. *For a matrix $A \in M(n \times n, \mathbb{C})$, define the exponential of A by*

$$e^A = \sum_{k=0}^{\infty} \frac{A^k}{k!}.$$

If $A, B \in M(n \times n, \mathbb{C})$ commute, then $e^{A+B} = e^A e^B$. In particular $\exp(A)\exp(-A) = I$, hence $\exp(A) \in \mathrm{GL}\ (n, \mathbb{C})$. If $S \in \mathrm{GL}\ (n, \mathbb{C})$, then $e^{SAS^{-1}} = Se^A S^{-1}$.

PROPOSITION 1.73. *For all $B \in \mathrm{GL}\ (n, \mathbb{C})$, there exists $A \in M(n \times n, \mathbb{C})$ such that $e^A = B$.*

Proof. We may assume that B is in Jordan normal form, and we may consider the blocks of B separately. If $B = \mathrm{diag}((\Lambda_1, ..., \Lambda_n))$ is diagonal, then we can take $A = \mathrm{diag}((\log \Lambda_1, ..., \log \Lambda_n))$. If B is a Jordan block of the form

$$B = \begin{pmatrix} \Lambda & 1 & 0 & \dots & 0 \\ 0 & \Lambda & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ 0 & 0 & 0 & \dots & \Lambda \end{pmatrix},$$

then we can take

$$A = \mu I_n + \log\left(I_n + \frac{1}{\Lambda}N\right) = \mu I_n + \sum_{k=1}^{n-1} \frac{(-1)^{k-1}}{k\Lambda^k} N^k$$

where μ is a logarithm of Λ and N is the nilpotent matrix given by

$$N = \begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ 0 & 0 & 0 & \dots & 0 \end{pmatrix}.$$

Then

$$e^A = e^{\mu I_n} e^{\log(I_n + \frac{1}{\Lambda}N)} = \Lambda\Big(I_n + \frac{1}{\Lambda}N\Big) = B.$$

∎

THEOREM 1.74. *Let $X = B_R(0)^*$. Let $T \in \mathrm{GL}\ (n, \mathbb{C})$ and $B \in M(n \times n, \mathbb{C})$ such that $\exp(2\pi i B) = T$, then $\Phi_0 := \exp(B\log) \in \mathrm{GL}\ \big(n, \mathcal{O}(\tilde{X})\big)$ is a fundamental system of solutions of*

$$w' = \frac{B}{z}w$$

on the universal covering $p : \tilde{X} \to B_R(0)^*$ such that $\sigma\Phi_0 = \Phi_0 T$.

Proof. Note that

$$\Phi_{0'} = \frac{B}{z}\ \exp(B\log) = \frac{B}{z}\Phi_0,$$

and

$$\sigma\Phi_0 = \sigma\ \exp(B\log) = \exp(B\sigma\log) = \exp(B(\log + 2\pi i)) = \exp(B\log)\ \exp(2\pi i B) = \Phi_0 T,$$

hence proved. ∎

THEOREM 1.75. *Let $X = B_R(0)^*$. Let $A \in M(n \times n, \mathcal{O}(X))$, then $w' = Aw$ has a fundamental system of solutions $\Phi \in \mathrm{GL}\ (n, \mathcal{O}(X))$ of the form*

$$\Phi = \Psi\Phi_0$$

where $\Phi_0 = \exp(B\log)$ for a constant matrix B and Ψ is invariant under heck transformations, i.e. Ψ defines an element in $\mathrm{GL}\ (n, \mathcal{O}(X))$.

Proof. Suppose Φ is a fundamental system of solutions of $w' = Aw$, and $\sigma\Phi = \Phi T$ where $T \in \mathrm{GL}\ (n, \mathbb{C})$. By the previous theorem exists $B \in M(n \times n, \mathbb{C})$ such that $\exp(2\pi i B) = T$ and $\Phi_0 = \exp(B\log)$ such that $\sigma\Phi_0 = \Phi_0 T$. Let $\Psi = \Phi\Phi_0^{-1}$, then

$$\sigma\Psi = \sigma\big(\Phi\Phi_0^{-1}\big) = (\sigma\Phi)\big(\sigma\Phi_0\big)^{-1} = (\Phi T)\big(\Phi_0 T\big)^{-1} = \Phi\Phi_0^{-1} = \Psi,$$

hence Ψ is invariant under deck transformations and defines an element in $\mathrm{GL}\ (n, \mathcal{O}(X))$. ∎

1.10. Big Theorem

Recall that $X = B_R(0)^*$, and universal covering $p : \tilde{X} \to X$, $A \in M(n \times n, \mathcal{O}(X))$, then $w' = Aw$ has a fundamental system of solutions $\Phi = \Psi\Phi_0$ where $\Phi_0 = \exp(B\log)$ for a constant matrix B and Ψ is invariant under deck transformations.

DEFINITION 1.76. *The origin is called a regular singular point of the equation $w' = Aw$ if Ψ has at most a pole (not necessarily a simple pole) at the origin.*

LEMMA 1.77. *Suppose $K \geq 0$ is a constant and $F : (0, r_0] \to R_+$ is a continuously partialerentiable function such that*

$$|F'(r)| \leq \frac{K}{r}F(r)$$

for every $0 < r \leq r_0$, then

$$F(r) \leq F(r_0)\left(\frac{r}{r_0}\right)^{-K}$$

for every $0 < r \leq r_0$.

Proof. Write

$$\begin{aligned}
&\frac{d}{dr}\log F(r) = \frac{F'(r)}{F(r)} \geq -\frac{K}{r},\\
\Rightarrow &\int_r^{r_0} \frac{d}{dr}\log F(r)dr \geq -\int_r^{r_0}\frac{K}{r}dr,\\
\Rightarrow &\log F(r_0) - \log F(r) \geq -K\log r_0 + K\log r,\\
\Rightarrow &\log\frac{F(r_0)}{F(r)} \geq K\log\frac{r}{r_0},\\
\Rightarrow &F(r) \leq F(r_0)\left(\frac{r}{r_0}\right)^{-K}.
\end{aligned}$$

∎

LEMMA 1.78. *If* $f \in \mathcal{O}(X)$, *then*

$$\left|\frac{\partial}{\partial r}\ |f|^2\right| \leq 2\ |f||f'|.$$

Proof. Note that $f' = d\frac{f}{d}z = \frac{1}{e^{i\theta}}\frac{\partial F}{\partial}r$, hence

$$\left|\frac{\partial F}{\partial r}\right| = |e^{i\theta}f'| = |f'|,$$

and

$$\left|\frac{\partial \overline{f}}{\partial r}\right| = \left|\overline{\frac{\partial f}{\partial r}}\right| = |f'|.$$

Hence

$$\left|\frac{\partial}{\partial r}\ |f|^2\right| = \left|\frac{\partial f}{\partial r}\overline{f} + f\frac{\partial \overline{f}}{\partial r}\right| \leq \left|\frac{\partial f}{\partial r}\right|\ |\overline{f}| + |f|\ \left|\frac{\partial \overline{f}}{\partial r}\right| = 2\ |f||f'|.$$

∎

THEOREM 1.79. *If A has at most a pole of order 1 at the origin, then the origin is a regular singular point of the equation $w' = Aw$.*

Proof. Write $\Phi = \Psi\Phi_0$ where $\Phi_0 = \exp(B\log)$ and Ψ is invariant under deck transformations, then

$$\Phi' = \Psi'\Phi_0 + \Psi\Phi_{0'} = \Psi'\Phi_0 + \Psi\frac{B}{z}\Phi_0,$$

and

$$\Phi' = A\Phi = A\Psi\Phi_0.$$

Apply Φ_0^{-1} on the right, we get

$$\Psi' + \Psi \frac{B}{z} = A\Psi.$$

By assumption, we can write $A = \frac{A_1}{z}$ for some $A_1 \in M(n \times n, B_R(0))$. Hence

$$\Psi' = \frac{1}{z}(A_1 - B)\Psi.$$

Define the matrix norm $|M| = \left(\sum_{i,j} |M_{ij}|^2\right)^{\frac{1}{2}}$, then

$$|\Psi'| \le \frac{1}{|z|}\ |A_1 - B|\ |\Psi|.$$

If for all $0 < r_0 < R$, there exists integer $M \ge 0$ such that for all $0 < r \le r_0$,

$$r^M\ |\Psi(re^{i\theta})| \le r_0^M\ |\Psi(r_0 e^{i\theta})| < \infty,$$

then Ψ has at most a pole of order M at the origin. It remains to show that such an integer M exists.

Let $M = \max\limits_{|z|\ \le r_0} |A_1(z) - B|$, then

$$|\Psi'| \le \frac{M}{r}\ |\Psi|$$

for every $0 < r \le r_0$. Suppose ψ_{jk} are the entries of Ψ. For fixed θ, let

$$F(r) = |\Psi(re^{i\theta})|^2 = \sum_{j,k} |\psi_{jk}(re^{i\theta})|^2,$$

then by Lemma 1.78,

$$\begin{aligned}|F'(r)| = |\left(\sum_{j,k} |\ \frac{\partial}{\partial r}\ |\psi_{jk}(re^{i\theta})|^2\right) &\le \sum_{j,k} |\left(\frac{\partial}{\partial r}\ |\psi_{jk}(re^{i\theta})|^2\right) \\ &\le \sum_{j,k} 2\ |\psi_{jk}(re^{i\theta})|\ |\psi_{(jk)'}(re^{i\theta})| \le 2\ |\Psi|\ |\Psi'| \le \frac{2M}{r}\ |\Psi|^2 = \frac{2M}{r} F(r).\end{aligned}$$

From Lemma 1.77, we have

$$F(r) \le F(r_0)\left(\frac{r}{r_0}\right)^{-2M},$$

which implies that $r^M\ |\Psi(re^{i\theta})| \le r_0^M\ |\Psi(r_0 e^{i\theta})| < \infty$ for every $0 < r \le r_0$. Hence Ψ has at most a pole of order M at the origin, which completes the proof. ∎

Example 1.2. Let $X = B_R(0)^*$ and $a, b \in \mathcal{O}(B_R(0))$ and consider the equation

$$w'' + \frac{a}{z} w' + \frac{b}{z^2} w = 0$$

Set $w_1 = w$ and $w_2 = zw'$ and let $w = (w_1, w_2)^t$, then the above equation is equivalent to

$$dw = \begin{pmatrix} 0 & -\frac{b}{z} \\ \frac{1}{z} & \frac{1-a}{z} \end{pmatrix} w.$$

On $p : \tilde{X} \to X$, $\Phi(z) = z^n \psi(z) \exp(B \log)$. By a change of basis we may assume that B is in Jordan normal form. In the first case,

$$B = \begin{pmatrix} \alpha_1 & 0 \\ 0 & \alpha_2 \end{pmatrix} \rightsquigarrow \exp(B\log) = \begin{pmatrix} z^{\alpha_1} & 0 \\ 0 & z^{\alpha_2} \end{pmatrix},$$

hence

$$\Phi(z) = \begin{pmatrix} \phi_1(z) & \phi_2(z) \\ z\phi_{1'}(z) & z\phi_{2'}(z) \end{pmatrix} = z^n \begin{pmatrix} \psi_1(z) & \psi_2(z) \\ \psi_3(z) & \psi_4(z) \end{pmatrix} \begin{pmatrix} z^{\alpha_1} & 0 \\ 0 & z^{\alpha_2} \end{pmatrix},$$

hence $\phi_1(z) = z^{n+\alpha_1}\psi_1(z)$ and $\phi_2(z) = z^{n+\alpha_2}\psi_2(z)$. In the second case,

$$B = \begin{pmatrix} \alpha & 1 \\ 0 & \alpha \end{pmatrix} \rightsquigarrow \exp(B\log) = \begin{pmatrix} z^\alpha & z^\alpha \log z \\ 0 & z^\alpha \end{pmatrix},$$

hence

$$\Phi(z) = \begin{pmatrix} \phi_1(z) & \phi_2(z) \\ z\phi_{1'}(z) & z\phi_{2'}(z) \end{pmatrix} = z^n \begin{pmatrix} \psi_1(z) & \psi_2(z) \\ \psi_3(z) & \psi_4(z) \end{pmatrix} \begin{pmatrix} z^\alpha & z^\alpha \log z \\ 0 & z^\alpha \end{pmatrix},$$

hence $\phi_1(z) = z^{n+\alpha}\psi_1(z)$ and $\phi_2(z) = z^{n+\alpha}(\psi_1(z)\log z + \psi_2(z))$.

Example 1.3. [Bessel's equation] Consider the equation

$$w'' + \frac{1}{z}w' + \left(1 - \frac{p^2}{z^2}\right)w = 0$$

By the above theorem, it has at least one solution of the form

$$\phi(z) = z^\rho \sum_{n=0}^{\infty} c_n z^n, \quad c_0 \neq 0.$$

then partialerentiating term by term we get

$$\phi'(z) = z^\rho \sum_{n=0}^{\infty} (\rho + n) c_n z^{n-1},$$

$$\phi''(z) = z^\rho \sum_{n=0}^{\infty} (\rho + n)(\rho + n - 1) c_n z^{n-2}.$$

Comparing the coefficients when $n = 0, 1, 2$, we get

$$(\rho^2 - p^2)c_0 = 0,$$

$$((\rho + 1)^2 - p^2)c_1 = 0,$$

$$((\rho + n)^2 - p^2)c_n + c_{n-2} = 0 \quad \text{for every} \quad n \geq 2.$$

Since $c_0 \neq 0$, we have $\rho = \pm p$, so $(2\rho + 1)c_1 = 0$ and for $n = 2k$,

$$4k(\rho + k)c_{2k} + c_{2k-2} = 0.$$

If $p \in \mathbb{C} \setminus \mathbb{Z}$, let $c_1 = 0$, $c_{2k+1} = 0$, then

$$c_{2k} = (-1)^k \left(\frac{1}{2}\right)^{2k} \frac{1}{\Gamma(k+1)\Gamma(\rho + k + 1)},$$

define

$$J_{p(z)} = \left(\frac{z}{2}\right)^p \sum_{k=0}^{\infty} \frac{(-1)^k}{\Gamma(k+1)\Gamma(\rho + k + 1)} \left(\frac{z}{2}\right)^{2k},$$

which is called the Bessel function of order ρ. If $p \notin \mathbb{Z}$ then J_p and J_{-p} are linearly independent, hence they form a fundamental system of solutions. If $p \in \mathbb{Z}$, then J_p and J_{-p} are linearly dependent, hence we need to find another solution. Let

$$\psi(z) = J_{p(z)} \log z + g(z),$$

where g is a holomorphic function on X to be determined. Then

$$\psi'(z) = J_{p'}(z) \log z + \frac{J_{p(z)}}{z} + g'(z),$$

$$\psi''(z) = J_{p''}(z)\log z + \frac{2J_{p'}(z)}{z} - \frac{J_{p(z)}}{z^2} + g''(z).$$

Substitute into the equation, we get

$$g''(z) + \frac{1}{z}g'(z) + \left(1 - \frac{p^2}{z^2}\right)g(z) = -\frac{2}{z}J_{p'}(z).$$

The equation can be solved by a power series of the form

$$g(z) = z^{-p}\sum_{n=0}^{\infty} a_n z^n.$$

Compact Riemann Surfaces

Chapter 2

2.1. *Vanishing Theorems*

Let X be a differential manifold, and $\mathfrak{U} = (U_i)_{i\in I}$ be an open cover of X. A family $(g_i)_{i\in I}$ of functions $g_i : X \to \mathbb{R}$ is called a partition of unity subordinate to $\mathfrak{U}$ if

(i) $0 \leq g_{i(x)} \leq 1$ for every $i \in I$ and $x \in X$;
(ii) supp $g_i \subset U_i$ for every $i \in I$;
(iii) for every $x \in X$, there exists a neighborhood V of x such that $g_i|_V = 0$ for all but finitely many $i \in I$, and $\sum_{i\in I} g_{i(x)} = 1$ for every $x \in X$.

Theorem 2.1. *If X is a differential manifold, with a countable basis, then for every open cover $\mathfrak{U}$ of X, there exists a partition of unity subordinate to $\mathfrak{U}$.*

Lemma 2.2. *Let $A \subset X$ be clased and $A \subset U$ for some open set U, then there exists a function infinitely partialerentiable $\phi : X \to \mathbb{R}$ such that $0 \leq \phi(x) \leq 1$ for every $x \in X$, $\phi|_A = 1$ and $\phi|_{X\setminus U} = 0$.*

Proof. Since $\{U, X \setminus A\}$ is an open cover of X, there exists a partition of unity (g_1, g_2) subordinate to $\{U, X \setminus A\}$, then we can take $\phi = g_1$. ∎

Let X be a Riemann surface, $\mathfrak{U} = (U_i)_{i\in I}$ an open cover of X and $\mathcal{F}$ a sheaf of abelian groups on X. Let $\mathcal{O}, \mathcal{E}, \mathcal{E}^{(1)}$ denote the sheaves of holomorphic functions, infinitely partialerentiable functions and infinitely partialerentiable 1-forms on X respectively. $\mathbb{C}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$ denote the locally constant sheaves with stalks $\mathbb{C}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$ respectively.

Define the 0-th, 1-st and 2-nd cochain groups by

$$C^0(\mathfrak{U}, \mathcal{F}) = \prod_{i\in I} \mathcal{F}(U_i),$$
$$C^1(\mathfrak{U}, \mathcal{F}) = \prod_{i,j\in I} \mathcal{F}\big(U_i \cap U_j\big),$$
$$C^2(\mathfrak{U}, \mathcal{F}) = \prod_{i,j,k\in I} \mathcal{F}\big(U_i \cap U_j \cap U_k\big).$$

Define the coboundary operators by

$$\delta : C^0(\mathfrak{U}, \mathcal{F}) \to C^1(\mathfrak{U}, \mathcal{F}), \quad (f_i)_{i\in I} \mapsto \big(f_i - f_j\big)_{i,j\in I},$$
$$\delta : C^1(\mathfrak{U}, \mathcal{F}) \to C^2(\mathfrak{U}, \mathcal{F}), \quad \big(f_{ij}\big)_{i,j\in I} \mapsto \big(f_{ij} - f_{ik} + f_{jk}\big)_{i,j,k\in I}.$$

We can easily verify that $\delta^2 = 0$, hence we can define the 0-th and 1-st cohomology groups by

$$Z^0(\mathfrak{U}, \mathcal{F}) = \ker\, \delta : C^0(\mathfrak{U}, \mathcal{F}) \to C^1(\mathfrak{U}, \mathcal{F}) \cong \mathcal{F}(X),$$
$$B^0(\mathfrak{U}, \mathcal{F}) = \mathrm{Im}\ \delta : C^{-1}(\mathfrak{U}, \mathcal{F}) \to C^0(\mathfrak{U}, \mathcal{F}) = 0,$$

$$H^0(\mathfrak{U},\mathcal{F}) = Z^0(\mathfrak{U},\mathcal{F})/B^0(\mathfrak{U},\mathcal{F}) \cong \mathcal{F}(X),$$

and

$$Z^1(\mathfrak{U},\mathcal{F}) = \ker\,\delta : C^1(\mathfrak{U},\mathcal{F}) \to C^2(\mathfrak{U},\mathcal{F}),$$
$$B^1(\mathfrak{U},\mathcal{F}) = \mathrm{Im}\,\delta : C^0(\mathfrak{U},\mathcal{F}) \to C^1(\mathfrak{U},\mathcal{F}),$$
$$H^1(\mathfrak{U},\mathcal{F}) = Z^1\frac{\mathfrak{U},\mathcal{F}}{B^1}(\mathfrak{U},\mathcal{F}).$$

The elements in $Z^1(\mathfrak{U},\mathcal{F})$ are called 1-cocycles and the elements in $B^1(\mathfrak{U},\mathcal{F})$ are called 1-coboundaries. Concretely, a 1-cocycle is a family $\left(f_{ij}\right)_{i,j\in I}$ of sections $f_{ij}\in\mathcal{F}\left(U_i\cap U_j\right)$ such that

$$f_{ij} - f_{ik} + f_{jk} = 0$$

for every $i,j,k\in I$, and a 1-coboundary is a family $\left(f_{ij}\right)_{i,j\in I}$ of sections $f_{ij}\in\mathcal{F}\left(U_i\cap U_j\right)$ such that there exists a 0-cochain $(g_i)_{i\in I}$ of sections $g_i\in\mathcal{F}(U_i)$ such that

$$f_{ij} = g_i - g_j$$

for every $i,j\in I$.

However, the above definition of cohomology groups depends on the choice of open cover $\mathfrak{U}$. We define a direct partial order on the set of open covers of X by $\mathfrak{V} = (V_k)_{k\in K} < \mathfrak{U} = (U_i)_{i\in I}$ if there exists an assignment $\tau : K\to I$ such that $V_k\subset U_{\tau(k)}$ for every $k\in K$. For any two covering $\mathfrak{V}' = (V_{k'})_{k\in K'}$ and $\mathfrak{V}'' = (V_{k''})_{k\in K''}$, we can take

$$\mathfrak{V} = \left(V_{k'}\cap V_{(k'')''}\right)_{(k',k'')\in K'\times K''} < \mathfrak{V}',\mathfrak{V}'',$$

then $\mathfrak{V} < \mathfrak{V}',\mathfrak{V}''$. This gives a directed system of cohomology groups $\{H^1(\mathfrak{U},\mathcal{F})\}_{\mathfrak{U}}$: if $\mathfrak{V} < \mathfrak{U}$, then there exists a homomorphism

$$t^{\mathfrak{U}}_{\mathfrak{V}} : H^1(\mathfrak{U},\mathcal{F}) \to H^1(\mathfrak{U},\mathcal{F}),\quad f_{kl}\mapsto f_{\tau(k)\tau(l)}|_{V_k\cap V_l}.$$

Note that $t^{\mathfrak{U}}_{\mathfrak{V}}$ is independent of the choice of τ: if $\tau' : K\to I$ is another assignment such that $V_k\subset U_{\tau'(k)}$ for every $k\in K$, then

$$\begin{aligned} f_{\tau(k)\tau(l)} - f_{\tau'(k)\tau'(l)} &= \left(f_{\tau(k)\tau(l)} + f_{\tau(l)\tau'(k)}\right) - \left(f_{\tau(l)\tau'(k)} - f_{\tau'(k)\tau'(l)}\right) \\ &= f_{\tau(k)\tau'(k)} - f_{\tau(l)\tau'(l)}, \end{aligned}$$

thus we can define that $h_k = f_{\tau(k)\tau'(k)}$ for every $k\in K$, then h_k is a 0-cochain hence $f_{\tau(k)\tau(l)} - f_{\tau'(k)\tau'(l)} = h_k - h_l\in B^1(\mathfrak{V},\mathcal{F})$, which implies that $f_{\tau(k)\tau(l)}$ and $f_{\tau'(k)\tau'(l)}$ represent the same element in $H^1(\mathfrak{V},\mathcal{F})$, hence $t^{\mathfrak{U}}_{\mathfrak{V}}$ is independent of the choice of τ.

Suppose that $\left(f_{\tau(k)\tau(l)}\right)_{k,l\in K}$ is a 1-coboundary, then there exists a 0-cochain $(g_k)_{k\in K}$ such that $f_{\tau(k)\tau(l)} = g_k - g_l$ for every $k,l\in K$. Note that $(U_i\cap V_k)_{k\in K}$ is an open cover of U_i. On $U_i\cap V_k\cap V_l$, we have

$$g_k - g_l = f_{\tau(k)\tau(l)} = f_{\tau(k)i} + f_{i\tau(l)} \rightsquigarrow g_k + f_{i\tau(k)} = g_l + f_{i\tau(l)}.$$

By sheaf axiom, there exists a section $h_i\in\mathcal{F}(U_i)$ such that $h_i|_{U_i\cap V_k} = g_k + f_{i\tau(k)}$ for every $k\in K$. Then for every $i,j\in I$ and $k\in K$, we have

$$f_{ij} = f_{i\tau(k)} - f_{j\tau(k)} = (h_i - g_k) - \left(h_j - g_k\right) = h_i - h_j,$$

hence $\left(f_{ij}\right)_{i,j\in I}$ is a 1-coboundary, which implies that $t^{\mathfrak{U}}_{\mathfrak{V}}$ is injective.

If $\mathfrak{W} < \mathfrak{V} < \mathfrak{U}$, then $t^{\mathfrak{U}}_{\mathfrak{W}} = t^{\mathfrak{V}}_{\mathfrak{W}}\circ t^{\mathfrak{U}}_{\mathfrak{V}}$, hence $\{H^1(\mathfrak{U},\mathcal{F})\}_{\mathfrak{U}}$ is a directed system of abelian groups. Define the first cohomology group of X with coefficients in $\mathcal{F}$ by

$$H^1(X,\mathscr{F}) = \varinjlim_{\mathfrak{U}} H^1(\mathfrak{U},\mathscr{F}) = \coprod_{\mathfrak{U}} H^1(\mathfrak{U},\mathscr{F})/\sim,$$

where $\sim$ is the equivalence relation defined by $\xi \sim \xi'$ if $\xi \in H^1(\mathfrak{U},\mathscr{F})$, $\xi' \in H^1(\mathfrak{U}',\mathscr{F})$ and there exists a common refinement $\mathfrak{V} < \mathfrak{U},\mathfrak{U}'$ such that $t^{\mathfrak{U}}_{\mathfrak{V}}(\xi) = t^{\mathfrak{U}'}_{\mathfrak{V}}(\xi')$. Addition can be defined by $\xi + \xi' = t^{\mathfrak{U}}_{\mathfrak{V}}(\xi) + t^{\mathfrak{U}'}_{\mathfrak{V}}(\xi')$ for some common refinement $\mathfrak{V} < \mathfrak{U},\mathfrak{U}'$, which is well-defined and makes $H^1(X,\mathscr{F})$ an abelian group. The natural embedding $H^1(\mathfrak{U},\mathscr{F}) \to H^1(X,\mathscr{F})$ is injective for every open cover $\mathfrak{U}$, in particular $H^1(X,\mathscr{F}) = 0$ if and only if $H^1(\mathfrak{U},\mathscr{F}) = 0$ for all open cover $\mathfrak{U}$.

THEOREM 2.3 (Dolbeault's lemma). *Suppose $X = B_R(0)$ with $0 < R \le \infty$ and $g \in \mathcal{E}(X)$, then there exists $f \in \mathcal{E}(X)$ such that $\frac{\partial F}{\partial}\overline{z} = g$.*

Proof. We first assume that g has compact support and $X = \mathbb{C}$. For $\zeta \in \mathbb{C}$, define

$$f(\zeta) = \frac{1}{2\pi i}\iint_{\mathbb{C}} \frac{g(z)}{z-\zeta} dz \wedge d\overline{z}.$$

Fix $\zeta \in \mathbb{C}$, and let $z = \zeta + re^{i\theta}$, then $dz \wedge d\overline{z} = -2irdr \wedge d\theta$. Since g has compact support, there exists $r_0 > 0$ such that $g(z) = 0$ for every $r \ge r_0$. Hence

$$f(\zeta) = \frac{1}{2\pi i}\iint_{0\le r\le r_0} \frac{g(\zeta + re^{i\theta})}{re^{i\theta}}(-2ir)dr \wedge d\theta = -\frac{1}{\pi}\iint_{0\le r\le r_0} g(\zeta + re^{i\theta})e^{-i\theta}dr \wedge d\theta.$$

Since g is infinitely partialerentiable, we can partialerentiate under the integral sign and get

$$\begin{aligned}\frac{(\partial F)}{\partial\overline{\zeta}} &= -\frac{1}{\pi}\iint_{0\le r\le r_0} \frac{\partial g}{\partial\overline{\zeta}}(\zeta + re^{i\theta})e^{-i\theta}dr \wedge d\theta \\ &= \frac{1}{2\pi i}\lim_{\varepsilon\to 0}\iint_{\varepsilon\le\ |z-\zeta|\ \le r_0} \frac{\partial g}{\partial\overline{z}}(z)\frac{1}{z-\zeta}dz \wedge d\overline{z}.\end{aligned}$$

Let

$$w(z) = \frac{1}{2\pi i}\frac{g(z)}{z-\zeta}dz \rightsquigarrow dw = \frac{1}{2\pi i}\frac{\partial g}{\partial\overline{z}}\frac{1}{z-\zeta}d\overline{z} \wedge dz,$$

then by Stokes' theorem,

$$\begin{aligned}\lim_{\varepsilon\to 0}\iint_{\varepsilon\le\ |z-\zeta|\ \le r_0} \frac{\partial g}{\partial\overline{z}}\frac{1}{z-\zeta}dz \wedge d\overline{z} &= -\lim_{\varepsilon\to 0}\iint_{\varepsilon\le\ |z-\zeta|\ \le r_0} dw \\ &= -\lim_{\varepsilon\to 0}\int_{|z-\zeta|\ = r_0} w + \lim_{\varepsilon\to 0}\int_{|z-\zeta|\ =\varepsilon} w \\ &= 0 + \lim_{\varepsilon\to 0}\frac{1}{2\pi}\int_0^{2\pi} g(\zeta + \varepsilon e^{i\theta})d\theta = g(\zeta).\end{aligned}$$

For general g, let $0 < R_0 < R_1 < \ldots$ such that $\lim_{n\to\infty} R_n = R$ and let $\psi_n \in \mathcal{E}(X)$ such that supp $(\psi_n) \subset B_{R_{n+1}}(0)$ and $\psi_n|_{B_{R_n}(0)} = 1$ for every $n \ge 0$. Define $g_n = \psi_n g$, then g_n has compact support, hence there exists $f_n \in \mathcal{E}(X)$ such that $\overline{\partial} f_n = \partial\frac{f_n}{\partial}\overline{z} = g_n$.

Let $|f|_K = \sup_{z\in K}|f(z)|$ be the supremum norm. Set $\tilde{f}_1 = f_1$. Since $\overline{\partial}\big(f_2 - \tilde{f}_1\big) = g_2 - g_1 = 0$ on $B_{R_1}(0)$, $f_2 - \tilde{f}_1$ is holomorphic on $B_{R_1}(0)$, hence there exists a Taylor polynomial P_1 of $f_2 - \tilde{f}_1$ at the origin such that $|f_2 - \tilde{f}_1 - P_1|_{B_{R_0}(0)} < \frac{1}{2}$. Define $\tilde{f}_2 = f_2 - P_1$, then $|\tilde{f}_2 - \tilde{f}_1|_{B_{R_0}(0)} < \frac{1}{2}$. Inductively, we can define $\tilde{f}_n$ such that $|\tilde{f}_{n+1} -$

$\tilde{f}_n|_{B_{R_{n-1}}(0)} < \frac{1}{2^n}$ and $\overline{\partial}\tilde{f}_n = g_n$ for every $n \geq 1$. Let $f = \lim_{n\to\infty} \tilde{f}_n$. On $B_{R_n}(0)$, we may write

$$f = \tilde{f}_n + \sum_{k=n}^{\infty}\left(f_{k+1} - \tilde{f}_k\right),$$

since $\tilde{f}_{k+1} - \tilde{f}_k \in \mathcal{O}\left(B_{R_{n-1}}(0)\right)$ and $|f_{k+1} - \tilde{f}_k|_{B_{R_n}(0)} < \frac{1}{2^k}$ for every $k \geq n$, the above series converges uniformly on $B_{R_n}(0)$, hence f is infinitely partialerentiable on X. Moreover, $\overline{\partial} f = \overline{\partial}\tilde{f}_n = g_n$ on $B_{R_n}(0)$ for every $n \geq 1$, hence $\overline{\partial} f = g$ on X, which completes the proof. ∎

THEOREM 2.4. *If X is a Riemann surface, then $H^1(X, \mathcal{E}) = 0$.*

Proof. Given $\mathfrak{U} = (U_i)_{i\in I}$, exists a partition of unity $(\psi_i)_{i\in I}$ subordinate to $\mathfrak{U}$. Let $\left(f_{ij}\right)_{i,j\in I}$ be a 1-cocycle of $\mathcal{E}$ with respect to $\mathfrak{U}$, then for every $i, j \in I$, we have

$$f_{ij} = \sum_{k\in I}\psi_k f_{ij} = \sum_{k\in I}\psi_k\left(f_{ik} - f_{jk}\right) = \sum_{k\in I}\psi_k f_{ik} - \sum_{k\in I}\psi_k f_{jk} = g_i - g_j,$$

where $g_i = \sum_{k\in I}\psi_k f_{ik}$ for every $i \in I$. Hence $\left(f_{ij}\right)_{i,j\in I}$ is a 1-coboundary, which implies that $H^1(\mathfrak{U}, \mathcal{E}) = 0$ for every open cover $\mathfrak{U}$, hence $H^1(X, \mathcal{E}) = 0$. ∎

THEOREM 2.5. *If X is a simply connected Riemann surface, then $H^1(X, \mathbb{C}) = 0$.*

Proof. For $\mathfrak{U} = (U_i)_{i\in I}$ and $\left(c_{ij}\right) \in Z^1(\mathfrak{U}, \mathbb{C}) \subseteq Z^1(\mathfrak{U}, \mathcal{E})$, since $H^1(X, \mathcal{E}) = 0$, there exists a 0-cochain $(f_i)_{i\in I}$ of $\mathcal{E}$ such that $c_{ij} = f_i - f_j$ for every $i, j \in I$. Note that $0 = dc_{ij} = df_i - df_j$ for every $i, j \in I$, hence there exists a 1-form ω on X such that $\omega|_{U_i} = df_i$ for every $i \in I$. Note that $d\omega|_{U_i} = ddf_i = 0$ for every $i \in I$, hence $d\omega = 0$ on X. Since X is simply connected, there exists a function f on X such that $df = \omega$, hence $df|_{U_i} = df_i$ for every $i \in I$, which implies that $c_i := f_i - f$ is a constant function on U_i for every $i \in I$. Hence for every $i, j \in I$, we have

$$c_{ij} = f_i - f_j = (f_i - f) - \left(f_j - f\right) = c_i - c_j,$$

which implies that $\left(c_{ij}\right)_{i,j\in I}$ is a 1-coboundary, hence $H^1(\mathfrak{U}, \mathbb{C}) = 0$ for every open cover $\mathfrak{U}$, hence $H^1(X, \mathbb{C}) = 0$. ∎

THEOREM 2.6. *If X is a simply connected Riemann surface, then $H^1(X, \mathbb{Z}) = 0$.*

Proof. For $\mathfrak{U} = (U_i)_{i\in I}$ and $\left(a_{jk}\right) \in Z^1(\mathfrak{U}, \mathbb{Z}) \subseteq Z^1(\mathfrak{U}, \mathbb{C})$, since $H^1(X, \mathbb{C}) = 0$, there exists a 0-cochain $\left(c_j\right)_{j\in I}$ of $\mathbb{C}$ such that $a_{jk} = c_j - c_k$ for every $j, k \in I$. Note that

$$1 = e^{2\pi i a_{jk}} = e^{2\pi i\left(c_j - c_k\right)} \rightsquigarrow e^{2\pi i c_j} = e^{2\pi i c_k}$$

for every $j, k \in I$. Since X is connected, there exists a constant $b \in \mathbb{C}^*$ such that $e^{2\pi i c_j} = b$ for every $j \in I$. Let $b = e^{2\pi i c}$ for some $c \in \mathbb{C}$, then $e^{2\pi i\left(c_j - c\right)} = 1$ for every $j \in I$, hence $a_j := c_j - c$ is an integer for every $j \in I$. Hence for every $j, k \in I$, we have

$$a_{jk} = c_j - c_k = \left(c_j - c\right) - (c_k - c) = a_j - a_k,$$

which implies that $\left(a_{jk}\right)_{j,k\in I}$ is a 1-coboundary, hence $H^1(\mathfrak{U}, \mathbb{Z}) = 0$ for every open cover $\mathfrak{U}$, hence $H^1(X, \mathbb{Z}) = 0$. ∎

THEOREM 2.7. *Let $X = B_R(0)$ with $0 < R \leq \infty$, then $H^1(X, \mathcal{O}) = 0$.*

Proof. For $\mathfrak{U} = (U_i)_{i\in I}$ and $(f_{ij}) \in Z^1(\mathfrak{U}, \mathcal{O}) \subseteq Z^1(\mathfrak{U}, \mathcal{E})$, since $H^1(X, \mathcal{E}) = 0$, there exists a 0-cochain $(f_i)_{i\in I}$ of $\mathcal{E}$ such that $f_{ij} = f_i - f_j$ for every $i, j \in I$. Note that

$$0 = \overline{\partial} f_{ij} = \overline{\partial} f_i - \overline{\partial} f_j \rightsquigarrow \overline{\partial} f_i = \overline{\partial} f_j$$

for every $i, j \in I$, hence there exists a function $h \in \mathcal{E}(X)$ such that $h|_{U_i} = \overline{\partial} f_i$ for every $i \in I$. By Dolbeault's lemma, there exists a function $g \in \mathcal{E}(X)$ such that $\overline{\partial} g = h$, hence $\overline{\partial}(f_i - g) = 0$ for every $i \in I$, which implies that $f_i - g$ is holomorphic on U_i for every $i \in I$. Hence for every $i, j \in I$, we have

$$f_{ij} = f_i - f_j = (f_i - g) - (f_j - g),$$

which implies that $(f_{ij})_{i,j\in I}$ is a 1-coboundary, hence $H^1(\mathfrak{U}, \mathcal{O}) = 0$ for every open cover $\mathfrak{U}$, hence $H^1(X, \mathcal{O}) = 0$. ∎

THEOREM 2.8 (Leray's theorem). *Let X be a Riemann surface and $\mathcal{F}$ a sheaf of abelian groups on X. If there exists an open cover $\mathfrak{U}$ of X such that $H^1(U_i, \mathcal{F}) = 0$ for every $i \in I$, then $H^1(X, \mathcal{F}) \cong H^1(\mathfrak{U}, \mathcal{F})$.*

Proof. It suffices to show that for all refinements $\mathfrak{V} < \mathfrak{U}$, the homomorphism $t^{\mathfrak{U}}_{\mathfrak{V}} : H^1(\mathfrak{U}, \mathcal{F}) \to H^1(\mathfrak{V}, \mathcal{F})$ is an isomorphism. We've alread y shown that $t^{\mathfrak{U}}_{\mathfrak{V}}$ is injective.

Let $\tau : K \to I$ be an assignment such that $V_k \subset U_{\tau(k)}$ for every $k \in K$. Fix $(f_{kl}) \in Z^1(\mathfrak{V}, \mathcal{F})$. By assumption, $H^1(U_i \cap \mathfrak{V}, \mathcal{F}) = 0$ for every $i \in I$, hence there exists a 0-cochain $(g_{ik})_{k\in K}$ of $\mathcal{F}$ such that $f_{kl} = g_{ik} - g_{il}$ for every $k, l \in K$. On $U_i \cap U_j \cap V_k \cap V_l$, we have

$$f_{kl} = g_{ik} - g_{il} = g_{jk} - g_{jl} \rightsquigarrow g_{jk} - g_{ik} = g_{jl} - g_{il},$$

hence there exists a section $F_{ij} \in \mathcal{F}(U_i \cap U_j)$ such that $F_{ij} = g_{jk} - g_{ik}$ for every $k \in K$. Then for every $i, j \in I$ and $k \in K$, we have

$$F_{ij} + F_{jt} = g_{ik} - g_{jk} + g_{jk} - g_{tk} = g_{ik} - g_{tk} = F_{it},$$

which implies that $(F_{ij})_{i,j\in I}$ is a 1-cocycle of $\mathcal{F}$ with respect to $\mathfrak{U}$. On $V_k \cap V_l$, we have

$$F_{\tau(k)\tau(l)} - f_{kl} = \left(g_{\tau(l)k} - g_{\tau(k)k}\right) - \left(g_{\tau(l)k} - g_{\tau(l)l}\right) = g_{\tau(l)l} - g_{\tau(k)k},$$

which implies that $t^{\mathfrak{U}}_{\mathfrak{V}}\left([F_{ij}]\right) = [f_{kl}]$, hence $t^{\mathfrak{U}}_{\mathfrak{V}}$ is surjective, which completes the proof. ∎

2.2. *Finiteness Theorems*

Write $Y_1 \Subset Y_2$ if $\overline{Y_1}$ is a compact subset of Y_2.

THEOREM 2.9 (Finiteness theorem). *Let X be a Riemann surface and $Y_1 \Subset Y_2 \subset X$ be open subsets of X, then the restriction map $H^1(Y_2, \mathcal{O}) \to H^1(Y_1, \mathcal{O})$ has a finite-dimensional image. In particular, if X is compact, then $H^1(X, \mathcal{O})$ is finite-dimensional.*

Proof. We choose finitely many charts $\{(U_i, z_i)\}_{i=1,\dots,n}$ and relatively compact subsets $W_i \Subset V_i \Subset U_i \Subset U_{i'}$ such that

(1) $Y_1 \subset \bigcup_{i=1}^n W_i =: Y' \Subset Y'' := \bigcup_{i=1}^n U_i \subset Y_2$;
(2) $z_{i(W_i)}, z_{i(U_i)}, z_{i(U_{i'})}$ are disks in $\mathbb{C}$.

Let $\mathfrak{U}' = (U_{i'})$, $\mathfrak{U} = (U_i)$ and $\mathfrak{W} = (W_i)$, then by the key proposition (to be proved later), there exists a finite-dimensional subspace $S \subset Z^1(\mathfrak{U}, \mathcal{O})$ such that for all $\xi \in$

$Z^1(\mathfrak{U},\mathcal{O})$, there exists $\sigma \in S$, $\eta \in C^0(\mathfrak{W},\mathcal{O})$ such that $\sigma = \xi + \delta\eta$ on $\mathfrak{W}$. Hence the image of $H^1(\mathfrak{U},\mathcal{O}) \to H^1(\mathfrak{W},\mathcal{O})$ is finite-dimensional (equals to the image of S). Since $H^1(\mathfrak{U}_i,\mathcal{O}) = 0$ and $H^1(\mathfrak{W}_i,\mathcal{O}) = 0$ for every i, by Leray's theorem,

$$H^1(U,\mathcal{O}) \cong H^1(Y'',\mathcal{O}), \quad H^1(W,\mathcal{O}) \cong H^1(Y',\mathcal{O}).$$

Since the restriction map $H^1(Y_2,\mathcal{O}) \to H^1(Y_1,\mathcal{O})$ factors through $H^1(Y'',\mathcal{O}) \to H^1(Y',\mathcal{O})$, the image of $H^1(Y_2,\mathcal{O}) \to H^1(Y_1,\mathcal{O})$ is finite-dimensional. In particular, if X is compact, then $H^1(X,\mathcal{O})$ is finite-dimensional. ∎

DEFINITION 2.10. *Let X be a Riemann surface, then the genus of X is defined by*

$$g(X) = \dim H^1(X,\mathcal{O}).$$

LEMMA 2.11. *Let X be a Riemann surface and $Y \Subset X$, $a \in Y$, then exists a meromorphic function $f \in \mathcal{M}(Y)$ such that f has a pole at a and $f \in \mathcal{O}(Y \setminus \{a\})$.*

Proof. Let (U_1, z) be a coordinate neighborhood of a such that $z(a) = 0$. Set $U_2 = X \setminus \{a\}$, then $\mathfrak{U} = (U_1, U_2)$ is an open cover of X. By the finiteness theorem,

$$k := \dim \operatorname{Im} \left(H^1(\mathtt{cal}\ X,\mathcal{O}) \to H^1(\mathtt{cal}\ Y,\mathcal{O})\right) < \infty.$$

Since $H^1(\mathfrak{U},\mathcal{O}) \hookrightarrow H^1(\mathtt{cal}\ X,\mathcal{O})$ and $H^1(\mathfrak{U} \cap Y,\mathcal{O}) \hookrightarrow H^1(\mathtt{cal}\ Y,\mathcal{O})$ are injective, we have

$$\dim \operatorname{Im} \left(H^1(\mathfrak{U},\mathcal{O}) \to H^1(\mathfrak{U} \cap Y,\mathcal{O})\right) \le k.$$

Observe that $z^{-j} \in \mathcal{O}(U_1 \cap U_2)$ for every $j = 1, ..., k+1$, so let $\xi_j = \overline{z^{-j}} \in H^1(\mathfrak{U},\mathcal{O})$, then $\xi_1, ..., \xi_{k+1}$ are linearly dependent in $H^1(\mathfrak{U} \cap Y,\mathcal{O})$, hence there exists $c_1, ..., c_{k+1} \in \mathbb{C}$, not all zero and $f_1 \in \mathcal{O}(U_1 \cap Y)$, $f_2 \in \mathcal{O}(U_2 \cap Y)$ such that

$$\sum_{j=1}^{k+1} c_j z^{-j} = f_2 - f_1 \quad \text{on} \quad U_1 \cap U_2 \cap Y.$$

Let f be the meromorphic function on Y defined by $f|_{U_1 \cap Y} = f_1 + \sum_{j=1}^{k+1} c_j z^{-j}$ and $f|_{U_2 \cap Y} = f_2$, then f has a pole at a and $f \in \mathcal{O}(Y \setminus \{a\})$, which completes the proof. ∎

LEMMA 2.12. *Let X be a non-compact Riemann surface and $Y \Subset X$, then there exists a non-constant holomorphic function on Y.*

Proof. Since X is non-compact, there exists Y_1 such that $Y \Subset Y_1 \Subset X$. Pick $a \in Y_1 \setminus Y$, then by the previous corollary, there exists $f_1 \in \mathcal{O}(Y_1 \setminus \{a\})$ such that f_1 has a pole at a. Let f be the holomorphic function on Y defined by $f = f_1|_Y$, then f is non-constant, which completes the proof. ∎

On a open subset $\mathfrak{U}$, we can always write a 1-form $\omega \in \mathcal{E}^1(\mathfrak{U})$ locally as $\omega = f dx + g dy = \phi dz + \psi d\overline{z}$, where $dz = dx + i dy$ and $d\overline{z} = dx - i dy$. In this case, we say $\phi dz \in \mathcal{E}^{1,0}(\mathfrak{U})$ is the $(1,0)$-part of ω and $\psi d\overline{z} \in \mathcal{E}^{0,1}(\mathfrak{U})$ is the $(0,1)$-part of ω. We can also write

$$df = \frac{(\partial F)}{\partial z} dz + \frac{(\partial F)}{\partial \overline{z}} d\overline{z}$$

for every $f \in \mathcal{E}(\mathfrak{U})$, then $\frac{\partial F}{\partial} z dz$ is the $(1,0)$-part of df and $\frac{\partial F}{\partial} \overline{z} d\overline{z}$ is the $(0,1)$-part of df. We denote $\frac{\partial F}{\partial} z dz$ by $d'f$ and $\frac{\partial F}{\partial} \overline{z} d\overline{z}$ by $d''f$, then we have $df = d'f + d''f$. Note that $d'^2 = 0$, $d''^2 = 0$.

THEOREM 2.13 (General form for Dolbeault's lemma). *Let X be a non-compact Riemann surface and $Y \Subset Y' \Subset X$, $\omega \in \mathcal{E}^{0,1}(Y')$, then there exists $g \in \mathcal{E}(Y)$ such that $d''g = \omega$ on Y.*

Proof. By Dolbeault's lemma, exists $\mathfrak{U} = (U_i)_{i\in I}$ such that each U_i is isomorphic to a disk and $\omega|_{U_i} = d''f_i$ for some $f_i \in \mathcal{E}(U_i)$ for every $i \in I$ (Since we can write $\omega = \phi d\overline{z}$, and by Dolbeault's lemma, there exists f_i such that $\partial\frac{f_i}{\partial}\overline{z} = \phi$ on U_i).

Note that

$$d''\left(f_i - f_j\right) = 0 \rightsquigarrow f_i - f_j \in \mathcal{O}\left(U_i \cap U_j\right) \rightsquigarrow \left(f_i - f_j\right)_{i,j\in I} \in Z^1(\mathfrak{U}, \mathcal{O}).$$

If we can prove that $L := \mathrm{Im}\,\left(H^1(Y', \mathcal{O}) \to H^1(Y, \mathcal{O})\right) = 0$, then there exists a 0-cochain $(g_i)_{i\in I}$ of $\mathcal{O}$ such that $f_i - f_j = g_i - g_j$ for every $i, j \in I$, hence $f_i - g_i = f_j - g_j$ on $U_i \cap U_j \cap Y$ for every $i, j \in I$, which implies that there exists a function g on Y such that $g|_{U_i\cap Y} = f_i - g_i$ for every $i \in I$, hence $d''g = d''f_i = \omega$ on Y, which completes the proof.

We now prove that $L = 0$. Let $\xi_1, ..., \xi_n \in H^1(Y', \mathcal{O})$ such that $\xi_1|_Y, ..., \xi_n|_Y$ is a basis of L. Let $f \in \mathcal{O}(Y')$ be a non-constant holomorphic function. Since $f\xi_i \in H^1(Y', \mathcal{O})$ for every i, we can write $f\xi_i|_Y = \sum_{j=1}^n c_{ij}\xi_j|_Y$ for some $c_{ij} \in \mathbb{C}$. Using the determinant trick, we have $F := \det\left(c_{ij} - f\delta_{ij}\right) \in \mathcal{O}(Y')$ and $F\xi_i|_Y = 0$ for every i, hence $F\xi|_Y = 0$ for every $\xi \in L$. We can pick a covering $\mathfrak{V} = (V_i)_{i\in I}$ of Y such that each zero is contained in a single V_i, then $F|_{V_i\cap V_j}$ contains no zero, hence there exists $g_{ij} := \left(F|_{V_i\cap V_j}\right)^{-1} f_{ij}$, i.e. $f_{ij} = Fg_{ij}$ for every $i, j \in I$. Let $\zeta, \xi \in H^1(\mathfrak{V}, \mathcal{O})$ be the classes of $\left(f_{ij}\right), \left(g_{ij}\right)$ respectively, then $\zeta = F\xi$ and $\zeta|_Y = F\xi|_Y = 0$, hence proved that $L = 0$. ∎

2.3. *Proof of Finiteness by L^2-norm Estimates*

Recall that

$$|f|_{L^2(D)} := \left(\iint_D |f(x+iy)|^2\ dxdy\right)^{\frac{1}{2}} \in R_{\geq 0} \cup \{\infty\}$$

$$L^2(D, \mathcal{O}) := \{f \in \mathcal{O}(D) : |f|_{L^2(D)} < \infty\}.$$

We have

$$|f|_{L^2(D)} \leq \sqrt{\pi}R\ |f|_{L^\infty(D)},$$

where $|f|_{L^\infty(D)} = \sup_{z\in D}|f(z)|$. If $f, g \in L^2(D, \mathcal{O})$, then write their inner product as

$$\langle f, g\rangle_{L^2(D)} := \iint_D f(z)\overline{g(z)}dxdy,$$

then $L^2(D, \mathcal{O})$ is a Hilbert space with respect to $\langle\cdot,\cdot\rangle_{L^2(D)}$. For $B = B_{r(a)}$, the family of functions

$$\psi_{n(z)} = (z-a)^n, \quad n = 0, 1, 2, ...$$

forms an orthogonal basis of $L^2(B, \mathcal{O})$. We can verify that

$$\langle\psi_n, \psi_m\rangle_{L^2(B)} = \iint_B (z-a)^n\overline{(z-a)^m}dxdy$$

$$= \iint_{0\le\theta<2\pi, 0\le\rho<r} \rho^{n+m} e^{i(n-m)\theta} \rho d\rho d\theta = \begin{cases} 0 & \text{if } n \neq m \\ \frac{\pi r^{2n+2}}{n+1} & \text{otherwise} \end{cases}$$

Therefore, $|\psi_n|_{L^2(B)} = \sqrt{\pi}\frac{r^{n+1}}{\sqrt{n+1}}$ for every n. For all $a \in B_{\rho(0)}$, we can write $f(z) = \sum_{n=0}^{\infty} c_n(z-a)^n$ for every $f \in L^2(B, \mathcal{O})$, then

$$|f(a)| = |c_0| \le \frac{1}{\sqrt{\pi}r} \; |f|_{L^2(B)} \le \frac{1}{\sqrt{\pi}(r-\rho)} \; |f|_{L^2\left(B_{\rho(0)}, \mathcal{O}\right)}.$$

DEFINITION 2.14. *Let X be a Riemann surface and $\mathcal{U}' = (U_{i'}, z_i)_{i=1,\dots,n}$ be a family of charts such that $z_{i(U_{i'})}$ is a disk in $\mathbb{C}$ for every i. Suppose $U_i \subset U_{i'}$ and $\mathcal{U} = (U_i)_{i=1,\dots,n}$ is an open cover of X. For $\eta = (f_i) \in C^0(\mathcal{U}, \mathcal{O})$, we define*

$$|\eta|^2_{L^2(\mathcal{U})} := \sum_{i=1}^{n} |\eta_i|^2_{L^2(U_i)}.$$

For $\xi = \left(f_{ij}\right) \in C^1(\mathcal{U}, \mathcal{O})$, we define

$$|\xi|^2_{L^2(\mathcal{U})} := \sum_{i,j} |f_{ij}|^2_{L^2(U_i \cap U_j)}.$$

Let $C^0_{L^2}(\mathcal{U}, \mathcal{O})$ be the set of 0-cochains having finite L^2-norm and $C^1_{L^2}(\mathcal{U}, \mathcal{O})$ be the set of 1-cochains having finite L^2-norm, then these spaces are Hilbert spaces since $|f|_{D_r} \le \frac{1}{\sqrt{\pi}r} \; |f|_{L^2(D_r)}$ so the convergence in L^2-norm implies the convergence in L^∞-norm, i.e. converge uniformly on compact subsets.

PROPOSITION 2.15. *Let $\mathcal{U} = (U_i)_{i=1,\dots,n}$ and $\mathtt{cal}\ B = (V_i)_{i=1,\dots,n}$ be two open covers of X such that $V_i \Subset U_i$ for every i. Then for all $\xi \in C^1(\mathcal{U}, \mathcal{O})$,*

(i) $|\xi|_{L^2(\mathtt{cal}\ B)} < \infty$;
(ii) *for all $\varepsilon > 0$, there exists a finite codimensional subspace $A \subset Z^1_{L^2}(\mathcal{U}, \mathcal{O})$ such that for all $\xi \in A$,*

$$|\xi|_{L^2(\mathtt{cal}\ B)} < \varepsilon \; |\xi|_{L^2(\mathcal{U})}.$$

Proof.

(1) Since f_{ij} is bounded on $\overline{V_i \cap V_j}$ for every i, j, we have $|f_{ij}|_{L^2(V_i \cap V_j)} < \infty$ for every i, j, hence $|\xi|_{L^2(\mathtt{cal}\ B)} < \infty$.
(2) We need a finite codimensional A such that

$$|f_{ij}|_{L^2(V_i \cap V_j)} < \varepsilon \; |f_{ij}|_{L^2(U_i \cap U_j)}$$

for every i, j and $\xi = \left(f_{ij}\right) \in A$. Fix a pair of i, j and let $V = V_i \cap V_j$, $U = U_i \cap U_j$, $f = f_{ij}$.

Since $\overline{V}$ is compact, exists r and $a_1, \dots, a_k \in U$ such that $\overline{B_{r(a_j)}} \subset U$ for every j and $V \subset \bigcup_{j=1}^{k} B_{\frac{r}{2}}\left(a_j\right)$. Pick $n \gg 0$ such that $2^{-n-1}k \le \varepsilon$. Let

$$A = \{f \in L^2(U, \mathcal{O}) \mid f(z) = \sum_{i=n\}^\infty c_{ij}\left(z-a_j\right)^i \text{ for every } j=1,\dots,k} .$$

Then A has codimension at most kn in $L^2(U, \mathcal{O})$ (forcing zeros of order at least n at a_j for every j). For $\rho \le r$, we have

$$|f|^2_{L^2\left(B_{\rho\left(a_j\right)}\right)} = \sum_{i=n}^{\infty} |c_{ij}|^2 \; |\psi_i|^2_{L^2\left(B_{\rho\left(a_j\right)}\right)} = \sum_{i=n}^{\infty} \frac{\pi\rho^{2i+2}}{i+1} \; |c_{ij}|^2,$$

in particular

$$|f|^2_{L^2\left(B_{\frac{r}{2}}(a_j)\right)} = \sum_{i=n}^{\infty} \frac{\pi r^{2i+2}}{i+1} \frac{1}{2^{2i+2}} \; |c_{ij}|^2 \leq \frac{1}{2^{2n+2}} \sum_{i=n}^{\infty} \frac{\pi r^{2i+2}}{i+1} \; |c_{ij}|^2 = \frac{1}{2^{2n+2}} \; |f|^2_{L^2\left(B_{r\left(a_j\right)}\right)}.$$

Hence

$$|f|^2_{L^2(V)} \leq \sum_{j=1}^{k} |f|^2_{L^2\left(B_{\frac{r}{2}}(a_j)\right)} \leq \frac{1}{2^{2n+2}} \sum_{j=1}^{k} |f|^2_{L^2\left(B_{r\left(a_j\right)}\right)} \leq \frac{k}{2^{2n+2}} \; |f|^2_{L^2(U)} \leq \varepsilon^2 \; |f|^2_{L^2(U)},$$

which completes the proof.

∎

LEMMA 2.16. *For $\mathtt{cal}\ W \Subset \mathtt{cal}\ B \Subset \mathcal{U} \Subset \mathcal{U}'$ with open sets isomorphic to disks, there exists $c > 0$ such that for all $\xi \in Z^1_{L^2}(\mathtt{cal}\ B, \mathcal{O})$, exists $\zeta \in Z^1_{L^2}(\mathcal{U}, \mathcal{O})$, $\eta \in C^0_{L^2}(\mathcal{U}, \mathcal{O})$ such that*

$$\zeta = \xi + \delta\eta \quad \text{on} \quad \mathtt{cal}\ W,$$

and

$$\max\left(|\zeta|_{L^2(\mathtt{cal}\ W)}, |\eta|_{L^2(\mathtt{cal}\ W)}\right) \leq c \; |\xi|_{L^2(\mathtt{cal}\ B)}.$$

Proof. Since $H^1(\mathtt{cal}\ B, \mathcal{E}) = 0$, there exists $(g_i) \in C^0(\mathtt{cal}\ B, \mathcal{E})$ such that $f_{ij} = g_j - g_i$ on every $V_i \cap V_j$, then

$$d''g_j - d''g_i = d''f_{ij} = 0 \rightsquigarrow d''g_i = d''g_j \quad \text{on} \quad V_i \cap V_j,$$

hence there exists $\omega \in \mathcal{E}^{0,1}(|\mathtt{cal}\ B|)$ such that $\omega|_{V_i} = d''g_i$ for every i.

Since $|\mathtt{cal}\ W| \Subset |\mathtt{cal}\ B|$, by partition of unity, exists $\psi \in C^{\infty(|\mathtt{cal}\ B|)}$ such that $\psi|_{|\mathtt{cal}\ W|} = 1$ and $\operatorname{supp}(\psi) \subset |\mathtt{cal}\ B|$, then $\psi\omega \in \mathcal{E}^{0,1}(|\mathcal{U}'|)$. By Dolbeault's lemma, $\exists h_i \in \mathcal{E}(U_{i'})$ such that $d''h_i = \psi\omega$ on each $U_{i'}$. Since $d''\left(h_i - h_j\right) = 0$ on $U_{i'} \cap U_{j'}$, let $F_{ij} \in \mathcal{O}\left(U_{i'} \cap U_{j'}\right)$ such that $F_{ij} = h_j - h_i$ on $U_{i'} \cap U_{j'}$, so $F_{kl} = F_{ki} + F_{il}$, impling that $F|_{\mathcal{U}} \in Z^1_{L^2}(\mathcal{U}, \mathcal{O})$. On W_i, $d''h_i = \psi\omega = \omega = d''g_i$, hence $d''(h_i - g_i) = 0$ on W_i, so there exists $\eta \in C^0_{L^2}(\mathtt{cal}\ W, \mathcal{O})$ such that $\eta|_{W_i} = h_i - g_i$ for every i. Now,

$$F_{ij} - f_{ij} = \left(h_j - h_i\right) - \left(g_j - g_i\right) = \left(h_j - g_j\right) - \left(h_i - g_i\right) = \eta_j - \eta_i \quad \text{on} \quad W_i \cap W_j,$$

hence $\zeta := F|_{\mathcal{U}}$ and η satisfy the required conditions.

For the estimate, let

$$H := Z^1_{L^2}(\mathcal{U}, \mathcal{O}) \times Z^1_{L^2}(\mathtt{cal}\ B, \mathcal{O}) \times C^0_{L^2}(\mathtt{cal}\ W, \mathcal{O})$$

with the norm

$$|(\zeta, \xi, \eta)|_H := \left(|\zeta|^2_{L^2(\mathcal{U})} + |\xi|^2_{L^2(\mathtt{cal}\ B)} + |\eta|^2_{L^2(\mathtt{cal}\ W)}\right)^{\frac{1}{2}}.$$

Let $L \subset H$ be the subspace

$$L := \{(\zeta, \xi, \eta) \in H : \zeta = \xi + \delta\eta \quad \text{on} \quad \mathtt{cal}\ W\}.$$

Then L is a closed subspace of H, hence also a Hilbert space. Consider the projection map $\pi : L \to Z^1_{L^2}(\mathtt{cal}\ B, \mathcal{O})$ defined by $\pi(\zeta, \xi, \eta) = \xi$. By the first part, π is surjective, hence by the open mapping theorem, there exists $c > 0$ such that for all $\xi \in Z^1_{L^2}(\mathtt{cal}\ B, \mathcal{O})$, there exists $(\zeta, \xi, \eta) \in L$ such that $|(\zeta, \xi, \eta)|_H \leq c \; |\xi|_{L^2(\mathtt{cal}\ B)}$. In particular, $\max\left(|\zeta|_{L^2(\mathtt{cal}\ W)}, |\eta|_{L^2(\mathtt{cal}\ W)}\right) \leq c \; |\xi|_{L^2(\mathtt{cal}\ B)}$, which completes the proof. ∎

PROPOSITION 2.17. *$\exists$ a finite-dimensional subspace $S \subset Z^1(\mathcal{U}, \mathcal{O})$ such that for all $\xi \in Z^1(\mathcal{U}, \mathcal{O})$, there exists $\sigma \in S$, $\eta \in C^0(\mathtt{cal}\ W, \mathcal{O})$ such that $\sigma = \xi + \delta\eta$ on $\mathtt{cal}\ W$.*

Proof. Let c be given in the previous lemma and $\varepsilon = \frac{1}{2}c$. By the previous proposition, there exists a finite codimensional subspace $A \subset Z^1_{L^2}(\mathcal{U}, \mathcal{O})$ such that for all $\xi \in A$,

$$|\xi|_{L^2(\mathtt{cal}\ B)} < \varepsilon\ |\xi|_{L^2(\mathcal{U})}.$$

Pick S to be the orthogonal complement of A in $Z^1_{L^2}(\mathcal{U}, \mathcal{O})$, then S is finite-dimensional. For $\xi \in Z^1(\mathcal{U}, \mathcal{O})$, let $M = |\xi|_{L^2(\mathtt{cal}\ B)} < \infty$. Then by the previous lemma, there exists $\zeta_0 \in Z^1_{L^2}(\mathcal{U}, \mathcal{O})$, $\eta_0 \in C^0_{L^2}(\mathtt{cal}\ W, \mathcal{O})$ such that $\zeta_0 = \xi + \delta\eta_0$ on $\mathtt{cal}\ W$ and $\max\left(|\zeta_0|_{L^2(\mathtt{cal}\ W)}, |\eta_0|_{L^2(\mathtt{cal}\ W)}\right) \le cM$. Write $\zeta_0 = \xi_0 + \sigma_0$ for some $\xi_0 \in A$ and $\sigma_0 \in S$, then

$$|\xi_0|_{L^2(\mathtt{cal}\ B)} \le \varepsilon\ |\xi_0|_{L^2(\mathcal{U})} \le \varepsilon c M = \frac{M}{2}.$$

By key lemma again, there exists $\zeta_1 \in Z^1_{L^2}(\mathcal{U}, \mathcal{O})$, $\eta_1 \in C^0_{L^2}(\mathtt{cal}\ W, \mathcal{O})$ such that $\zeta_1 = \xi_0 + \delta\eta_1$ on $\mathtt{cal}\ W$ and $\max\left(|\zeta_1|_{L^2(\mathtt{cal}\ W)}, |\eta_1|_{L^2(\mathtt{cal}\ W)}\right) \le c\frac{M}{2}$. Write $\zeta_1 = \xi_1 + \sigma_1$ for some $\xi_1 \in A$ and $\sigma_1 \in S$, then

$$|\xi_1|_{L^2(\mathtt{cal}\ B)} \le \varepsilon\ |\xi_1|_{L^2(\mathcal{U})} \le \varepsilon c\frac{M}{2} = \frac{M}{4}.$$

Continue this process, we have $\xi_n \in A$ such that $|\xi_n|_{L^2(\mathtt{cal}\ B)} \le \frac{M}{2^n}$ for every n, and

$$\sum_{i=0}^{k} \xi_i + \sum_{i=0}^{k} \sigma_i = \sum_{i=0}^{k} \zeta_i = \sum_{i=0}^{k} \xi + \delta \sum_{i=0}^{k} \eta_i \quad \text{on} \quad \mathtt{cal}\ W,$$

impling that

$$\xi_k + \sum_{i=0}^{k} \sigma_i = \xi + \delta \sum_{i=0}^{k} \eta_i \quad \text{on} \quad \mathtt{cal}\ W$$

for every k. Define the series

$$\sigma := \sum_{i=0}^{\infty} \sigma_i, \quad \eta := \sum_{i=0}^{\infty} \eta_i,$$

then σ converges since S is finite-dimensional and $|\sigma_i|_{L^2(\mathcal{U})} \le |\zeta_i|_{L^2(\mathcal{U})} \le c\frac{M}{2^i}$ for every i, and η converges since $|\eta_i|_{L^2(\mathcal{U})} \le c\frac{M}{2^i}$ for every i. Moreover, $\sigma = \xi + \delta\eta$ on $\mathtt{cal}\ W$, which completes the proof. ∎

We say that a sequence of sheaf homomorphisms

$$\mathcal{F} \xrightarrow{\alpha} \mathtt{cal}\ G \xrightarrow{\beta} \mathtt{cal}\ H$$

is exact at $\mathtt{cal}\ G$ if for each $x \in X$, the sequence of stalks

$$\mathcal{F}_x \xrightarrow{\alpha_x} \mathtt{cal}\ G_x \xrightarrow{\beta_x} \mathtt{cal}\ H_x$$

is exact at $\mathtt{cal}\ G_x$, i.e. $\ker \beta_x = \operatorname{Im} \alpha_x$.

THEOREM 2.18. *Suppose X is a topological space and*

$$0 \to \mathcal{F} \xrightarrow{\alpha} \mathtt{cal}\ G \xrightarrow{\beta} \mathtt{cal}\ H \to 0$$

is an exact sequence of sheaves on X. Then the induced sequence of cohomology groups

$$0 \to H^0(X, \mathcal{F}) \xrightarrow{\alpha_*} H^0(X, \mathtt{cal}\ G) \xrightarrow{\beta_*} H^0(X, \mathtt{cal}\ H)$$

$$\xrightarrow{\delta} H^1(X,\mathcal{F}) \xrightarrow{\alpha_*} H^1(X, \mathtt{cal}\ G) \xrightarrow{\beta_*} H^1(X, \mathtt{cal}\ H)$$

is exact.

Example 2.1. By Dolbeault's lemma, we have an exact sequence of sheaves

$$0 \to \mathcal{O} \to \mathcal{E} \xrightarrow{d''} \mathcal{E}^{0,1} \to 0,$$

hence we have an exact sequence of cohomology groups

$$0 \to H^0(X,\mathcal{O}) \to H^0(X,\mathcal{E}) \xrightarrow{d''} H^0(X,\mathcal{E}^{0,1})$$

$$\xrightarrow{\delta} H^1(X,\mathcal{O}) \to H^1(X,\mathcal{E}) \xrightarrow{d''} H^1(X,\mathcal{E}^{0,1}).$$

Since $H^1(X,\mathcal{E}) = 0$ by the partition of unity, we have an isomorphism

$$H^1(X,\mathcal{O}) \cong \mathcal{E}^{0,1}\frac{X}{d''}\mathcal{E}(X).$$

2.4. Riemann-Roch Theorem

DEFINITION 2.19.

(i) *Let X be a Riemann surface. A divisor on X is a mapping $D : X \to \mathbb{Z}$ such that for all $K \subset X$ with K compact, we have $D(x) = 0$ for all but finitely many $x \in K$.*

(ii) $(D_1 + D_2)(x) = D_1(x) + D_2(x)$ *for every $x \in X$, hence the set of divisors* Div (X) *on X forms an abelian group.*

(iii) *For $D_1, D_2 \in$ Div (X), we say $D_1 \geq D_2$ if $D_1(x) \geq D_2(x)$ for every $x \in X$.*

(iv) *For $f \in \mathcal{M}(X)$, $a \in X$, define the order of f at a by*

$$\mathrm{ord}_{a(f)} = \begin{cases} 0 & \text{if } f \text{ is holomorphic at } a \text{ and } f(a) \neq 0 \\ k & \text{if } f \text{ has a zero of order } k \text{ at } a \\ -k & \text{if } f \text{ has a pole of order } k \text{ at } a \\ \infty & \text{if } f \text{ is identically zero on a neighborhood of } a \end{cases}$$

The principal divisor associated to f is defined by

$$(f)(x) = \mathrm{ord}_{x(f)} \quad \text{for every} \quad x \in X.$$

(i) *$D_1 \sim D_2$ are linearly equivalent if $D_1 - D_2$ is a principal divisor, i.e. there exists $f \in \mathcal{M}(X)$ such that $D_1 - D_2 = (f)$, then $\sim$ is an equivalence relation on* Div (X)*, and the set of equivalence classes* Pic $(X) :=$ Div $\frac{X}{\sim}$ *is called the Picard group of X.*

(ii) *On compact Riemann surfaces, define the degree of a divisor D by* deg $D = \sum_{x\in X} D(x)$*, then* deg $(f) = 0$ *for every $f \in \mathcal{M}(X)$, hence* deg *induces a group homomorphism* deg : Pic $(X) \to \mathbb{Z}$.

(iii) *For $D \in$ Div (X), $U \subset X$ open, define*

$$\mathcal{O}_{D(U)} = \{f \in \mathcal{M}(U) : (f) + D \geq 0 \ \text{ on } \ U\}.$$

Then $\mathcal{O}_D$ is a sheaf on X. In particular, when $D = 0$, $\mathcal{O}_D = \mathcal{O}$.

(i) *For linearly equivalent divisors $D_1 \sim D_2$, say $D_1 = D_2 + (g)$ for some $g \in \mathcal{M}(X)$, then we have an isomorphism of sheaves*

$$\mathcal{O}_{D_1} \to \mathcal{O}_{D_2}, \quad f \mapsto fg,$$

(i) *On compact Riemann surfaces, if D is a divisor with* $\deg D < 0$, *then* $H^0(X, \mathcal{O}_D) = \mathcal{O}_{D(X)} = 0$ *since on compact Riemann surfaces, a meromorphic function has as many zeros as poles counting multiplicity, hence if such* $(f) + D \geq 0$ *for some* $f \in \mathcal{M}(X)$, *then* f *must be identically zero.*

DEFINITION 2.20. *The skyscraper sheaf* $\mathbb{C}_p$ *at* $p \in X$ *is defined by*

$$\mathbb{C}_{p(U)} = \begin{cases} \mathbb{C} \text{ if } p \in U \\ 0 \text{ if } p \notin U \end{cases}$$

PROPOSITION 2.21. $H^0(X, \mathbb{C}_p) = \mathbb{C}$ *and* $H^1(X, \mathbb{C}_p) = 0$ *for every* $p \in X$.

Proof. Clearly $H^0(X, \mathbb{C}_p) = \mathbb{C}_{p(X)} = \mathbb{C}$. Given an open cover $\mathcal{U} = (U_i)_{i \in I}$ and $(f_{ij}) \in Z^1(\mathcal{U}, \mathbb{C}_p)$, we can find a refinement $\texttt{cal}\ V = (V_i)_{i \in I}$ of $\mathcal{U}$ such that $p \in V_1$ and $p \notin V_i$ for every $i \neq 1$, then $p \notin V_i \cap V_j$ for every $i \neq j$, hence $f_{ij} = 0$ for every i, j, so $\xi = (f_{ij})$ is a coboundary, i.e. $\xi \in B^1(\texttt{cal}\ V, \mathbb{C}_p)$, hence $H^1(X, \mathbb{C}_p) = 0$. ∎

Define a sheaf homomorphism $\beta : \mathcal{O}_{D+p} \to \mathbb{C}_p$ as follows. Let $U \subset X$ be an open set. If $p \notin U$, then $\beta_U = 0$ is the zero map. If $p \in U$ and $f \in \mathcal{O}_{D+p}(U)$, then f has a Laurent expansion at p of the form

$$f(z) = \sum_{n=-k-1}^{\infty} c_n (z-p)^n$$

where $k = D(p)$. Define $\beta_{U(f)} = c_{-k-1}$. Clearly β_U is a sheaf homomorphism, and we have an exact sequence of sheaves

$$0 \to \mathcal{O}_D \to \mathcal{O}_{D+p} \xrightarrow{\beta} \mathbb{C}_p \to 0.$$

This exact sequence induces a long exact sequence of cohomology groups

$$0 \to H^0(X, \mathcal{O}_D) \to H^0(X, \mathcal{O}_{D+p}) \xrightarrow{\beta_*} \mathbb{C}$$
$$\xrightarrow{\delta} H^1(X, \mathcal{O}_D) \to H^1(X, \mathcal{O}_{D+p}) \to 0$$

THEOREM 2.22 (Riemann-Roch theorem). *Let X be a compact Riemann surface and D be a divisor on X. Then*

$$\dim H^0(X, \mathcal{O}_D) - \dim H^1(X, \mathcal{O}_D) = \deg D - g + 1.$$

Proof. The long exact sequence of cohomology groups can be split into three short exact sequences

$$0 \to H^0(X, \mathcal{O}_D) \to H^0(X, \mathcal{O}_{D+p}) \xrightarrow{\beta_*} \operatorname{Im} \beta_* \to 0,$$
$$0 \to \operatorname{Im} \beta_* \xrightarrow{\delta} \mathbb{C} \to \operatorname{Im} \delta \to 0,$$
$$0 \to \operatorname{Im} \delta \to H^1(X, \mathcal{O}_D) \to H^1(X, \mathcal{O}_{D+p}) \to 0.$$

Therefore,

$$\dim H^0(X, \mathcal{O}_{D+p}) = \dim H^0(X, \mathcal{O}_D) + \dim \operatorname{Im} \beta_*,$$
$$1 = \dim \operatorname{Im} \beta_* + \dim \operatorname{Im} \delta,$$

$$\dim H^1(X, \mathcal{O}_D) = \dim \operatorname{Im} \delta + \dim H^1\big(X, \mathcal{O}_{D+p}\big).$$

Summing up the first and third equations and subtracting the second equation, we have

$$\dim H^0\big(X, \mathcal{O}_{D+p}\big) + \dim H^1\big(X, \mathcal{O}_{D+p}\big) = \dim H^0(X, \mathcal{O}_D) + \dim H^1(X, \mathcal{O}_D) + 1$$

$$\Rightarrow \dim H^0\big(X, \mathcal{O}_{D+p}\big) - \dim H^1\big(X, \mathcal{O}_{D+p}\big) = \dim H^0(X, \mathcal{O}_D) - \dim H^1(X, \mathcal{O}_D) + 1.$$

Hence if the result holds for D, then it also holds for $D + p$, and conversely if the result holds for $D + p$, then it also holds for D. We can verify that the result holds for $D = 0$ since $H^0(X, \mathcal{O}) = \mathbb{C}$ and $\dim H^1(X, \mathcal{O}) = g$ by definition, hence the result holds for every divisor D by induction on $\deg D$. ∎

LEMMA 2.23. *Let X be a compact Riemann surface of genus g and $a \in X$. Then there exists a non-constant meromorphic function on X with only pole at a of order at most $g + 1$.*

Proof. Let $D = (g + 1)a$. By Riemann-Roch theorem, we have

$$\dim H^0(X, \mathcal{O}_D) \geq (g + 1) - g + 1 = 2,$$

hence there exists a non-constant meromorphic function $f \in \mathcal{O}_{D(X)}$, which has only pole at a of order at most $g + 1$. ∎

LEMMA 2.24. *Suppose X is a compact Riemann surface of genus g. Then there exists a holomorphic covering map $X \to \mathbb{P}^1$ with at most $g + 1$ sheets.*

Proof. Let $a \in X$ and f be a non-constant meromorphic function on X with only pole at a of order at most $g + 1$ given by the previous corollary, then $f : X \to \mathbb{P}^1$ is a holomorphic covering map with at most $g + 1$ sheets. ∎

LEMMA 2.25. *Every compact Riemann surface of genus* 0 *is isomorphic to* $\mathbb{P}^1$.

Proof. By the previous corollary, there exists a holomorphic covering map $f : X \to \mathbb{P}^1$ with at most 1 sheet, hence f is an isomorphism. ∎

2.5. *Serre Duality*

Recall that ω is the sheaf of holomorphic 1-forms on X, $\mathcal{E}^{1,0}$ is the sheaf of smooth 1-forms of type $(1, 0)$ on X, and $\mathcal{E}^{(2)}$ is the sheaf of smooth 2-forms on X. For each $\omega \in \mathcal{E}^{(2)}(U)$, we can write $\omega = f dx \wedge dy$ for some $f \in \mathcal{E}(U)$ locally. By Dolbeault's lemma, there exists $g \in \mathcal{E}(U)$ such that

$$\frac{\partial}{\partial \overline{z}} g = -f \rightsquigarrow d(g dz) = f dz \wedge d\overline{z} = \omega,$$

hence we have an exact sequence of sheaves

$$0 \to \omega \to \mathcal{E}^{1,0} \xrightarrow{d} \mathcal{E}^{(2)} \to 0,$$

inducing a long exact sequence of cohomology groups

$$0 \to H^0(X, \omega) \to H^0\big(X, \mathcal{E}^{1,0}\big) \xrightarrow{d} H^0\big(X, \mathcal{E}^{(2)}\big)$$

$$\xrightarrow{\delta} H^1(X, \omega) \to H^1\big(X, \mathcal{E}^{1,0}\big) \xrightarrow{d} H^1\big(X, \mathcal{E}^{(2)}\big).$$

Similar to how $H^1(X,\mathcal{E})=0$, we also have $H^1(X,\mathcal{E}^{1,0})=0$ and $H^1\big(X,\mathcal{E}^{(2)}\big)=0$ by the partition of unity, hence we have an isomorphism

$$H^1(X,\omega)\cong \mathcal{E}^{(2)}\frac{X}{d}\mathcal{E}^{1,0}(X).$$

Let $\omega\in\mathcal{M}^{(1)}(X)$ be a meromorphic 1-form on X. For $x\in X$ and a neighborhood U of x, write $\omega|_U=fdz$ for some $f\in\mathcal{M}(U)$, then define the order of ω at x by

$$\mathrm{ord}_{x(\omega)}=\mathrm{ord}_{x(f)}.$$

This is well-defined since if $\omega|_U=gdw$ for some $g\in\mathcal{M}(U)$ and local coordinate w, then $fdz=gdw$, hence $f=g\frac{dw}{dz}$, so $\mathrm{ord}_{x(f)}=\mathrm{ord}_{x(g)}+\mathrm{ord}_{x(\frac{dw}{dz})}=\mathrm{ord}_{x(g)}$ since $\frac{dw}{dz}$ is holomorphic and non-vanishing at x. For $\omega\in\mathcal{M}^{(1)}(X)^*$, the principal divisor associated to ω is defined by

$$(\omega)(x)=\mathrm{ord}_{x(\omega)}\quad\text{for every}\quad x\in X,$$

called the canonical divisor associated to ω. For any $\omega_1,\omega_2\in\mathcal{M}^{(1)}(X)^*$, locally write

$$\omega_1|_U=f_1dz,\quad \omega_2|_U=f_2dz,$$

and set $f|_U=\frac{f_1}{f_2}\in\mathcal{M}(U)^*$ and thus $\omega_1=f\omega_2$ on U. By sheaf axiom, we can glue $f|_U$ to a global meromorphic function f on X, hence $\omega_1=f\omega_2$ on X, so $(\omega_1)-(\omega_2)=(f)$ is a principal divisor, hence the canonical divisor is unique up to linear equivalence, denoted by K.

Define

$$\omega_{-D}(U)=\{\omega\in\mathcal{M}^{(1)\}(U)^*:(\omega)-D\geq 0\ \text{ on }\ U}$$

for every divisor D on X, then ω_{-D} is a sheaf on X. In particular, when $D=0$, $\omega_0=\omega$.

PROPOSITION 2.26. *There is a natural isomorphism*

$$\mathcal{O}_{K-D}\xrightarrow{\sim}\omega_{-D},\quad f\mapsto f\omega,$$

where K is the canonical divisor associated to ω.

Proof. Follows from

$$(f)+(K)-D\geq 0\Longleftrightarrow (f\omega)+(-D)\geq 0$$

for every $f\in\mathcal{M}(X)^*$ and $\omega\in\mathcal{M}^{(1)}(X)^*$. ∎

PROPOSITION 2.27. *The degree of the canonical divisor K is $2g-2$, where g is the genus of X.*

Proof. By Riemann-Roch theorem, we have

$$\begin{aligned}1-g+\deg\,K&=\dim H^0(X,\mathcal{O}_K)-\dim H^1(X,\mathcal{O}_K)\\&=\dim H^0(X,\omega)-\dim H^1(X,\omega)\\&=\dim H^1(X,\mathcal{O})^*-\dim H^0(X,\mathcal{O})^*=g-1,\end{aligned}$$

hence $\deg\,K=2g-2$. ∎

THEOREM 2.28 (Serre duality). *Let X be a compact Riemann surface. Then there is a natural isomorphism*

$$H^0(X,\omega_{-D})\cong H^1(X,\mathcal{O}_D)^*$$

for every divisor D on X.

Proof. *Our goal is to construct a natural bilinear pairing*

$$H^0(X,\omega_{-D})\times H^1(X,\mathcal{O}_D)\to\mathbb{C}$$

such that the induced map $H^0(X,\omega_{-D})\to H^1(X,\mathcal{O}_D)^*$ *is an isomorphism. For* $\omega\in H^0(X,\omega_{-D})$ *and* $\xi=\left(f_{ij}\right)\in Z^1(\mathcal{U},\mathcal{O}_D)$ *for some open cover* $\mathcal{U}=(U_i)_{i\in I}$*, their product* $\omega\cdot\xi$ *is* $\left(f_{ij}\omega\right)\in Z^1(\mathcal{U},\omega)$*. Recall that*

$$H^1(X,\omega)\cong\mathcal{E}^{(2)}\frac{X}{d}\mathcal{E}^{1,0}(X),$$

so we define the residue map from $H^1(X,\omega)$ *to* $\mathbb{C}$ *as follows. For* $\eta\in H^1(X,\omega)$*, let* τ *be a representative of* η *in* $\mathcal{E}^{(2)}(X)$*, then define*

$$\mathrm{Res}(\eta)=\iint_X\tau.$$

In practice, to compute the RHS, we need to find a finite covering of X*, then find a partition of unity subordinate to the covering, and compute the integral of* τ *on each open set in the covering, then sum up the integrals. To verify that the residue map is well-defined, let* $\tau'=\tau+df$ *for some* $f\in\mathcal{E}^{1,0}(X)$*, then by Stokes' theorem,*

$$\iint_X\tau'=\iint_X\tau+\iint_X df=\iint_X\tau+\int_{\partial X}f=\iint_X\tau,$$

hence the residue map is well-defined. Now we define the bilinear pairing by

$$\langle\omega,\xi\rangle=\mathrm{Res}(\omega\cdot\xi)\quad\text{for every}\quad\omega\in H^0(X,\omega_{-D})\quad\text{and}\quad\xi\in H^1(X,\mathcal{O}_D).$$

The induced map ι_D *is* $\omega\mapsto\langle\omega,\cdot\rangle$*, which is a linear map from* $H^0(X,\omega_{-D})$ *to* $H^1(X,\mathcal{O}_D)^*$*. To verify that* ι_D *is injective, we need to show that for all nonzero meromorphic 1-form* $\omega\in H^0(X,\omega_{-D})$*, there exists* $\xi\in H^1(X,\mathcal{O}_D)$ *such that* $\langle\omega,\xi\rangle\neq 0$*. Since* D *is finite, we can find* $a\in X$ *such that* $D(a)=0$*. Let* U_0 *be a neighborhood of* a *such that* $U_0\cong B_R(0)\subset\mathbb{C}$ *and* a *corresponds to* 0*. By shrinking* U_0 *if necessary, we can assume that* $\omega|_{U_0}=f\,dz$ *with* f *holomorphic and non-vanishing on* U_0*, except possibly a zero at* a*. Let* $U_1=X\setminus\{a\}$*, then* $\mathcal{U}=(U_0,U_1)$ *is an open cover of* X*. Let* $f_0=\frac{1}{zf}\in\mathcal{O}_{D(U_0)}$ *and* $f_1=0\in\mathcal{O}_{D(U_1)}$*, then* $\mu=(f_0,f_1)\in C^0(\mathcal{U},\mathcal{O}_D)$ *and* $\delta\mu=\left(f_{ij}\right)\in Z^1(\mathcal{U},\mathcal{O}_D)$ *with* $f_{01}=-f_{10}=\frac{1}{zf}$ *on* $U_0\cap U_1$*. Let* $\xi=\overline{\delta\mu}\in H^1(X,\mathcal{O}_D)$*, then*

$$\langle\omega,\xi\rangle=\mathrm{Res}(\omega\cdot\xi)=\mathrm{Res}(\delta(\omega\mu)).$$

Let $\zeta=\omega\mu=(\omega_0,\omega_1)=\left(d\frac{z}{z},0\right)$*. We need to find a representative of* $\delta\zeta$ *in* $\mathcal{E}^{(2)}(X)$ *to compute the residue. Since* $\delta\zeta\in Z^1(\mathcal{U},\omega)\subset Z^1(\mathcal{U},\mathcal{E}^{1,0})=B^1(\mathcal{U},\mathcal{E}^{1,0})$*, there exists* $(\sigma_0,\sigma_1)\in C^0(\mathcal{U},\mathcal{E}^{1,0})$ *such that* $\omega_0-\omega_1=\sigma_0-\sigma_1$ *on* $U_0\cap U_1$*. Since* $\omega_0-\omega_1=d\frac{z}{z}$ *is holomorphic on* $U_0\cap U_1$*, we have*

$$d(\omega_0-\omega_1)=d''(\omega_0-\omega_1)=0\rightsquigarrow d\sigma_0=d\sigma_1\quad\text{on}\quad U_0\cap U_1,$$

hence we can glue $d\sigma_0$ *and* $d\sigma_1$ *to a global smooth 2-form* τ *on* X*, which is a representative of* $\delta\zeta$ *in* $\mathcal{E}^{(2)}(X)$*. Pick a smaller* $V_0\subset U_0$ *such that* $V_0\cong B_\varepsilon(0)$ *for some* $\varepsilon<R$*, then by partition of unity exists* $\psi\in\mathcal{E}(X)$ *such that* $\psi|_{V_0}=1$ *and* $\psi|_{X\setminus U_0}=0$*. Then*

$$\begin{aligned}\iint_X d(\psi\sigma_1)=\iint_{U_0}d(\psi\sigma_1)&=\iint_{U_0}d\psi\sigma_0-\iint_{U_0}d\psi\omega_0-\iint_{U_0}\psi d\omega_1\\&=\iint_{U_0}d\psi\omega_0=-\lim_{\varepsilon'\to 0}\iint_{\varepsilon'\leq\,|z|\,\leq R}d\psi\omega_0\end{aligned}$$

$$= -\iint_{\varepsilon \le |z| \le R} d\psi\omega_0 = \int_{|z|=\varepsilon} \psi\omega_0 = 2\pi i.$$

Set $\phi = 1 - \psi$, then $\phi|_{V_0} = 0$ and $\phi|_{X \setminus U_0} = 1$, hence

$$\iint_X \tau = \iint_X d(\psi\sigma_1) + \iint_X d(\phi\sigma_0)$$

$$= \iint_X d(\psi\sigma_1) + \iint_{U_0} d\phi\sigma_0 = 2\pi i.$$

Hence $\langle \omega, \xi \rangle = 2\pi \frac{i}{2}\pi i = 1 \neq 0$, which implies that ι_D is injective.

LEMMA 2.29. Let $D, B \in \operatorname{Div}(X)$, then $\psi \in H^0(X, \omega_B)$ induces an isomorphism $H^1(X, \mathcal{O}_{D-B}) \to H^1(X, \mathcal{O}_D)$. Taking dual, we have mapping

$$H^1(X, \mathcal{O}_D)^* \xrightarrow{\psi^*} H^1(X, \mathcal{O}_{D-B})^*.$$

If $\psi \neq 0$, then ψ^* is injective.

Proof. *Let $A := (\psi)$ be the principal divisor associated to ψ. If $A + B \geq 0$, we have*

$$\mathcal{O}_{D-B} \hookrightarrow \mathcal{O}_{D+A} \cong \mathcal{O}_D$$

since $A + B \geq 0$.

If $D + A = (D - B) + p_1 + \dots + p_m$, then

$$0 \to \mathcal{O}_{(D-B)+p_1+\dots+p_{m-1}} \to \mathcal{O}_{D+A} \to \mathbb{C}_{p_m} \to 0$$

is an exact sequence of sheaves, inducing a long exact sequence of cohomology groups

$$0 \to H^0\left(X, \mathcal{O}_{(D-B)+p_1+\dots+p_{m-1}}\right) \to H^0(X, \mathcal{O}_{D+A}) \to \mathbb{C}$$

$$\xrightarrow{\delta} H^1\left(X, \mathcal{O}_{(D-B)+p_1+\dots+p_{m-1}}\right) \to H^1(X, \mathcal{O}_{D+A}) \to 0.$$

In particular, we have a surjection $H^1(X, \mathcal{O}_{D-B}) \to H^1(X, \mathcal{O}_{D+A}) \cong H^1(X, \mathcal{O}_D)$, hence ψ^* is injective. ∎

Suppose $\Lambda \in H^1(X, \mathcal{O}_D)^*$ is a nonzero linear functional, we need to show that there exists $\omega \in H^0(X, \omega_{-D})$ such that $\iota_{D(\omega)} = \Lambda$.

For $p \in X$, let $D_n = D - np$, then $\psi \in H^0\left(X, \mathcal{O}_{np}\right)$ induces an injective map

$$H^1(X, \mathcal{O}_D)^* \xrightarrow{\psi^*} H^1\left(X, \mathcal{O}_{D_n}\right)^*.$$

Let

$$\Lambda = \{\psi\Lambda \mid \psi \in H^0\left(X, \mathcal{O}_{np\}}\right) \subset H^1\left(X, \mathcal{O}_{D_n}\right)^*,$$

then Λ is a linear subspace of $H^1\left(X, \mathcal{O}_{D_n}\right)^*$. By Riemann-Roch theorem, we have $\dim \Lambda = \dim H^0\left(X, \mathcal{O}_{np}\right) \geq n - g + 1$, hence $\dim \Lambda \to \infty$ as $n \to \infty$. On the other hand, by Riemann-Roch theorem again, we have

$$\dim \operatorname{Im}\left(\iota_{D_n}\right) = \dim H^0\left(X, \omega_{-D_n}\right) = \dim H^1\left(X, \mathcal{O}_{K-D_n}\right) + 1 - g + \deg\,(K - D_n).$$

For $n > \deg D$, we have $H^0\left(X, \mathcal{O}_{D_n}\right) = 0$, hence $\dim H^1\left(X, \mathcal{O}_{D_n}\right)^* = g - 1 +$ $\deg D_n = g - 1 + \deg D - n$. We can pick n large enough such that

$$\dim \Lambda + \dim \operatorname{Im}\left(\iota_{D_n}\right) > \dim H^1\left(X, \mathcal{O}_{D_n}\right)^*,$$

hence $\Lambda \cap \operatorname{Im}\left(\iota_{D_n}\right) \neq 0$, *so there exists* $\psi\Lambda \in \iota_{D_n}(\omega_n)$ *with* $\omega_n \in H^0\left(X, \omega_{-D_n}\right)$. *Since* $\psi \in H^0\left(X, \mathcal{O}_{np}\right)$, *we have* $A + np \geq 0$ *and* $\frac{1}{\psi} \in H^0(X, \mathcal{O}_A)$. *Let* $D' = D_n - A = D - np - A$, *then* $D' \leq D$ *so* $0 \to \mathcal{O}_{D'} \hookrightarrow \mathcal{O}_D$ *is an injection of sheaves, inducing a surjection*

$$\pi : H^1(X, \mathcal{O}_{D'}) \to H^1(X, \mathcal{O}_D).$$

Define $\iota_{D'}^{D(\Lambda)} := \Lambda \circ \pi \in H^1(X, \mathcal{O}_{D'})^*$. *Since* $\psi\Lambda = \iota_{D_n}(\omega_n)$, *functoriality of the pairing gives*

$$\iota_{D'}^{D(\Lambda)} = \iota_{D'}!\left(\frac{1}{\psi}\omega_n\right).$$

We claim that $\omega_0 := \frac{1}{\psi}\omega_n \in H^0(X, \omega_{-D})$. *If not, then there exists* $a \in X$ *such that* $\operatorname{ord}_{a(\omega_0)} < D(a)$. *Let* U_0 *be a neighborhood around* a *isomorphic to a disk in* $\mathbb{C}$ *such that* a *corresponds to* 0, *and* $\omega_0|_{U_0} = f\,dz$ *for some* f *with no zeros or poles in* $U_0 \setminus \{a\}$, *and* $D|_{U_0 \setminus \{a\}} = 0$ *and* $D'|_{U_0 \setminus \{a\}} = 0$. *Set* $U_1 = X \setminus \{a\}$, *then* $\mathcal{U} = (U_0, U_1)$ *is an open cover of* X. *Let* $f_0 = \frac{1}{zf} \in \mathcal{O}_{M(U_0)}$ *and* $f_1 = 0 \in \mathcal{O}_{M(U_1)}$, *then* $\eta = (f_0, f_1) \in C^0(\mathcal{U}, \mathcal{M})$, *then*

$$\begin{aligned}
\omega_0\eta = (\omega_0 f_0, \omega_0 f_1) = \left(d\frac{z}{z}, 0\right) &\in C^0\left(\mathcal{U}, \mathcal{M}^{(1)}\right) \\
\Rightarrow \operatorname{ord}_{a(\omega_0\eta)} &= -1 \\
\Rightarrow \operatorname{ord}_{a(\eta)} &= -1 - \operatorname{ord}_{a(\omega_0)} \geq -D(a) \\
\Rightarrow \eta &\in C^0(\mathcal{U}, \mathcal{O}_D) \\
\Rightarrow \delta\eta &\in Z^1(\mathcal{U}, \mathcal{O}_D) \\
\Rightarrow \xi = [\delta\eta] &= 0 \in H^1(X, \mathcal{O}_D).
\end{aligned}$$

Note that $\delta\eta \in Z^1(\mathcal{U}, \mathcal{O}_D) = Z^1(\mathcal{U}, \mathcal{O}_{D'})$. *Let* $\zeta' := [\delta\eta] \in H^1(X, \mathcal{O}_{D'})$. *Therefore,*

$$\langle \omega_0, \zeta' \rangle = \iota_{D'}^{D}(\Lambda)(\xi) = \Lambda(\xi) = 0 \rightsquigarrow \langle \omega_0, \zeta' \rangle = \operatorname{Res}(\omega_0 \cdot \eta) = 1.$$

This is a contradiction, hence $\omega_0 \in H^0(X, \omega_{-D})$. ∎

2.6. Mittag-Leffler Problem

Let X be a compact Riemann surface of genus $g \geq 1$. Recall that for $p \in X$, by Riemann-Roch, there exists a holomorphic covering map $f : X \to \mathbb{P}^1$ with at most $g + 1$ sheets. We would like to know whether we can find a holomorphic covering map $f : X \to \mathbb{P}^1$ with at most g sheets. If such mapping exists, then on a local coordinate z around p with $z(p) = 0$, the principal part of f at p is of the form

$$\sum_{i=0}^{g-1} \frac{c_i}{z^{i+1}}$$

with $\left(c_0, ..., c_{g-1}\right) \neq 0$.

DEFINITION 2.30. *Let* $\mu = (f_i) \in C^0(\mathfrak{U}, \mathcal{M})$ *for some open cover* $\mathfrak{U} = (U_i)_{i \in I}$ *of* X. *We say* μ *is a Mittag-Leffler distribution if* $\delta\mu \in Z^1(\mathfrak{U}, \mathcal{O})$, *i.e.* $f_i - f_j$ *is holomorphic on* $U_i \cap U_j$ *for every* i, j. *We say* $f \in \mathcal{M}(X)$ *is a solution to* μ *if* $f - f_i \in \mathcal{O}(U_i)$ *for every* i.

PROPOSITION 2.31. $\mu \in C^0(\mathfrak{U}, \mathcal{M})$ *has a solution if and only if* $\delta\mu = 0$ *in* $H^1(X, \mathcal{O})$.

Proof.

(1) "$\Rightarrow$": If f is a solution to μ, then set $g_i = f_i - f \in \mathcal{O}(U_i)$, then $f_j - f_i = g_j - g_i$ on $U_i \cap U_j$, hence $\delta\mu = 0$ in $H^1(X, \mathcal{O})$.
(2) "$\leq$": If $\delta\mu = 0$ in $H^1(X, \mathcal{O})$, then there exists $g = (g_i) \in C^0(\mathfrak{U}, \mathcal{O})$ such that $f_j - f_i = g_j - g_i$ on $U_i \cap U_j$, hence we can glue $f_i - g_i$ to a global meromorphic function f on X, which is a solution to μ.

∎

Theorem 2.32. *μ has a solution if and only if* $\mathrm{Res}(\omega\mu) = 0$ *for every* $\omega \in H^0(X, \omega)$.

Proof.

$$\begin{aligned}\mu \text{ has a solution} &\iff \delta\mu = 0 \ \text{ in } \ H^1(X,\mathcal{O}) \\ &\iff \Lambda(\delta\mu) = 0 \ \text{ for every } \ \Lambda \in H^1(X,\mathcal{O})^* \\ &\iff \langle \omega, \delta\mu \rangle = 0 \ \text{ for every } \ \omega \in H^0(X,\omega) \\ &\iff \mathrm{Res}(\omega\mu) = 0 \ \text{ for every } \ \omega \in H^0(X,\omega).\end{aligned}$$

∎

Let $\omega_1, ..., \omega_g$ be a basis of $\omega(X)$. On a coordinate neighborhood U around p with coordinate z such that $z(p) = 0$, we can write

$$\omega_k = f_k dz = \sum_{i=0}^{\infty} a_{ki} z^i dz \ \text{ for every } \ k = 1, ..., g.$$

Consider the covering $\mathfrak{U} = (U, X \setminus \{p\})$ and the Mittag-Leffler distribution $\mu = (h, 0) \in C^0(\mathfrak{U}, \mathcal{M})$ with $h = \sum_{i=0}^{g-1} \frac{c_i}{z^{i+1}}$ for some $\left(c_0, ..., c_{g-1}\right) \neq 0$. By the previous theorem, μ has a solution if and only if

$$\mathrm{Res}(\omega_k\mu) = \mathrm{Res}(\omega_k h) = \sum_{i=0}^{g-1} a_{ki} c_i = 0$$

for every $k = 1, ..., g$. Therefore, μ has a solution if and only if the matrix

$$\prod_{k=0}^{g-1}(k!) \cdot \begin{pmatrix} a_{10} & a_{11} & \dots & a_{1,g-1} \\ a_{20} & a_{21} & \dots & a_{2,g-1} \\ \vdots & \vdots & \ddots & \vdots \\ a_{g0} & a_{g1} & \dots & a_{g,g-1} \end{pmatrix} = \begin{pmatrix} f_1(0) & f_{1'}(0) & \dots & f_1^{(g-1)}(0) \\ f_2(0) & f_{2'}(0) & \dots & f_2^{(g-1)}(0) \\ \vdots & \vdots & \ddots & \vdots \\ f_{g(0)} & f_{g'}(0) & \dots & f_g^{(g-1)}(0) \end{pmatrix}$$

is singular, i.e. has zero determinant. Define the above matrix to be the Wronskian matrix of $f_1, ..., f_g$ at p, and its determinant

$$W\left(f_1, ..., f_g\right)(p) = \det \begin{pmatrix} f_1(0) & f_{1'}(0) & \dots & f_1^{(g-1)}(0) \\ f_2(0) & f_{2'}(0) & \dots & f_2^{(g-1)}(0) \\ \vdots & \vdots & \ddots & \vdots \\ f_{g(0)} & f_{g'}(0) & \dots & f_g^{(g-1)}(0) \end{pmatrix}$$

is called the Wronskian of $f_1, ..., f_g$ at p. For a basis $\omega_1, ..., \omega_g$ of $\omega(X)$, the Wronskian $W\left(\omega_1, ..., \omega_g\right)(p)$ is defined to be $W\left(f_1, ..., f_g\right)(p)$, where $\omega_k = f_k dz$ on U for every k. p is called a Weierstrass point if $W\left(\omega_1, ..., \omega_g\right)(p) = 0$ for some (or equivalently, any) basis

$\omega_1, ..., \omega_g$ of $\omega(X)$, and the weight of p is defined to be the order of zero of $W(\omega_1, ..., \omega_g)$ at p.

We need to verify a few things about the Wronskian.

PROPOSITION 2.33. *If $f_1, ..., f_g$ are linearly independent over $\mathbb{C}$, then $W(f_1, ..., f_g)$ is not identically zero.*

Proof. Proceed by induction on g. The case $g = 1$ is trivial. For $g > 1$, assume that $W(f_1, ..., f_{g-1})$ is not identically zero, then consider the differential equation in variable w:

$$\det \begin{pmatrix} f_1 & f_{1'} & \dots & f_1^{(g-1)} \\ f_2 & f_{2'} & \dots & f_2^{(g-1)} \\ \vdots & \vdots & \ddots & \vdots \\ f_{g-1} & f_{(g-1)'} & \dots & f_{g-1}^{(g-1)} \\ w & w' & \dots & w^{(g-1)} \end{pmatrix} = W(f_1, ..., f_{g-1})w^{(g-1)} + ... = 0.$$

Since $W(f_1, ..., f_{g-1})$ is not identically zero, the above equation is a linear differential equation with meromorphic coefficients, hence its solution space has dimension $g - 1$. Since $f_1, ..., f_{g-1}$ are linearly independent solutions to the above equation, they form a basis of the solution space. Since f_g is not a linear combination of $f_1, ..., f_{g-1}$, f_g is not a solution to the above equation, hence $W(f_1, ..., f_g)$ is not identically zero. ∎

PROPOSITION 2.34. *If (U, z) and $(\tilde{U}, \tilde{z})$ are two charts, then on $U \cap \tilde{U}$, we have*

$$W_{z(\omega_1, ..., \omega_g)} = \left(\frac{d\tilde{z}}{dz}\right)^N W_{\tilde{z}}(\omega_1, ..., \omega_g),$$

where $N = \frac{g(g+1)}{2}$.

Proof. Write $\omega_k = f_k dz = \tilde{f}_k d\tilde{z}$ on $U \cap \tilde{U}$, then $f_k = \psi \tilde{f}_k$ with $\psi = \frac{d\tilde{z}}{dz} \in \mathcal{O}^*(U \cap \tilde{U})$. Therefore,

$$\frac{d^m f_k}{dz^m} = \psi^{m+1} \frac{d^m \tilde{f}_k}{d\tilde{z}^m} + \text{lower order terms in } \; \psi, \frac{d\psi}{dz}, ..., \frac{d^{m-1}\psi}{dz^{m-1}},$$

hence

$$W_{z(\omega_1, ..., \omega_g)} = \left(\frac{d\tilde{z}}{dz}\right)^{1+2+...+g} W_{\tilde{z}}(\omega_1, ..., \omega_g).$$

∎

PROPOSITION 2.35. *If $\tilde{\omega}_1, ..., \tilde{\omega}_g$ is another basis of $\omega(X)$, then $W(\tilde{\omega}_1, ..., \tilde{\omega}_g) = cW(\omega_1, ..., \omega_g)$ for some $c \in \mathbb{C}^*$.*

Proof. Write $\tilde{\omega}_k = \sum_{j=1}^g c_{jk}\omega_j$ for some $C = (c_{jk}) \in \mathrm{GL}_{g(\mathbb{C})}$, then on the local coordinate z around p with $z(p) = 0$, we have $\tilde{\omega}_k = \tilde{f}_k dz$ with $\tilde{f}_k = \sum_{j=1}^g c_{jk} f_j$, hence

$$W(\tilde{\omega}_1, ..., \tilde{\omega}_g) = W(\tilde{f}_1, ..., \tilde{f}_g) = \det C \cdot W(f_1, ..., f_g) = \det C \cdot W(\omega_1, ..., \omega_g).$$

∎

THEOREM 2.36. *p is a Weierstrass point if and only if there exists a non-constant meromorphic function on X with only pole at p of order at most g.*

Proof. Straightforward from the discussion before the definition of Wronskian. ∎

THEOREM 2.37. *The weighted sum of Weierstrass points on X is $(g-1)g(g+1)$.*

Proof. Let $(U_i, z_i)_{i\in I}$ be a covering of X and $\psi_{ij} = \frac{dz_j}{dz_i} \in \mathcal{O}^*(U_i \cap U_j)$. Choose a basis $\omega_1, ..., \omega_g$ of $\omega(X)$. Let W_i be the Wronskian of $\omega_1, ..., \omega_g$ on U_i, then by the previous proposition, we have $W_j = \psi_{ij}^N W_i$ on $U_i \cap U_j$ with $N = \frac{g(g+1)}{2}$. Define the divisor D by $D(x) = \mathrm{ord}_{x(W_i)}$ for $x \in U_i$, then D is well-defined since ψ_{ij} is non-vanishing, and the weighted sum of Weierstrass points is precisely deg D. Write $\omega_1 = f_{1i} dz_i$ on U_i for every i, then $f_{1i} = \psi_{ij} f_{1j}$ on $U_i \cap U_j$, hence

$$f_{1i}^N = \psi_{ij}^N f_{1j}^N \rightsquigarrow W_i f_{1i}^{-N} = W_j f_{1j}^{-N} \quad \text{on} \quad U_i \cap U_j.$$

Therefore, we can glue $W_i f_{1i}^{-N}$ to a global meromorphic function f on X, hence

$$D = (f) + N(\omega_1) \rightsquigarrow \deg D = N \cdot \deg(\omega_1) = N \cdot (2g-2) = (g-1)g(g+1).$$

∎

THEOREM 2.38. *Let X be a compact Riemann surface of genus $g \geq 2$, then exists a holomorphic covering $f : X \to \mathbb{P}^1$ with at most g sheets.*

Proof. For surfaces of genus ≥ 2, the previous theorem gurantees the existence of a Weierstrass point, hence there exists a non-constant meromorphic function on X with only pole at p of order at most g, i.e. a holomorphic covering $f : X \to \mathbb{P}^1$ with at most g sheets. ∎

LEMMA 2.39. *Every compact Riemann surface of genus 2 is hyperelliptic, i.e. there exists a holomorphic covering $f : X \to \mathbb{P}^1$ with at most 2 sheets.*

Proof. By the previous theorem, there exists a holomorphic covering $f : X \to \mathbb{P}^1$ with at most 2 sheets. If f is a holomorphic covering with only one sheet, then f is an isomorphism, hence $X \cong \mathbb{P}^1$, which contradicts the assumption that $g = 2$. Therefore, f is a holomorphic covering with exactly two sheets, so X is hyperelliptic. ∎

Example 2.2. Let $\Gamma_1, \Gamma_2 \in \mathbb{C}$ be two linearly independent complex numbers over $\mathbb{R}$, and let $P = \{t_1\Gamma_1 + t_2\Gamma_2 \mid 0 \leq t_1 < 1, 0 \leq t_2 < 1\}$. Suppose at $a_1, ..., a_n \in P$, we have prescribed principal parts

$$h_j = \sum_{k=-r_j}^{-1} c_k^{(j)} (z - a_j)^k \quad \text{for every} \quad j = 1, ..., n.$$

Then there exists a global meromorphic function f on $\mathcal{M}(\mathbb{C})$ doubly periodic with respect to $\Gamma = \mathbb{Z}\Gamma_1 + \mathbb{Z}\Gamma_2$ with the prescribed principal part h_j at a_j for every j if and only if

$$\sum_{j=1}^{n} c_{-1}^{(j)} = 0.$$

This is straightforward since the existence of such f is equivalent to the existence of a solution to the Mittag-Leffler distribution $\mu = (h_1, ..., h_n, 0) \in C^0(\mathfrak{U}, \mathcal{M})$ with $\mathfrak{U} =$

$(U_1, ..., U_n, \mathbb{C} \setminus P)$ and U_j a neighborhood of a_j for every j, which is equivalent to $\mathrm{Res}(\omega\mu) = 0$ for every $\omega \in \omega(X)$, where $X = \mathbb{C}/\Gamma$. Since $\omega(X)$ is generated by dz, we have $\mathrm{Res}(\omega\mu) = \mathrm{Res}(dz \cdot \mu) = \sum_{j=1}^{n} c_{-1}^{(j)}$, hence the existence of such f is equivalent to $\sum_{j=1}^{n} c_{-1}^{(j)} = 0$.

PROPOSITION 2.40. *If D is a divisor of degree $\deg D > 2g - 2$, then $H^1(X, \mathcal{O}_D) = 0$.*

Proof. By Serre duality, we have

$$H^1(X, \mathcal{O}_D)^* \cong H^0(X, \omega_{-D}) \cong H^0(X, \mathcal{O}_{K-D}),$$

which equals zero since $\deg\,(K - D) < 0$. ∎

PROPOSITION 2.41. *Let X be a compact Riemann surface, then $H^1(X, \mathcal{M}) = 0$.*

Proof. Fix a 1-cocycle $\left(f_{ij}\right) \in Z^1(\mathfrak{U}, \mathcal{M})$. We may assume that $\mathfrak{U}$ is finite and the total number of poles of $\left(f_{ij}\right)$ is finite by passing to a refinement. Thus we can find a divisor D such that $f_{ij} \in \mathcal{O}_{D(U_i \cap U_j)}$ for every i, j and $\deg D > 2g - 2$. By the previous proposition, we have $H^1(X, \mathcal{O}_D) = 0$, hence $\left(f_{ij}\right) \in B^1(\mathfrak{U}, \mathcal{O}_D) \subset B^1(\mathfrak{U}, \mathcal{M})$, which implies that $\left(f_{ij}\right) = 0$ in $H^1(X, \mathcal{M})$. ∎

2.6.1. Riemann-Hurwitz Formula

DEFINITION 2.42. *Let X, Y be compact Riemann surfaces, and $f : X \to Y$ be a holomorphic covering map. For $x \in X$, the ramification index of f at x is defined to be the positive integer k such that f can be written as $f(z) = z^k$ on a local coordinate z around x with $z(x) = 0$, denoted by $nu(f, x)$. If $nu(f, x) > 1$, then x is called a ramification point of f, and $f(x)$ is called a branch point of f. The branching number $b(f, x)$ is defined to be $nu(f, x) - 1$. The total branching number of f is defined to be $b = \sum_{x \in X} b(f, x)$.*

THEOREM 2.43. *Let X, Y be compact Riemann surfaces, and $f : X \to Y$ be a n-sheeted holomorphic covering map, then*

$$g = \frac{b}{2} + n(g' - 1) + 1,$$

where g and g' are the genera of X and Y respectively, and b is the total branching number of f.

Proof. Let ω be a non-vanishing meromerphic 1-form on Y and $\omega|_V = g(z')dz'$ on a local coordinate z' around $y \in Y$ with $z'(y) = 0$. For $x \in f^{-1}(y)$, let z be a local coordinate around x with $z(x) = 0$ such that $f(z) = z^k$ with $k = nu(f, x)$, then

$$f^*\omega = g\left(z^k\right)d\left(z^k\right) = kg\left(z^k\right)z^{k-1}dz,$$

hence

$$\mathrm{ord}_{x(f^*\omega)} = (k - 1) + k \cdot \mathrm{ord}_{y(\omega)} = b(f, x) + nu(f, x) \cdot \mathrm{ord}_{y(\omega)}.$$

Summing over all $x \in f^{-1}(y)$ gives

$$\begin{aligned}\sum_{x \in f^{-1}(y)} \mathrm{ord}_{x(f^*\omega)} &= \sum_{x \in f^{-1}(y)} b(f, x) + \sum_{x \in f^{-1}(y)} nu(f, x) \cdot \mathrm{ord}_{y(\omega)} \\ &= b(f, y) + n \cdot \mathrm{ord}_{y(\omega)}.\end{aligned}$$

Summing over all $y \in Y$ gives

$$\deg(f^*\omega) = \sum_{x\in X} \mathrm{ord}_{x(f^*\omega)} = \sum_{y\in Y} b(f,y) + n \cdot \sum_{y\in Y} \mathrm{ord}_{y(\omega)} = b + n \cdot \deg\,(\omega).$$

Since $\deg\,(\omega) = 2g' - 2$ and $\deg\,(f^*\omega) = 2g - 2$, we have

$$2g - 2 = b + n(2g' - 2) \rightsquigarrow g = \frac{b}{2} + n(g' - 1) + 1).$$

$\blacksquare$

2.7. deRham-Hodge Theorem

Recall that

$$0 \to \mathbb{C} \to \mathcal{E}(X) \xrightarrow{d} \mathcal{E}^{(1)}(X) \xrightarrow{d} \mathcal{E}^{(2)}(X)$$

is a complex of sheaves on X, hence we can define the first deRham cohomology group of X to be

$$\mathrm{Rh}^1(X) = \frac{\ker\left(\mathcal{E}^{(1)}(X) \xrightarrow{d} \mathcal{E}^{(2)}(X)\right)}{\mathrm{Im}\left(\mathcal{E}(X) \xrightarrow{d} \mathcal{E}^{(1)}(X)\right)}.$$

If X is simply connected, then every closed 1-form on X is exact, hence $\mathrm{Rh}^1(X) = 0$. If we let $\texttt{cal}\ L(U) = \ker\left(\mathcal{E}^{(1)}(U) \xrightarrow{d} \mathcal{E}^{(2)}(U)\right)$ for every open set U, then $\texttt{cal}\ L$ is the sheaf of closed 1-forms on X, and we have a short exact sequence of sheaves

$$0 \to \mathbb{C} \to \mathcal{E} \xrightarrow{d} \texttt{cal}\ L \to 0,$$

which induces a long exact sequence of cohomology groups

$$0 \to H^0(X,\mathbb{C}) \to H^0(X,\mathcal{E}) \xrightarrow{d} H^0(X,\texttt{cal}\ L) \to H^1(X,\mathbb{C}) \to H^1(X,\mathcal{E}) \to \ldots.$$

Since $H^1(X,\mathcal{E}) = 0$, we have

$$H^1(X,\mathbb{C}) \cong \frac{\texttt{cal}\ L(X)}{d\mathcal{E}(X)} = \mathrm{Rh}^1(X).$$

DEFINITION 2.44.

(i) *For each $\omega \in \mathcal{E}^{(1)}(X)$, we can write $\omega|_U = fdz + gd\,\texttt{overline}\ z$ for some local coordinate z on U, then we define $\texttt{overline}\ \omega|_U = \texttt{overline}\ fd\,\texttt{overline}\ z + \texttt{overline}\ gdz$.*

(ii) *If $\omega \in \omega(X)$ is a holomorphic 1-form on X, then $\texttt{overline}\ \omega$ is called an anti-holomorphic 1-form on X. We denote the space of anti-holomorphic 1-forms on X by $\texttt{overline}\ \omega(X)$.*

(iii) *Define the Hodge star operator $* : \mathcal{E}^{(1)}(X) \to \mathcal{E}^{(1)}(X)$ by*

$$fdz + gd\,\texttt{overline}\ z \mapsto i(\texttt{overline}\ fd\,\texttt{overline}\ z - \texttt{overline}\ gdz).$$

We have $**\omega = -\omega$, hence $*$ is an isomorphism. Moreover, we can define a positive-definite inner product on $\mathcal{E}^{(1)}(X)$ by

$$\langle \omega_1, \omega_2 \rangle = \iint_X \omega_1 \wedge *\,\texttt{overline}\ \omega_2.$$

We can verify that

$$\langle \omega, \omega \rangle = \iint_X \omega \wedge * \,\texttt{overline}\ \omega = 2 \iint_X (|f|^2 + |g|^2) dx \wedge dy \geq 0,$$

and $\langle \omega, \omega \rangle = 0$ *if and only if* $\omega = 0$. *Also,*

$$\overline{\langle \omega_1, \omega_2 \rangle} = \iint_X \overline{\omega_1} \wedge * \,\omega_2 = \iint_X * \,\omega_2 \wedge \overline{\omega_1} = \iint_X \omega_2 \wedge * \,\overline{\omega_1} = \langle \omega_2, \omega_1 \rangle,$$

Hence $\mathcal{E}^{(1)}(X)$ *is a unitary space with respect to the above inner product.*

DEFINITION 2.45. *A 1-form* $\omega \in \mathcal{E}^{(1)}(X)$ *is called harmonic if* $d\omega = 0$ *and* $d(* \,\omega) = 0$. *Locally, if* $\omega|_U = fdz + gd\ \texttt{overline}\ z$, *then* ω *is harmonic if and only if*

$$\frac{\partial g}{\partial z} = \frac{(\partial F)}{\partial\ \texttt{overline}\ z}, \quad \frac{\partial\ \texttt{overline}\ f}{\partial z} = -\frac{\partial\ \texttt{overline}\ g}{\partial\ \texttt{overline}\ z}.$$

Let $\mathrm{Harm}^1(X)$ *be the space of harmonic 1-forms on* X.

PROPOSITION 2.46. *Let* $\omega \in \mathcal{E}^{(1)}(X)$. *The following are equivalent:*

(i) ω *is harmonic;*
(ii) $d'\omega = d''\omega = 0$*;*
(iii) $\omega = \omega_1 + \omega_2$ *for some* $\omega_1 \in \omega(X)$ *and* $\omega_2 \in \texttt{overline}\ \omega(X)$*;*
(iv) *For all* $a \in X$, *there exists a neighborhood* U *around* a *such that* $\omega = df$ *for some harmonic function* f *on* U.

Proof.

(1) "(i) $\Rightarrow$ (ii)": Locally, if $\omega|_U = fdz + gd\ \texttt{overline}\ z$, then

$$d'\omega = \frac{\partial g}{\partial z} dz \wedge d\ \texttt{overline}\ z, \quad d''\omega = -\frac{(\partial F)}{\partial\ \texttt{overline}\ z} dz \wedge d\ \texttt{overline}\ z,$$

and

$$\frac{\partial g}{\partial z} = \frac{\partial \overline{g}}{\partial \overline{z}} = \frac{\partial \overline{f}}{\partial z} = -\frac{(\partial F)}{\partial \overline{z}} = -\frac{\partial g}{\partial z} \rightsquigarrow \frac{\partial g}{\partial z} = \frac{(\partial F)}{\partial\ \texttt{overline}\ z} = 0.$$

(1) "(ii) $\Rightarrow$ (iii)": Locally, if $\omega|_U = fdz + gd\ \texttt{overline}\ z$, then $d'\omega = 0$ implies $\frac{\partial g}{\partial z} = 0$, hence g is anti-holomorphic on U, and $d''\omega = 0$ implies $\frac{(\partial F)}{\partial\ \texttt{overline}\ z} = 0$, hence f is holomorphic on U.
(2) "(i)(ii) $\Rightarrow$ (iv)": Since ω is closed, locally $\omega|_U = df$ for some function f on U. We have

$$d'\omega = d'(df) = d'(d' + d'')f = d'd''f = 0,$$

hence f is harmonic on U.

(1) "(iv) $\Rightarrow$ (ii)": Locally, if $\omega|_U = df$ for some harmonic function f on U, then $d'\omega = d'(df) = d'(d' + d'')f = d'd''f = 0$ and $d''\omega = d''(df) = d''(d' + d'')f = d''d'f = 0$.

∎

LEMMA 2.47.

$$\mathcal{E}^{0,1}(X) = d''\mathcal{E}(X) \oplus \texttt{overline}\ \omega(X),$$

where $\oplus$ *is orthogonal direct sum with respect to the inner product defined above.*

Proof. Suppose $f \in \mathcal{E}(X)$, $\omega \in \omega(X)$, then

$$\overline{\omega} \wedge * \, d'' f = -i\left(\overline{\omega} \wedge d' \overline{f}\right) = -i\overline{\omega} \wedge d \ \texttt{overline} \ f = id\left(\overline{f}\overline{\omega}\right)$$

hence

$$\langle \overline{\omega}, d'' f \rangle = \iint_X d\left(\overline{f}\overline{\omega}\right) = 0.$$

Therefore, $d''\mathcal{E}(X)$ is orthogonal to $\texttt{overline} \ \omega(X)$.

Clearly $\mathcal{E}^{0,1}(X) \supseteq d''\mathcal{E}(X) + \texttt{overline} \ \omega(X)$. Since

$$H^1(X, \mathcal{O}) = \frac{\mathcal{E}^{0,1}(X)}{d''\mathcal{E}(X)} \rightsquigarrow \dim \frac{\mathcal{E}^{0,1}(X)}{d''\mathcal{E}(X)} = g = \dim \texttt{overline} \ \omega(X),$$

we have $\mathcal{E}^{0,1}(X) = d''\mathcal{E}(X) \oplus \texttt{overline} \ \omega(X)$. ∎

Remark. Similarly, we can show that $\mathcal{E}^{1,0}(X) = d'\mathcal{E}(X) \oplus \omega(X)$.

Lemma 2.48.

$$d'\mathcal{E}(X) \oplus d''\mathcal{E}(X) = d\mathcal{E}(X) \oplus * \, d\mathcal{E}(X).$$

Proof. We have

$$d' f \wedge * \, d'' g = -id' f \wedge d' \texttt{overline} \ g = 0$$

and

$$df \wedge * \, (* \, dg) = -df \wedge dg = -d(fdg) \rightsquigarrow \langle df, * \, dg \rangle = -\iint_X d(fdg) = 0,$$

hence $d'\mathcal{E}(X) \perp d''\mathcal{E}(X)$ and $d\mathcal{E}(X) \perp * \, d\mathcal{E}(X)$.

Clearly $d\mathcal{E}(X) \subseteq d'\mathcal{E}(X) \oplus d''\mathcal{E}(X)$ and $* \, d\mathcal{E}(X) \subseteq d'\mathcal{E}(X) \oplus d''\mathcal{E}(X)$. Write

$$d' f = \frac{1}{2}\left(df + * \, d\left(-i\overline{f}\right)\right) \in d\mathcal{E}(X) \oplus * \, d\mathcal{E}(X),$$

$$d'' f = \frac{1}{2}\left(df + * \, d\left(i\overline{f}\right)\right) \in d\mathcal{E}(X) \oplus * \, d\mathcal{E}(X),$$

hence $d'\mathcal{E}(X) \oplus d''\mathcal{E}(X) \subseteq d\mathcal{E}(X) \oplus * \, d\mathcal{E}(X)$. ∎

Theorem 2.49.

$$\texttt{cal} \ L(X) := \ker\left(\mathcal{E}^{(1)}(X) \xrightarrow{d} \mathcal{E}^{(2)}(X)\right) = d\mathcal{E}(X) \oplus \mathrm{Harm}^1(X).$$

Proof. Clearly RHS is contained in LHS. Conversely, since $* \, d\mathcal{E}(X)$, $d\mathcal{E}(X)$ and $\mathrm{Harm}^1(X)$ are mutually orthogonal, it suffices to show that $\texttt{cal} \ L(X) \perp * \, d\mathcal{E}(X)$. Suppose $\omega \in \texttt{cal} \ L(X)$ and $f \in \mathcal{E}(X)$, then

$$\omega \wedge * * df = -\omega \wedge df = d(f\omega) \rightsquigarrow \langle \omega, * \, df \rangle = \iint_X d(f\omega) = 0.$$

∎

Theorem 2.50.

$$\mathcal{E}^{(1)}(X) = d\mathcal{E}(X) \oplus * \, d\mathcal{E}(X) \oplus \mathrm{Harm}^1(X).$$

Proof. By the previous two lemmas, we have

$$\mathcal{E}^{(1)}(X) = d'\mathcal{E}(X) \oplus d''\mathcal{E}(X) \oplus \omega(X) \oplus \texttt{overline} \ \omega(X) = d\mathcal{E}(X) \oplus * \, d\mathcal{E}(X) \oplus \mathrm{Harm}^1(X).$$

∎

THEOREM 2.51 (deRham-Hodge).
$$H^1(X,\mathbb{C}) \cong \mathrm{Rh}^1(X) \cong \mathrm{Harm}^1(X).$$

Proof. By the discussion before the definition of harmonic 1-forms, we have $H^1(X,\mathbb{C}) \cong \mathrm{Rh}^1(X)$. By the previous theorem, we have $\texttt{cal}\ L(X) = d\mathcal{E}(X) \oplus \mathrm{Harm}^1(X)$, hence
$$\mathrm{Rh}^1(X) = \frac{\texttt{cal}\ L(X)}{d\mathcal{E}(X)} \cong \mathrm{Harm}^1(X).$$
∎

PROPOSITION 2.52. *If $\sigma \in \mathrm{Harm}^1(X)$ with $\sigma = \overline{\sigma}$, i.e. σ is real, then exists a unique $\omega \in \omega(X)$ such that $\sigma = \mathrm{Re}\ \omega = \frac{\omega+\overline{\omega}}{2}$.*

Proof. Write $\sigma = \omega_1 + \overline{\omega_2}$ for $\omega_1, \omega_2 \in \omega(X)$, then
$$\sigma = \texttt{overline}\ \sigma \rightsquigarrow \omega_1 + \overline{\omega_2} = \overline{\omega_1} + \omega_2 \rightsquigarrow \omega_1 = \omega_2 \rightsquigarrow \sigma = \mathrm{Re}\ (2\omega_1).$$
The uniqueness of ω is straightforward since if $\sigma = \mathrm{Re}\ \omega = \mathrm{Re}\ \omega'$, then $\mathrm{Re}\ (\omega - \omega') = 0$, hence $\omega - \omega' = 0$ since $\omega - \omega' \in \omega(X)$. ∎

PROPOSITION 2.53. *Let X be a compact Riemann surface. If $\sigma \in \mathrm{Harm}^1(X)$ is exact, then $\sigma = 0$. If $f \in \mathcal{E}(X)$ is harmonic, then f is constant.*

Proof. Since $d\mathcal{E}(X) \perp \mathrm{Harm}^1(X)$, if $\sigma \in \mathrm{Harm}^1(X)$ is exact, then $\sigma \perp \sigma$, hence $\sigma = 0$. Since $df \in d\mathcal{E}(X) \perp \mathrm{Harm}^1(X)$, if f is harmonic, then $df = 0$, hence f is constant. ∎

PROPOSITION 2.54. *Let X be a compact Riemann surface and $\sigma \in \mathrm{Harm}^1(X)$. If for every closed curve Γ on X (resp. $\omega \in \omega(X)$), we have*
$$\int_\Gamma \sigma = 0, \left(\text{resp. } \mathrm{Re}\left(\int_\Gamma \omega\right) = 0\right),$$
then $\sigma = 0$ (resp. $\omega = 0$).

Proof. Since σ (resp. Re ω) is exact, the result follows from the previous theorem. ∎

2.8. *Jacobi Inversion Problem*

THEOREM 2.55 (Residue Theorem). *Let X be a compact Riemann surface, $a_1, ..., a_n \in X$ be distinct points and let $X' = X \setminus \{a_1, ..., a_n\}$. If ω is a meromorphic 1-form on X which is holomorphic on X', then*
$$\sum_{k=1}^n \mathrm{Res}_{a_k}(\omega) = 0.$$

Proof. Pick disjoint charts (U_k, z_k) around a_k for every k such that $z_{k(a_k)} = 0$ and $U_k \cong B_1(0)$. By partition of unity, exists $f_k \in \mathcal{E}(X)$ such that $0 \le f_k \le 1$, supp $(f_k) \subset B_R(0) \subset B_1(0)$ for some $R < 1$ and $f_k = 1$ on $B_\varepsilon(0)$ for some $0 < \varepsilon < R$. Since $\omega \in \omega(X')$, $d\omega = 0$ on X'. On $U_{k'} \cap X'$, $f_k\omega = \omega$, hence
$$\iint_X d(f_k\omega) = \lim_{\varepsilon'\to 0} \iint_{\varepsilon' \le |z| \le R} d(f_k\omega) = \int_{|z|=R} f_k\omega - \lim_{\varepsilon'\to 0} \int_{|z|=\varepsilon'} f_k\omega$$

$$= -\lim_{\varepsilon' \to 0} \int_{|z| = \varepsilon'} \omega = -2\pi i \ \mathrm{Res}_{a_k}(\omega).$$

Let $g = 1 - \sum_{k=1}^{n} f_k$, then $g \in \mathcal{E}(X)$, $0 \le g \le 1$ and supp $(g) \subset X'$. Since ω is holomorphic on X', we have

$$\iint_X d(g\omega) = 0.$$

Therefore,

$$\sum_{k=1}^{n} \mathrm{Res}_{a_k}(\omega) = -\frac{1}{2\pi i} \sum_{k=1}^{n} \iint_X d(f_k\omega) = -\frac{1}{2\pi i} \iint_X d\left(\sum_{k=1}^{n} f_k\omega\right)$$
$$= -\frac{1}{2\pi i} \iint_X d((1-g)\omega) = -\frac{1}{2\pi i} \iint_X d(\omega - g\omega) = 0.$$

∎

Let X be a compact Riemann surface of genus $g \ge 1$ and $\omega_1, ..., \omega_g$ be a basis of $\omega(X)$.

DEFINITION 2.56. *Let $C_1(X)$ be the free abelian group generated by all closed curves on X, i.e.*

$$C_1(X) = \{ \sum_{k=1\}^n a_k \Gamma_k \mid n \in Z_{\ge 0}, a_k \in \mathbb{Z}, \Gamma_k \text{ is a curve on } X \text{ for every } k} .$$

Define the boundary map $\partial : C_1(X) \to \mathrm{Div}\ (X)$ by $\partial(\Gamma) = \Gamma(1) - \Gamma(0)$ for every curve Γ on X, and extend ∂ linearly to $C_1(X)$. Define the group of 1-cycles on X to be

$$Z_1(X) = \ker \left(C_1(X) \xrightarrow{\partial} \mathrm{Div}\ (X) \right),$$

and let $B_1(X)$ be the closed curves Γ on X such that $\int_\Gamma \omega = 0$ for every $\omega \in \omega(X)$. Define the first homology group of X to be

$$H_1(X) = \frac{Z_1(X)}{B_1(X)}.$$

DEFINITION 2.57. *Let V be a N-dimensional complex vector space over $\mathbb{R}$. A lattice in V is an additive subgroup Γ of V such that $\Gamma = \mathbb{Z}\Gamma_1 + ... + \mathbb{Z}\Gamma_N$ for some linearly independent vectors $\Gamma_1, ..., \Gamma_N$ in V.*

PROPOSITION 2.58. *[Lattice Criterion] Γ is a lattice in V if and only if*

(i) Γ *is discrete in V;*
(ii) $\Gamma \not\subseteq W$ *for every proper subspace W of V.*

Proof. Clearly a lattice satisfies (i) and (ii).

For the converse, we proceed by induction on N. When $N = 0$ it's trivial. For $N > 0$, by (ii), $\exists x_1, ..., x_N \in \Gamma$ which are linearly independent over $\mathbb{R}$. Let $V_1 = \langle x_1, ..., x_{N-1} \rangle_{\mathbb{R}}$ and $\Gamma_1 = \Gamma \cap V_1$. By induction hypothesis, Γ_1 is a lattice in V_1, hence $\exists \Gamma_1, ..., \Gamma_{N-1} \in \Gamma_1$ such that $\Gamma_1 = \mathbb{Z}\Gamma_1 + ... + \mathbb{Z}\Gamma_{N-1}$. Also $\Gamma_1, ..., \Gamma_{N-1}, x_N$ are linearly independent over $\mathbb{R}$.

Consider the parallelotope

$$P = \{t_1\Gamma_1 + ... + t_{N-1\}\Gamma_{N-1} + t_N x_N \mid 0 \le t_i < 1 \text{ for every } i}.$$

For all $x \in V$, we can uniquely write $x = c_1(x)\Gamma_1 + ... + c_{N-1}(x)\Gamma_{N-1} + c(x)x_N$ for some $c_{i(x)}, c(x) \in \mathbb{R}$. Pick

$$c(\Gamma_N) = m \in \{c(x) \mid x \in (\Gamma \cap P) \setminus \Gamma_1\}.$$

Since Γ is discrete, the minimum is attained and $c(\Gamma_N) > 0$. We claim that $\Gamma = \Gamma_1 + \mathbb{Z}\Gamma_N$. For all $x \in \Gamma$, exists $n \in \mathbb{Z}$ such that $c(x) = nc(\Gamma_N) + \Lambda$ for some $0 \leq \Lambda < c(\Gamma_N)$ and $c_{i(x)} - nc_{i(\Gamma_N)} = n_i + \Lambda_i$ for some $n_i \in \mathbb{Z}$ and $0 \leq \Lambda_i < 1$ for every i. Then

$$x - n\Gamma_N - \sum_{i=1}^{N-1} n_i\Gamma_i = \sum_{i=1}^{N-1} \Lambda_i\Gamma_i + \Lambda x_N \in \Gamma \cap P$$

hence $\Lambda = 0$, which implies that $x - n\Gamma_N - \sum_{i=1}^{N-1} n_i\Gamma_i \in \Gamma_1$, hence $x \in \Gamma_1 + \mathbb{Z}\Gamma_N$. This shows that Γ is generated by $\Gamma_1, ..., \Gamma_{N-1}, \Gamma_N$, hence Γ is a lattice in V. ∎

DEFINITION 2.59. *The period lattice of X with respect to $\omega_1, ..., \omega_g$ is defined to be*

$$\Lambda = \mathrm{Per}\ (\omega_1, ..., \omega_g) = \{\left(\int_\Gamma \omega_1, ..., \int_\Gamma \omega_g\right) \mid \Gamma \in H_1(X)\} \subseteq \mathbb{C}^g.$$

The Jacobi variety of X is defined to be Jac $(X) = \frac{\mathbb{C}^g}{\Lambda}$.

THEOREM 2.60. Γ *is a lattice in* $\mathbb{C}^g \cong \mathbb{R}^{2g}$.

Proof. We first show that there exists distinct $a_1, ...a_g$ in X such that for all $\omega \in (X)$, if $\omega(a_1) = ... = \omega(a_g) = 0$, then $\omega = 0$. For $a \in X$, define

$$H_a \coloneqq \{\omega \in \omega(X) \mid \omega(a) = 0\},$$

then either $H_a = \omega(X)$ or codim $H_a = 1$. Sine $\bigcap_{a \in X} H_a = \{0\}$, we can find $a_1 \in X$ such that $H_{a_1} \neq \omega(X)$, then find $a_2 \in X$ such that $H_{a_1} \cap H_{a_2} \neq H_{a_1}$, and so on, until we find $a_1, ..., a_g$ such that $\bigcap_{k=1}^g H_{a_k} = \{0\}$.

Let $\omega_1, ..., \omega_g$ be a basis of $\omega(X)$. Find disjoint charts (U_k, z_k) around a_k for every k such that $z_{k(a_k)} = 0$ and $U_k \cong B_1(0)$. Locally write $\omega_i|_{U_j} = \phi_{ij} dz_j$ for some holomorphic function $\phi_{ij} \in \mathcal{O}(U_j)$. Consider the matrix $A = \left(\phi_{ij}(a_j)\right)_{1 \leq i,j \leq g}$, then A is full-rank since $\{\omega_i\}$ is a basis of $\omega(X)$ and $\bigcap_{k=1}^g H_{a_k} = \{0\}$. Define a map

$$F : U_1 \times ... \times U_g \to \mathbb{C}^g, \quad (x_1, ..., x_g) \mapsto \left(\sum_{j=1}^g \int_{a_j}^{x_j} \omega_1, ..., \sum_{j=1}^g \int_{a_j}^{x_j} \omega_g\right).$$

The integral is independent of the choice of path since U_j is simply connected for every j. The Jacobian matrix of F at $(a_1, ..., a_g)$ is $\left(\frac{\partial F_i}{\partial z_j}\right)_{1 \leq i,j \leq g} = A$, which is invertible, hence F is a local biholomorphism around $(a_1, ..., a_g)$. Therefore, there exists an open neighborhood V of 0 in $\mathbb{C}^g$ such that F induces a biholomorphism between V and an open neighborhood of $(a_1, ..., a_g)$ in X^g. We claim that $\Gamma \cap V = \{0\}$. If not, say $F(x_1, ..., x_g) \in \Gamma \cap (V \setminus \{0\})$ for some $(x_1, ..., x_g) \in U_1 \times ... \times U_g$. Suppose that after reindexing, $x_j \neq a_j$ for $j = 1, ..., k$ and $x_j = a_j$ for $j = k + 1, ..., g$. By Abel's theorem, exists a meromorphic function f on X such that $(f) = x_1 + ... + x_k - a_1 - ... - a_k$, i.e. $\mathrm{Res}_{a_j}(f) = c_j \neq 0$ for $j = 1, ..., k$. By Residue theorem,

$$0 = \mathrm{Res}(f\omega_i) = \sum_{j=1}^k \mathrm{Res}_{a_j}(f\omega_i) = \sum_{j=1}^k c_j\phi_{ij}(a_j),$$

which implies that $A(c_1, ..., c_k)^T = 0$, contradicting the invertibility of A. Therefore, $\Gamma \cap V = \{0\}$, which implies that Γ is discrete in $\mathbb{C}^g$. If Γ is contained in a proper subspace W of $\mathbb{C}^g$, say

$$W = \{(x_1, y_1, ..., x_g, y_g) \in \mathbb{R}^{2g\} \mid \sum_{j=1}^{g} r_j x_j + r_{j'} y_j = 0}$$

for some $r_j, r_{j'} \in \mathbb{R}$ not all zero. Note that

$$\begin{aligned} 0 &= \sum_{j=1}^{g} r_j x_j + r_{j'} y_j \\ &= \operatorname{Re}\left(\sum_{j=1}^{g} (r_j - i r_{j'})(x_j + i y_j)\right) \\ &= \operatorname{Re}\left(\sum_{j=1}^{g} (r_j - i r_{j'}) \int_\Gamma \omega_j\right) \\ &= \operatorname{Re}\left(\int_\Gamma \sum_{j=1}^{g} (r_j - i r_{j'}) \omega_j\right) \end{aligned}$$

for every $\Gamma \in H_1(X)$. By the previous proposition, $\sum_{j=1}^{g} (r_j - i r_{j'}) \omega_j = 0$, which implies that $r_j = r_{j'} = 0$ for every j, contradicting the choice of $r_j, r_{j'}$. Therefore, Γ is not contained in any proper subspace of $\mathbb{C}^g$. By the lattice criterion, Γ is a lattice in $\mathbb{C}^g$. ∎

DEFINITION 2.61. *Let* $\mathrm{Div}_{p(X)}$ *be the group of principal divisors on* X*, and* $\mathrm{Div}_0(X)$ *be the group of degree zero divisors on* X*. Define the Picard group of* X *to be*

$$\mathrm{Pic}\ (X) = \frac{\mathrm{Div}\ (X)}{\mathrm{Div}_{p(X)}},$$

and

$$\mathrm{Pic}_0(X) = \frac{\mathrm{Div}_0(X)}{\mathrm{Div}_{p(X)}}.$$

We have a short exact sequence of abelian groups

$$0 \to \mathrm{Pic}_0(X) \to \mathrm{Pic}\ (X) \xrightarrow{\deg} \mathbb{Z} \to 0.$$

DEFINITION 2.62. *For any divisor* D *on* X *with degree zero, we can find a closed curve* Γ *on* X *such that* $\partial\Gamma = D$ *by connecting the points in* D *by curves. Define a map* $\varphi : \mathrm{Div}_0(X) \to \mathrm{Jac}\ (X)$ *by*

$$\varphi(D) = \left(\int_\Gamma \omega_1, ..., \int_\Gamma \omega_g\right) + \Lambda.$$

We can verify that φ *is well-defined since if* Γ' *is another closed curve such that* $\partial\Gamma' = D$*, then* $\Gamma - \Gamma' \in Z_1(X)$*, hence* $\int_{\Gamma-\Gamma'} \omega_k \in \Lambda$ *for every* k*, which implies that* $\int_\Gamma \omega_k + \Lambda = \int_{\Gamma'} \omega_k + \Lambda$ *for every* k*.*

Note that $D \in \ker\ \varphi$ if and only if there exists a closed curve Γ on X such that $\partial\Gamma = D$ and $\int_\Gamma \omega_k = 0$ for every k.

THEOREM 2.63 (Abel's Theorem). *Let X be a compact Riemann surface and D be a divisor of degree zero on X. Then there exists a meromorphic function f on X such that $D = (f)$ if and only if there exists a closed curve $\Gamma \in C_1(X)$ such that $\partial\Gamma = D$ and $\int_\Gamma \omega = 0$ for every $\omega \in \omega(X)$.*

By Abel's theorem, we have $\ker \varphi = \mathrm{Div}_{p(X)}$, hence φ induces an injective homomorphism $j : \mathrm{Pic}_0(X) \to \mathrm{Jac}\ (X)$. In fact, j is also surjective, hence an isomorphism $\mathrm{Pic}_0(X) \cong \mathrm{Jac}\ (X)$, which is the moduli space of degree zero line bundles on X.

THEOREM 2.64. $j : \mathrm{Pic}_0(X) \to \mathrm{Jac}\ (X)$ *is surjective.*

Proof. For all $p \in \mathrm{Jac}\ (X)$, we can find $\xi \in \mathbb{C}^g$ such that $p = \xi + \Lambda$. Let $N \gg 0$ such that $\frac{1}{N}\xi \in V$, where V is the open neighborhood of 0 in $\mathbb{C}^g$ such that F induces a biholomorphism between V and an open neighborhood of $(a_1, ..., a_g)$ in X^g. Then we can find $(x_1, ..., x_g) \in U_1 \times ... \times U_g$ and curves $\Gamma_j \in U_j$ connecting a_j and x_j for every j such that

$$\left(\int_c \omega_1, ..., \int_c \omega_g\right) = \frac{1}{N}\xi,$$

where $c = \sum_{j=1}^g \Gamma_j$, thus $D = \partial C$ is a divisor of degree zero such that $\varphi(D) = \frac{1}{N}\xi + \Lambda$. Let $\theta = ND + \mathrm{Pic}_{p(X)}$, then $\varphi(\theta) = \xi + \Lambda = p$, which implies that $j(\theta) = p$. Therefore, j is surjective. ∎

Fix $a_1, ..., a_g \in X$ and define

$$\psi : X^g \to \mathrm{Pic}_0(X), \quad (x_1, ..., x_g) \mapsto [x_1 + ... + x_g - a_1 - ... - a_g].$$

By the isomorphism $j : \mathrm{Pic}_0(X) \to \mathrm{Jac}\ (X)$, this induces a holomorphic map $J : X^g \to \mathrm{Jac}\ (X)$, defined by

$$J(x_1, ..., x_g) = \left(\sum_{k=1}^g \int_{a_k}^{x_k} \omega_1, ..., \sum_{k=1}^g \int_{a_k}^{x_k} \omega_g\right) + \Lambda.$$

To show that J is surjective, it suffices to show that ψ is surjective. Let D be a divisor of degree zero on X. Consider $D' = D + a_1 + ... + a_g$, then $\deg D' = g$. By Riemann-Roch theorem, we have $\ell(D') = g$, hence $h^0(X, \mathcal{O}_{D'}) \geq 1$, which implies that there exists a meromorphic function $f \in \mathcal{M}(X)$ such that $(f) + D' \geq 0$, i.e. $(f) + D' = x_1 + ... + x_g$ for some $x_1, ..., x_g \in X$. Therefore, $\psi(x_1, ..., x_g) = [D]$, which implies that ψ is surjective.

Example 2.3. When $g = 1$, then J is an isomorphism: if $x \in \ker J$, then there exists $f \in \mathcal{M}(X)$ such that $(f) = x - a_1$, hence $f : X \to \mathbb{P}^1$ sending a to ∞, which is an isomorphism, contradiction. Therefore, J is injective, hence J is an isomorphism, so $\mathrm{Jac}\ (X) \cong \frac{\mathbb{C}}{\Lambda}$ is a torus.

2.9. Abel's Theorem

Recall that $C_1(X)$ is the free abelian group generated by all closed curves on X, and $\partial : C_1(X) \to \mathrm{Div}\ (X)$ is the boundary map defined by $\partial(\Gamma) = \Gamma(1) - \Gamma(0)$ for every curve Γ on X.

DEFINITION 2.65. *Let X be a Riemann surface and $D \in \operatorname{Div}(X)$ and*

$$X_D := \{x \in X \mid D(x) \geq 0\}.$$

We say that $f \in \mathcal{E}(X_D)$ is a weak solution of D if for all $a \in X$, there exists a chart (U, z) with $z(a) = 0$ and $\psi \in \mathcal{E}(U)$ with $\psi(a) \neq 0$ such that $f = \psi z^k$ on $U \cap X_D$, where $k = D(a)$.

A weak solution is weak in the sense that it may not be a meromorphic function on X since it is only partialerentiable on X_D.

PROPOSITION 2.66. *Let $D = \sum_{i=1}^{n} k_i a_i$ be a divisor on X with $k_i \in \mathbb{Z}$ and $a_i \in X$ distinct for every i. If f is a weak solution of D, then for all $g \in \mathcal{E}(X)$, we have*

$$\frac{1}{2\pi i} \iint_X \frac{df}{f} \wedge dg = \sum_{i=1}^{n} k_i g(a_i).$$

Proof. Choose disjoint charts (U_j, z_j) around a_j for every j such that $z_{j(a_j)} = 0$ and $U_j \cong B_1(0)$ such that $f = \psi_j z_j^{k_j}$ with $\psi_j \in \mathcal{E}(U_j)$ and $\psi_{j(x)} \neq 0$ for every $x \in U_j$.

Suppose $0 < r < r' < 1$. Exists $\phi_j \in \mathcal{E}(X)$ such that $\operatorname{supp}(\phi_j) \subset B_{r'}(0)$ and $\phi_j = 1$ on $B_{r(0)}$ for every j. Let $g_j = \phi_j g \in \mathcal{E}(X)$, then

$$\begin{aligned}
\iint_{U_j} \frac{df}{f} \wedge dg_j &= \iint_{U_j} \left(k_j \frac{dz_j}{z_j} + \frac{d\psi_j}{\psi_j} \right) \wedge dg_j \\
&= \iint_{U_j} k_j \frac{dz_j}{z_j} \wedge dg_j + \iint_{U_j} \frac{d\psi_j}{\psi_j} \wedge dg_j \\
&= \iint_{U_j} k_j \frac{dz_j}{z_j} \wedge dg_j - \iint_{U_j} d\left(g_j \frac{d\psi_j}{\psi_j} \right) \\
&= \iint_{U_j} k_j \frac{dz_j}{z_j} \wedge dg_j = -\lim_{\varepsilon \to 0} k_j \iint_{\varepsilon \leq |z_j| \leq r'} d\left(g_j \frac{dz_j}{z_j} \right) \\
&= \lim_{\varepsilon \to 0} k_j \int_{|z_j| = \varepsilon} g_j \frac{dz_j}{z_j} = 2\pi i k_j g(a_j).
\end{aligned}$$

Set $g_0 = g - \sum_{j=1}^{n} g_j$, then $g_0|_{|zj| \leq r} = 0$ for every j, hence $\operatorname{supp}(g_0)$ is compact in $X' = X \setminus \{a_1, \ldots, a_n\}$. Therefore, g_0 can be extended to a smooth function on X with compact support in X', so

$$\begin{aligned}
\iint_X \frac{df}{f} \wedge dg &= \iint_X \frac{df}{f} \wedge dg_0 + \sum_{j=1}^{n} \iint_{U_j} \frac{df}{f} \wedge dg_j \\
&= -\iint_X d\left(g_0 \frac{df}{f} \right) + 2\pi i \sum_{j=1}^{n} k_j g(a_j) = 2\pi i \sum_{j=1}^{n} k_j g(a_j).
\end{aligned}$$

∎

LEMMA 2.67. *Let $c : [0, 1] \to X$ be a curve on X and $U \Subset X$ with $c([0, 1]) \subset U$. Then exists a weak solution f of ∂c with $f|_{X \setminus U} \equiv 1$ such that for all $\omega \in \mathcal{E}^{(1)}(X)$ with $d\omega = 0$, we have*

$$\int_c \omega = \frac{1}{2\pi i} \iint_X \frac{df}{f} \wedge \omega.$$

Proof. Suppose that $U \cong B_1(0) \subseteq \mathbb{C}$. Let $a = c(0)$ and $b = c(1)$. Let $c([0,1]) \subset B_{r(0)}$ for some $r < 1$. Note that $\frac{z-a}{z-b}$ is nowhere vanishing holomorphic function on $\{r < |z| < 1\}$, hence exists a well-defined holomorphic branch of $\log \frac{z-a}{z-b}$ on $\{r < |z| < 1\}$. Choose $\psi \in \mathcal{E}(U)$ such that $\psi = 1$ on $B_{r(0)}$ and supp $(\psi) \subset B_{r'}(0)$ for some $r < r' < 1$. Define

$$f(z) = \begin{cases} \exp\left(\psi(z) \log \frac{z-a}{z-b}\right) \\ r < |z| < 1 \\ \frac{z-a}{z-b} \\ |z| \le r \\ 1 \\ |z| \ge r' \end{cases}$$

which is a weak solution of ∂c since $f(z) = \psi(z)\frac{z-a}{z-b}$ on U and $f(z) = 1$ on $X \setminus U$. For $\omega = \mathcal{E}^{(1)}(X)$ with $d\omega = 0$, let $\omega|_U = dh$ for some $h \in \mathcal{E}(U)$. Choose $\phi \in \mathcal{E}(X)$ such that $\phi = 1$ on $B_{r'}(0)$ and supp $(\phi) \subset B_{r''}(0)$ for some $r' < r'' < 1$. Then $g = \phi h \in \mathcal{E}(X)$ and $\omega = dg$ on $\{|z| \le r'\}$, hence

$$\frac{1}{2\pi i} \iint_X \frac{df}{f} \wedge \omega = \frac{1}{2\pi i} \iint_{|z| \le r'} d\left(g \frac{df}{f}\right) = g(b) - g(a) = \int_c \omega.$$

In general, by compactness of $c([0,1])$, we can find a partition $0 = t_0 < t_1 < \dots < t_n = 1$ such that $c\big([t_{j-1}, t_j]\big) \subset U_j$ for some $U_j \cong B_1(0)$ for every j. Let $c_j = c|_{[t_{j-1}, t_j]}$ for every j. By the previous case, exists a weak solution f_j of ∂c_j with $f_j|_{X \setminus U_j} \equiv 1$ such that for all $\omega \in \mathcal{E}^{(1)}(X)$ with $d\omega = 0$, we have

$$\int_{c_j} \omega = \frac{1}{2\pi i} \iint_X \frac{df_j}{f_j} \wedge \omega.$$

Let $f = \prod_{j=1}^n f_j$, then f is a weak solution of $\sum_{j=1}^n \partial c_j = \partial c$ with $f|_{X \setminus \bigcup_{j=1}^n U_j} \equiv 1$ such that for all $\omega \in \mathcal{E}^{(1)}(X)$ with $d\omega = 0$, we have

$$\int_c \omega = \sum_{j=1}^n \int_{c_j} \omega = \sum_{j=1}^n \frac{1}{2\pi i} \iint_X \frac{df_j}{f_j} \wedge \omega = \frac{1}{2\pi i} \iint_X \frac{df}{f} \wedge \omega.$$

∎

THEOREM 2.68. *Let X be a compact Riemann surface, $D \in$ Div (X) with* deg $D = 0$. *Then D has a solution if and only if there exists a 1-chain $c \in C_1(X)$ with $\partial c = D$ such that $\int_c \omega = 0$ for every $\omega \in \omega(X)$.*

Proof.

"$\le$": Let $c = \sum_{j=1}^k n_j c_j$ for some $n_j \in \mathbb{Z}$ and curves c_j on X for every j. By the previous lemma, exists weak solutions f_j of ∂c_j with $f_j|_{X \setminus U_j} \equiv 1$ for some $U_j \Subset X$ such that for all $\omega \in \mathcal{E}^{(1)}(X)$ with $d\omega = 0$, we have

$$\int_{c_j} \omega = \frac{1}{2\pi i} \iint_X \frac{df_j}{f_j} \wedge \omega.$$

Let $f=\prod_{j=1}^{k} f_j^{n_j}$, then f is a weak solution of $\partial c = D$ with $f|_{X\setminus\bigcup_{j=1}^{k} U_j}\equiv 1$ such that for all $\omega\in\mathcal{E}^{(1)}(X)$ with $d\omega=0$, we have

$$\int_c \omega=\sum_{j=1}^{k} n_j\int_{c_j}\omega=\sum_{j=1}^{k} n_j\frac{1}{2\pi i}\iint_X \frac{df_j}{f_j}\wedge\omega=\frac{1}{2\pi i}\iint_X\frac{df}{f}\wedge\omega.$$

Since $\int_c\omega=0$ for every $\omega\in\omega(X)$, we have $\iint_X\frac{df}{f}\wedge\omega=0$ for every $\omega\in\omega(X)$. Note that

$$0=\iint_X\frac{df}{f}\wedge\omega=\iint_X\frac{d'f+d''f}{f}\wedge\omega=\iint_X\frac{d''f}{f}\wedge\omega.$$

Recal that $\mathcal{E}^{(0,1)}(X)=d''\mathcal{E}(X)\oplus\overline{\omega}(X)$, hence the above equation implies that $\langle\frac{d''f}{f},-*\omega\rangle=\langle\frac{d''f}{f},-i\overline{\omega}\rangle=0$ for every $\omega\in\omega(X)$, which implies that $\frac{d''f}{f}=d''g$ for some $g\in\mathcal{E}(X)$, hence fe^{-g} is a meromorphic function on X with $(fe^{-g})=D$, which implies that D has a solution.

"$\Rightarrow$": If $D=0$, then we can take $c=0$. If $D\neq 0$ and $D=(f)$ for some $f\in\mathcal{M}(X)$. For each holomorphic 1-form ω on X and $V\subseteq\mathbb{P}^1$ with $f^{-1}(V)=\bigsqcup_{j=1}^{n} U_j$ and $\phi_j: U_j\to V$ is a biholomorphism for every j, define

$$\text{Trace }(\omega)|_V=\sum_{j=1}^{n}\phi_j^*\omega.$$

Then Trace (ω) is a well-defined holomorphic 1-form on Y, where Y equals $\mathbb{P}^1$ excluding the critical values of f. By extension theorem on $\mathbb{P}^1$, Trace (ω) can be extended to a holomorphic 1-form on $\mathbb{P}^1$, which must be zero since $\omega(\mathbb{P}^1)=\{0\}$. Therefore, Trace $(\omega)=0$, which implies that $\sum_{f(x)=y}\omega(x)=0$ for every $y\in\mathbb{P}^1$. Let c be the curve on X defined by $c(t)=f^{-1}(e^{2\pi it})$ for every $t\in[0,1]$, then $\partial c=(f)$ and $\int_c\omega=\int_0^1\sum_{f(x)=e^{2\pi it}}\omega(x)dt=0$ for every $\omega\in\omega(X)$. ∎

Non-compact Riemann Surfaces

Chapter 3

3.1. ***Weyl's Lemma***

Definition 3.1. *Given an open set $X \subseteq \mathbb{C}$, define*

$$\mathcal{D}(X) = \{f \in \mathcal{E}(X) : f \text{ has compact support in } X\}.$$

For $(f_n) \subseteq \mathcal{D}(X)$, we say that $f_n \to f$ in $\mathcal{D}(X)$ if there exists a compact set $K \subseteq X$ such that supp $(f_n) \subseteq K$ *for n sufficiently large and* supp $(f) \subseteq K$ *and $D^\alpha f_n \to D^\alpha f$ uniformly on K for every multi-index $\alpha = (\alpha_1, \alpha_2) \in \mathbb{N}^2$. Denote by $f_n \xrightarrow{\mathcal{D}} f$ if $f_n \to f$ in $\mathcal{D}(X)$.*

Definition 3.2. *A distribution on X is a continuous linear mapping*

$$T : \mathcal{D}(X) \to \mathbb{C}, \quad f \mapsto T[f].$$

Continuous in the sense that if $f_n \xrightarrow{\mathcal{D}} f$, then $T[f_n] \to T[f]$. Let $\mathcal{D}'(X)$ be the collection of all distributions on X.

Given a continuous function $h \in \texttt{cal } C(X)$, we can define a distribution T_h by

$$T_h[f] = \int_X h(z)f(z)dxdy$$

with $z = x + iy$ for every $f \in \mathcal{D}(X)$. We show that $f \mapsto T_f$ is an embedding of $\texttt{cal } C(X)$ into $\mathcal{D}'(X)$. Suppose that $T_h = 0$, i.e. $\iint_X h(z)f(z)dxdy = 0$ for every $f \in \mathcal{D}(X)$. If $h(a) \neq 0$ for some $a \in X$, then we can find a neighborhood U of a such that Re $h(z) > 0$ for every $z \in U$. Choose $\phi \in \mathcal{D}(X)$ with supp $\phi \subset U$ and $0 \leq \phi \leq 1$ such that $\phi|_A \equiv 1$ for some neighborhood A of a, then

$$T_h[\phi] = \iint_X h(z)\phi(z)dxdy = \iint_A h(z)dxdy + \iint_{U\setminus A} h(z)\phi(z)dxdy > 0,$$

contradiction. Therefore, $h(a) = 0$ for every $a \in X$, hence $h = 0$. Therefore, $f \mapsto T_f$ is an embedding of $\texttt{cal } C(X)$ into $\mathcal{D}'(X)$, so we can view $\mathcal{D}'(X)$ as an extension of continuous functions on X.

Observe that for $h \in \mathcal{E}(X)$, $f \in \mathcal{D}(X)$, by integration by parts, we have

$$\iint_X h(z)\frac{(\partial F)}{\partial x}(z)dxdy = -\iint_X \frac{\partial h}{\partial x}(z)f(z)dxdy$$

$$\iint_X h(z)\frac{(\partial F)}{\partial y}(z)dxdy = -\iint_X \frac{\partial h}{\partial y}(z)f(z)dxdy.$$

More generally,

$$\iint_X h(z)D^\alpha f(z)dxdy = (-1)^{|\alpha|}\iint_X D^\alpha h(z)f(z)dxdy,$$

where $\alpha = (\alpha_1, \alpha_2) \in \mathbb{N}^2$ is a multi-index and $|\alpha| = \alpha_1 + \alpha_2$. Therefore, for $T \in \mathcal{D}'(X)$, we can define $D^\alpha T$ by

$$(D^\alpha T)[f] := (-1)^{|\alpha|} T[D^\alpha f]$$

for every $f \in \mathcal{D}(X)$, which is a distribution since if $f_n \xrightarrow{\mathcal{D}} f$, then $D^\alpha f_n \xrightarrow{\mathcal{D}} D^\alpha f$.

Proposition 3.3. *There exists $\rho \in \mathcal{D}(\mathbb{C})$ such that*

(i) supp $(\rho) \subseteq B_1(0)$;
(ii) $\rho(z) = \rho(|z|)$ *for every* $z \in \mathbb{C}$;
(iii) $\iint_{\mathbb{C}} \rho(x + iy) dxdy = 1$.

Proof. Let $\phi_0 \in \mathcal{E}(\mathbb{R})$ such that supp $(\phi_0) \subseteq (-1, 1)$ and $\phi_0(x) = \phi_0(-x)$ for every $x \in \mathbb{R}$ and $\phi_0|_{[-\frac{1}{2}, \frac{1}{2}]} \equiv 1$. Define $\phi(z) = \phi_0(z)$ on $\mathbb{C}$, and then define

$$\rho(z) = \frac{\phi(z)}{\iint_{\mathbb{C}} \phi(x + iy) dxdy}.$$

Then ρ satisfies the required properties. ∎

For $\varepsilon > 0$, define $\rho_\varepsilon(z) = \frac{1}{\varepsilon^2}\rho\left(\frac{z}{\varepsilon}\right)$ for every $z \in \mathbb{C}$. Then ρ_ε also satisfies the properties in the previous proposition, and supp $(\rho_\varepsilon) \subseteq B_\varepsilon(0)$ for every $\varepsilon > 0$.

Define

$$X^{(\varepsilon)} = \{z \in X \mid \overline{B_\varepsilon(z)} \subseteq X\}$$

is an open subset of X for every $\varepsilon > 0$. For $f \in$ `cal` $C(X)$, define the smoothing of f on $X^{(\varepsilon)}$ by

$$(\mathrm{sm}_\varepsilon f)(z) = \iint_X \rho_\varepsilon(z - \zeta) f(\zeta) d\xi d\eta$$

for every $z \in X^{(\varepsilon)}$ with $\zeta = \xi + i\eta$. Note that $\mathrm{sm}_\varepsilon f$ is well-defined since supp $(\rho_\varepsilon(z - \cdot)) \subseteq B_\varepsilon(z) \subseteq X$ for every $z \in X^{(\varepsilon)}$. Moreover, $\mathrm{sm}_\varepsilon f$ is partialerentiable on $X^{(\varepsilon)}$ since we can partialerentiate under the integral sign, so

$$\frac{d}{dz}(\mathrm{sm}_\varepsilon f)(z) = \iint_X \frac{d}{dz}\rho_\varepsilon(z - \zeta) f(\zeta) d\xi d\eta.$$

Note that

$$\begin{aligned} \mathrm{sm}_\varepsilon f(z) &= \iint_X \rho_\varepsilon(z - \zeta) f(\zeta) d\xi d\eta = \iint_{|z-\zeta| < \varepsilon} \rho_\varepsilon(z - \zeta) f(\zeta) d\xi d\eta \\ (w = z - \zeta = u + iv) \quad &= \iint_{|w| < \varepsilon} \rho_\varepsilon(w) f(z - w) dudv = \iint_X \rho_\varepsilon(w) f(z - w) dudv, \end{aligned}$$

as $\varepsilon \to 0$ and $w \to 0$, $f(z - w) \to f(z)$, hence $\mathrm{sm}_\varepsilon f(z) \to f(z)$ for every $z \in X$.

Write

$$\begin{aligned} (\mathrm{sm}_\varepsilon f)(z) &= \iint_{|\zeta - z| < \varepsilon} \rho_\varepsilon(z - \zeta) f(\zeta) d\xi d\eta \\ &= \iint_{|\zeta - z| < \varepsilon} \rho_\varepsilon(\zeta - z) f(\zeta) d\xi d\eta = \iint_{|w| < \varepsilon} \rho_\varepsilon(w) f(z + w) dudv \\ \Rightarrow D^\alpha(\mathrm{sm}_\varepsilon f)(z) &= \iint_{|w| < \varepsilon} \rho_\varepsilon(w) D^\alpha f(z + w) dudv, \end{aligned}$$

$$= \iint_{|\zeta - z| < \varepsilon} \rho_\varepsilon(z - \zeta) D^\alpha f(\zeta) d\xi d\eta = \mathrm{sm}_\varepsilon(D^\alpha f)(z).$$

PROPOSITION 3.4. *For $z \in X^{(\varepsilon)}$, if f is harmonic on $B_\varepsilon(z)$, then $\mathrm{sm}_\varepsilon f(z) = f(z)$.*

Proof. For $0 \le r < \varepsilon$, by Mean Value Property, we have

$$f(z) = \frac{1}{2\pi} \int_0^{2\pi} f\big(z + re^{i\theta}\big) d\theta.$$

Therefore,

$$\begin{aligned}
(\mathrm{sm}_\varepsilon f)(z) &= \iint_{|w| < \varepsilon} \rho_\varepsilon(w) f(z + w) du dv \\
&= \iint_{\substack{0 \le r < \varepsilon \\ 0 \le \theta < 2\pi}} \rho_\varepsilon\big(re^{i\theta}\big) f\big(z + re^{i\theta}\big) r dr d\theta \\
&= \int_0^\varepsilon \rho_\varepsilon(r) \left(\frac{1}{2\pi} \int_0^{2\pi} f\big(z + re^{i\theta}\big) d\theta \right) r dr \\
&= \int_0^\varepsilon \rho_\varepsilon(r) f(z) r dr \\
&= f(z) \int_0^\varepsilon \rho_\varepsilon(r) r dr = f(z).
\end{aligned}$$

∎

THEOREM 3.5 (Weyl's Lemma). *If $T \in \mathcal{D}'(X)$ and $\delta T = \left(\frac{\partial^2}{\partial} x^2 + \frac{\partial^2}{\partial} y^2\right) T = 0$, then T is a smooth function on X, i.e. there exists $h \in \mathcal{E}(X)$ such that $T = T_h$.*

Proof. For $\varepsilon > 0$ and $z \in X^{(\varepsilon)}$, define

$$h(z) = T[\rho_\varepsilon(\cdot - z)].$$

We have

$$\begin{aligned}
\frac{\partial h}{\partial x}(z) &= \lim_{\delta x \to 0} \frac{1}{\delta x} \big(T_\zeta[g(\zeta, x + \delta x + iy)] - T_\zeta[g(\zeta, x + iy)] \big) \\
&= \lim_{\delta x \to 0} T_\zeta \left[\frac{g(\zeta, x + \delta x + iy) - g(\zeta, x + iy)}{\delta x} \right].
\end{aligned}$$

Define

$$f_{\delta x}(\zeta) = \frac{g(\zeta, x + \delta x + iy) - g(\zeta, x + iy)}{\delta x},$$

then $f_{\delta x} \in \mathcal{D}(X)$ for every δx sufficiently small and $f_{\delta x} \xrightarrow{\mathcal{D}} \frac{\partial}{\partial x} \rho_\varepsilon(\cdot - z)$ as $\delta x \to 0$, hence

$$\frac{\partial h}{\partial x}(z) = \lim_{\delta x \to 0} T_{\zeta[f_{\delta x}(\zeta)]} = T_{\zeta\left[\frac{\partial}{\partial x} \rho_\varepsilon(\zeta - z)\right]}.$$

Similarly, we have

$$\frac{\partial h}{\partial y}(z) = T_{\zeta\left[\frac{\partial}{\partial y} \rho_\varepsilon(\zeta - z)\right]},$$

so $h(z) \in \mathcal{E}\big(X^{(\varepsilon)}\big)$.

We claim that $T[\mathrm{sm}_\varepsilon f] = \iint_{X^{(\varepsilon)}} h(z)f(z)dxdy$ for every $f \in \mathcal{D}(L)$ with $L \Subset X^{(\varepsilon)}$. Since $\rho_\varepsilon(\zeta - z)f(z)$ has compact suppport, pick a rectangle R such that supp $(\rho_\varepsilon(\zeta - \cdot)f) \subseteq R$. Partition R into n^2 small rectangles $R_1, ..., R_{n^2}$ of equal size and z_k be the center of R_k for every k. Let

$$G_{n(\zeta)} = \frac{|R|}{n^2}\sum_{k=1}^{n^2} \rho_\varepsilon(\zeta - z_k)f(z_k),$$

then $G_n \in \mathcal{D}(X)$ for every n and $G_n \xrightarrow{\mathcal{D}} \iint_X \rho_\varepsilon(\zeta - z)f(z)dxdy = (\mathrm{sm}_\varepsilon f)(\zeta)$ as $n \to \infty$, hence

$$\begin{aligned} T\left[\iint_X \rho_\varepsilon(\cdot - z)f(z)dxdy\right] &= \lim_{n\to\infty} T[G_n] \\ &= \lim_{n\to\infty} \frac{|R|}{n^2}\sum_{k=1}^{n^2} T[\rho_\varepsilon(\cdot - z_k)]f(z_k) \\ &= \iint_{X^{(\varepsilon)}} h(z)f(z)dxdy. \end{aligned}$$

Recall Dolbeault's lemma, for $X = B_R(0)$ and $g \in \mathcal{E}(X)$, we can find $f \in \mathcal{E}(X)$ such that $\frac{(\partial F)}{\partial \overline{z}} = g$. Since $\delta = \frac{\partial^2}{\partial}x^2 + \frac{\partial^2}{\partial}y^2 = 4\frac{\partial^2}{\partial}z\partial\overline{z}$, we can also find $f \in \mathcal{E}(X)$ such that $\delta f = g$ for every $g \in \mathcal{E}(X)$.

Now, we claim that $T[f] = T[\mathrm{sm}_\varepsilon f]$ for every $f \in \mathcal{D}(X)$. By Dolbeault's lemma, there exists $\psi \in \mathcal{E}(\mathbb{C})$ such that $\delta\psi = f$. Note that $\delta\psi = 0$ on $V := \mathbb{C} \setminus \mathrm{supp}\, f$, i.e. harmonic on V, hence $\psi = \mathrm{sm}_\varepsilon\psi$ on $V^{(\varepsilon)}$, so $\phi := \psi - \mathrm{sm}_\varepsilon\psi$ has compact support in $\mathbb{C}$, so $\phi \in \mathcal{D}(X)$. Therefore,

$$0 = \delta T[\phi] = T[\delta\phi] = T[\delta\psi - \delta(\mathrm{sm}_\varepsilon\psi)] = T[f - \mathrm{sm}_\varepsilon f].$$

Therefore, $T[f] = T[\mathrm{sm}_\varepsilon f]$ for every $f \in \mathcal{D}(X)$, hence $T = T_h$ on $X^{(\varepsilon)}$. Since $\varepsilon > 0$ is arbitrary, we have $T = T_h$ on X, so T is a smooth function on X. ∎

Lemma 3.6. *If $\frac{\partial T}{\partial \overline{z}} = 0$, then T is a holomorphic function on X.*

Proof. Since $\frac{\partial T}{\partial \overline{z}} = 0$, we have

$$\delta T = 4\frac{\partial^2 T}{\partial z \partial \overline{z}} = 0,$$

hence T is a smooth function on X by Weyl's lemma, and thus $T \in \mathcal{O}(X)$. ∎

3.2. Countable Topology

Note that if X is compact, then X is second countable.

Lemma 3.7 (Poincaré-Volterra). *Suppose Z' is a connected manifold, Z is a Hausdorff space with countable topology and $f : Z' \to Z$ is a continuous discrete mapping, then Z' has countable topology.*

Proof. Let $\mathfrak{U}$ be a countable basis of the topology of Z. Let $\mathfrak{V}$ be the collection of all open subsets V of Z' such that

(1) V is a connected component of $f^{-1}(U)$ for some $U \in \mathfrak{U}$;

(2) V has countable topology.

We claim that $\mathfrak{V}$ is a basis of the topology of Z'. Let $x \in Z'$ and D be an open neighborhood of x in Z'. Since $f^{-1}(f(x))$ is discrete, exists $x \in W \Subset D$ such that $\partial W \cap f^{-1}(f(x)) = \emptyset$ and $W \cap f^{-1}(f(x)) = \{x\}$. Also, $f(\partial W)$ is compact in Z with $f(x) \notin f(\partial W)$, hence exists $U \in \mathfrak{U}$ such that $f(x) \in U$ and $U \cap f(\partial W) = \emptyset$. Let V be the connected component of $f^{-1}(U)$ containing x, then $V \subseteq W$. Since $\overline{W}$ is compact, it has countable topology, hence V has countable topology, so $V \in \mathfrak{V}$. Therefore, $\mathfrak{V}$ is a basis of the topology of Z'.

We now claim that $\mathfrak{V}$ is countable. Fix $V^* \in \mathfrak{V}$. For $n \in \mathbb{N}$, define

$$\mathfrak{V}_n = \{V \in \mathfrak{V} : \exists V^* = V_0, V_1, ..., V_n = V \text{ in } \mathfrak{V} \text{ such that } V_{j-1\} \cap V_j \neq \emptyset \text{ for every } j}.$$

Fix $x_0 \in V^*$, for $x \in V \in \mathfrak{V}$, let $c : [0,1] \to Z'$ be a curve from x_0 to x, then since $[0,1]$ is compact, exists a partition $0 = t_0 < t_1 < ... < t_n = 1$ such that $c\big([t_{j-1}, t_j]\big) \subseteq V_j$ for some $V_j \in \mathfrak{V}$ for every j, hence $V \in \mathfrak{V}_n$. Therefore, $\mathfrak{V} = \bigcup_{n=0}^{\infty} \mathfrak{V}_n$.

We now claim that for every $V_0 \in \mathfrak{V}$, there exists at most countably many $V \in \mathfrak{V}$ such that $V \cap V_0 \neq \emptyset$. For all $U \in \mathfrak{U}$, the connected components of $f^{-1}(U)$ are pairwise disjoint. Since V_0 has a countable topology, $\exists$ at most countably many V which are connected components of $f^{-1}(U)$ such that $V \cap V_0 \neq \emptyset$: otherwise, if there are uncountably many, then V_0 has uncountable many disjoint open subsets, contradicting that V_0 has countable topology. Since $\mathfrak{U}$ is countable, there are at most countably many $V \in \mathfrak{V}$ such that $V \cap V_0 \neq \emptyset$.

We then show that $\mathfrak{V}_n$ is countable for every n. Proceed by induction on n. For $n = 0$, $\mathfrak{V}_0 = \{V^*\}$ is countable. Suppose that $\mathfrak{V}_{n-1}$ is countable for some $n \geq 1$. For $V \in \mathfrak{V}_n$, there exists $V' \in \mathfrak{V}_{n-1}$ such that $V \cap V' \neq \emptyset$, hence there are at most countably many V such that $V \cap V' \neq \emptyset$ for every $V' \in \mathfrak{V}_{n-1}$, so $\mathfrak{V}_n$ is countable. Therefore, $\mathfrak{V} = \bigcup_{n=0}^{\infty} \mathfrak{V}_n$ is countable. ∎

LEMMA 3.8. *Suppose Z, Z' are topological spaces and $f : Z' \to Z$ is a continuous open surjective mapping. If Z' has countable topology, then Z has countable topology.*

Proof. Let $\mathfrak{U}$ be a countable basis for the topology of Z' and $\mathfrak{V} = \{f(U) : U \in \mathfrak{U}\}$, which is a collection of open sets since f is an open mapping. Since f is surjective, $\mathfrak{V}$ is a basis of the topology of Z. Since $\mathfrak{U}$ is countable, $\mathfrak{V}$ is countable, hence Z has countable topology. ∎

THEOREM 3.9 (Rado). *Let X be a Riemann surface. Then X is second countable, i.e. there exists a countable basis of the topology of X.*

Proof. Let (U, z) be a chart on X and K_0, K_1 be two compact disks on U with $K_0 \cap K_1 = \emptyset$. Set $Y := X \setminus (K_0 \cup K_1)$. By Dirichlet boundary value problem, $\exists u : \overline{Y} \to \mathbb{R}$ such that u is harmonic on Y and $u|_{\delta K_0} = 0$ and $u|_{\delta K_1} = 1$. Let $\omega = du$, then $d''\omega = d''(d' + d'')u = 0$, hence ω is a non-constant holomorphic 1-form on Y. Consider the universal covering $p : \tilde{Y} \to Y$, then there exists a primitive $f \in \mathcal{O}(\tilde{Y})$ such that $df = p^*\omega$. Since ω is non-constant, f is a non-constant holomorphic function $f : \tilde{Y} \to \mathbb{C}$.

By Poincaré-Volterra lemma, $\tilde{Y}$ has countable topology. By the previous lemma, since covering maps are open surjective continuous mappings, Y has countable topology. Since X is the union of an open chart and Y, X has countable topology. ∎

3.3. *Density of Holomorphic Functions*

A Banach space is a complete normed vector space.

- A linear functional is continuous if and only if it is bounded, i.e. there exists $C > 0$ such that $|f(x)| \leq C \parallel x \parallel$ for every x in the Banach space.
- (Banach theorem) A continuous surjective linear functional is an open mapping
- (Hahn-Banach theorem) Let X be a Banach space, Y be a linear subspace of X and $f : Y \to \mathbb{C}$ be a continuous linear functional, then there exists a continuous linear functional $F : X \to \mathbb{C}$ such that $F|_Y = f$ and $\parallel F \parallel = \parallel f \parallel$.

Example 3.1.

- $C_{\mathbb{R}}([0,1])$ is a Banach space with the supremum norm $\parallel f \parallel_\infty = \sup_{x\in[0,1]} |f(x)|$.
- $C^k_{\mathbb{R}}([0,1])$ is a Banach space with the norm

$$\parallel f \parallel := \sum_{j=0}^{k} \sup_{x\in[0,1]} |f^j(x)|, \quad \parallel f \parallel_k = \sum_{j=0}^{k} \parallel f^j \parallel_\infty .$$

- $C^\infty_{\mathbb{R}}([0,1])$ and $C_{\mathbb{R}}((0,1))$ are not Banach spaces with the above norms since they are not complete.

3.3.1. Fréchet Spaces

DEFINITION 3.10. *A seminorm on a vector space X is a function $p : X \to \mathbb{R}$ such that for every $x, y \in X$ and $\lambda \in \mathbb{C}$, we have*

(i) $p(x+y) \leq p(x) + p(y)$;
(ii) $p(\lambda x) = |\lambda|\, p(x)$.
(iii) $p(x) \geq 0$.

Consider a countable family of seminormrs $\{p_n\}_{\{n\in\mathbb{N}\}}$ such that they form a Fréchet space which is complete, on which Banach and Hahn-Banach theorems hold, as we can define a metric by

$$d(x,y) = \sum_{n=1}^{\infty} 2^{-n} \cdot \frac{p_n(x-y)}{1+p_n(x-y)}$$

for every $x, y \in X$.

Let X be a Riemann surface and $Y \subset X$ be an open set. Since Y has countable topology, we can find a countable family of compact subsets $\{K_j\}_{\{j\in J\}}$ such that each K_j is contained in a chart (U_j, z_j) and

$$Y = \bigcup_{j\in J} K_j^\circ .$$

For $j \in J$ and $\alpha \in (\alpha_1, \alpha_2) \in (\mathbb{N}\cup\{0\})^2$, define a seminorm $p_{j,\alpha}$ on $\mathcal{E}(Y)$ by

$$p_{j\alpha}(f) = \sup_{z\in K_j} |D_j^\alpha f(z)|$$

for every $f \in \mathcal{E}(Y)$. A neighborhood basis of 0 is given by finite intersections of sets of the form

$$B(p_{j\alpha}, \varepsilon) = \{f \in \mathcal{E}(Y) : p_{j\alpha}(f) < \varepsilon\})$$

for some $j \in J$, $\alpha \in (\mathbb{N}\cup\{0\})^2$ and $\varepsilon > 0$. Convergence $f_n \to f$ under this topology is equivalent to uniform convergence of f_n and all its derivatives on each K_j.

Analogously, we can define a Fréchet space structure on $\mathcal{E}^{0,1}(Y)$ by writing $\omega \in \mathcal{E}^{0,1}(Y)$ locally as $\omega = f_j d\overline{z_j}$ on U_j for every $j \in J$ and defining

$$p_{j\alpha}(\omega) := \sup_{z \in K_j} |D_j^\alpha f_{j(z)}|$$

for every $\omega \in \mathcal{E}^{0,1}(Y)$, $j \in J$ and $\alpha \in (\mathbb{N} \cup \{0\})^2$.

LEMMA 3.11. *Suppose Y is an open subset of a Riemann surface X. Then every continuous linear map $T : \mathcal{E}(Y) \to \mathbb{C}$ has compact support, i.e. there exists a compact subset $K \subset Y$ such that $T[f] = 0$ for every $f \in \mathcal{E}(Y)$ with* supp $f \subseteq Y \setminus K$.

Proof. Since T is continuous, there exists $j_1, \dots, j_m$, $\alpha_1, \dots, \alpha_m$ and $\varepsilon > 0$ such that $|T[f]| < 1$ for all $f \in \bigcap_{k=1}^m B\left(p_{j_k\alpha_k}, \varepsilon\right)$. Let $K = \bigcup_{k=1}^m K_{j_k}$, then K is a compact subset of Y. For every $f \in \mathcal{E}(Y)$ with supp $f \subseteq Y \setminus K$, we have $f|_K \equiv 0$, hence $nf|_K \equiv 0$ for every $n \in \mathbb{N}$, so

$$p_{j_k\alpha_k}(nf) = \sup_{z \in K_{j_k}} |D_{j_k}^{\alpha_k} nf(z)| = 0$$

for every k and n, hence $nf \in \bigcap_{k=1}^m B\left(p_{j_k\alpha_k}, \varepsilon\right)$ for every n, so $|T[nf]| < 1$ for every n, which implies that $T[f] = 0$. ∎

LEMMA 3.12. *Suppose Y is an open subset of a Riemann surface X and $S : \mathcal{E}^{0,1}(X) \to \mathbb{C}$ is a continuous linear map such that $S[d''g] = 0$ for every $g \in \mathcal{E}(X)$ with* supp $g \Subset$ *Y. Then there exists a holomorphic* 1*-form $\sigma \in \Omega(X)$ such that*

$$S[\omega] = \iint_Y \sigma \wedge \omega$$

for every $\omega \in \mathcal{E}^{0,1}(X)$ with supp $\omega \Subset Y$.

Proof. Let $(U, z) \subset Y$ be a chart. For $\varphi \in \mathcal{D}(U)$, define

$$\tilde{\varphi} := \begin{cases} \varphi d\overline{z} & \text{on } U \\ 0 & \text{on } X \setminus U \end{cases}$$

which is a smooth $(0,1)$-form on X. Define a distribution on U by

$$S_U : \mathcal{D}(U) \to \mathbb{C}, \quad \varphi \mapsto S[\tilde{\varphi}].$$

By assumption, $S[d''g] = 0$ for every $g \in \mathcal{D}(U)$. Note that

$$\frac{\partial S_U}{\partial \overline{z}}[g] = S_U\left[\frac{\partial g}{\partial \overline{z}}\right] = S\left[\frac{\partial g}{\partial \overline{z}} d\overline{z}\right] = S[d''g] = 0$$

for all $g \in \mathcal{D}(U)$, hence $\Delta S_U = 4d'd''S_U = 0$, so by Weyl's lemma, exists $h \in \mathcal{O}(U)$ such that for all $g \in \mathcal{D}(U)$,

$$S[\tilde{\varphi}] = S_{U[\varphi]} = \iint_U h(z)\varphi(z) dz d\overline{z} = \iint_U (h(z)dz) \wedge (\varphi(z)d\overline{z}).$$

Let $\sigma_U = h(z)dz$ be the corresponding holomorphic 1-form on U so that $S[\omega] = \iint_U \sigma_U \wedge \omega$ for all $\omega \in \mathcal{E}^{0,1}(X)$ with supp $\omega \Subset U$.

On another chart (U', z'), we can similarly define a holomorphic 1-form $\sigma_{U'}$ on U'. For $\omega \in \mathcal{E}^{0,1}(X)$ with supp $\omega \Subset (U \cap U')$, we have

$$\iint_U \sigma_U \wedge \omega = S[\omega] = \iint_{U \cap U'} \sigma_{U'} \wedge \omega,$$

for all $\omega \in \mathcal{E}^{0,1}(X)$ with supp $\omega \Subset (U \cap U')$, hence $\sigma_U = \sigma_{U'}$ on $U \cap U'$. Therefore, we can glue σ_U for all charts $(U, z) \subset Y$ to get a holomorphic 1-form $\sigma \in \Omega(Y)$ such that $\sigma|_U = \sigma_U$, so

$$S[\omega] = \iint_Y \sigma \wedge \omega$$

for every $\omega \in \mathcal{E}^{0,1}(X)$ with supp ω relatively compact in some chart of Y. For an arbitrary $\omega \in \mathcal{E}^{0,1}(X)$ with supp $\omega \Subset Y$, we can find a finite cover of supp ω by charts (U_j, z_j) and a partition of unity $\{\phi_j\}_{\{j=1\}}^n$ subordinate to the cover such that $\phi_j \in \mathcal{D}(U_j)$ for every j, hence

$$S[\omega] = \sum_{j=1}^n S[\phi_j \omega] = \sum_{j=1}^n \iint_Y \sigma \wedge (\phi_j \omega) = \iint_Y \sigma \wedge \omega.$$

∎

3.3.2. Runge Domain

DEFINITION 3.13. *For a subset $K \subset X$, let $h_{X(K)}$ be the union of K and all relatively compact connected components of $X \setminus K$.*

Let X be a Riemann surface and $Y \subset X$ be an open set. We say that Y is a Runge domain if $h_{X(Y)} = Y$, i.e. no connected component of $X \setminus Y$ is compact.

Example 3.2. Consider

$$Y = \{z \in \mathbb{C} : 1 < |z| < 2\}.$$

Y is not Runge in $\mathbb{C}$ since $B_1(0)$ is a connected component of $\mathbb{C} \setminus Y$ that is compact. However, Y is Runge in $\mathbb{C}^*$ since $B_1(0)^*$ is not compact.

PROPOSITION 3.14. *Let X be a non-compact Riemann surface and $Y \Subset X$ be a Runge domain. If $Y \subset Y' \Subset X$, then the image of the restriction map $\beta : \mathcal{O}(Y') \to \mathcal{O}(Y)$ is dense in $\mathcal{O}(Y)$.*

Proof. It suffices to show that for every continuous linear map $T : \mathcal{E}(Y) \to \mathbb{C}$, $T|_{\beta(\mathcal{O}(Y'))} = 0$ implies $T|_{\mathcal{O}(Y)} = 0$: to see this, suppose the $\beta(\mathcal{O}(Y'))$ is not dense in $\mathcal{O}(Y)$, say $f_0 \in \mathcal{O}(Y) \setminus \overline{\beta(\mathcal{O}(Y'))}$, then let

$$\mathcal{E}_0 = \overline{\beta(\mathcal{O}(Y'))} \oplus \mathbb{C} f_0 \quad \text{and} \quad T_0 : \mathcal{E}_0 \to \mathbb{C}, \quad f + \lambda f_0 \mapsto \lambda$$

is a continuous linear map on $\mathcal{E}_0$. By Hahn-Banach theorem, we can extend T_0 to a continuous linear map $T : \mathcal{E}(Y) \to \mathbb{C}$. Then $T|_{\beta(\mathcal{O}(Y'))} = T_0|_{\beta(\mathcal{O}(Y'))} = 0$ but $T(f_0) = T_0(f_0) = 1$, contradiction.

To prove the claim, suppose that $T|_{\beta(\mathcal{O}(Y'))} = 0$, i.e. $T[f] = 0$ for every $f \in \mathcal{O}(Y')$. By the general form of Dolbeault's lemma, for all $\omega \in \mathcal{E}^{0,1}(X)$, exists $f \in \mathcal{E}(Y')$ such that $\omega|_{Y'} = d'' f$, so we can define

$$S : \mathcal{E}^{0,1}(X) \to \mathbb{C}, \quad \omega \mapsto T[f|_Y].$$

S is well-defined since if $d'' f = \omega = d'' g$ on Y', then $d''(f - g) = 0$ on Y', so $f - g \in \mathcal{O}(Y')$, hence $T[f - g] = 0$. Consider the space

$$V = \{(\omega, f) \in \mathcal{E}^{0,1}(X) \times \mathcal{E}(Y') : d'' f = \omega|_{Y'}\},$$

which is a closed subspace since d'' is continuous, so V is a Fréchet space. Note that the following diagram is commutative.

$$\begin{tikzcd} V \arrow[r, "\pi_2"] \arrow[d, two heads, "\pi_1"'] & \mathcal{E}(Y') \arrow[r, "\beta"] & \mathcal{E}(Y) \arrow[d, "T"] \\ \mathcal{E}^{0,1}(X) \arrow[rr, "S"'] & & \mathbb{C} \end{tikzcd}$$

Since $\pi_1 : V \to \mathcal{E}^{0,1}(X)$ is surjective, by Banach's theorem, π_1 is an open mapping, and since π_2, β, T are continuous, S is also continuous.

By Lemma 3.11, there exist compact subsets $K \subset Y$ and $L \subset X$ such that

(1) $T[f] = 0$ for every $f \in \mathcal{E}(Y)$ with supp $f \subseteq Y \setminus K$;
(2) $S[\omega] = 0$ for every $\omega \in \mathcal{E}^{0,1}(X)$ with supp $\omega \subseteq X \setminus L$.

We claim that $S[\omega] = 0$ for every $\omega \in \mathcal{E}^{0,1}(X)$ with supp $\omega \Subset X \setminus h_X(K)$. If $f \in \mathcal{E}(X)$ with supp $f \Subset X \setminus K$, then $S[d''f] = T[f|_Y] = 0$. By Lemma 3.12, there exists $\sigma \in \Omega(X \setminus K)$ such that

$$S[\omega] = \iint_{X\setminus K} \sigma \wedge \omega$$

for every $\omega \in \mathcal{E}^{0,1}(X)$ with supp $\omega \Subset X \setminus K$. By assumption, $\sigma \equiv 0$ on $(X \setminus K) \cap (X \setminus L) = X \setminus (K \cup L)$. Let U be a connected component of $X \setminus h_{X(K)}$, then $U \cap (X \setminus (K \cup L)) \neq \emptyset$ otherwise U is a non-compact connected component of $K \cup L$. Thus $\sigma|_U = 0$ for every connected component U of $X \setminus h_X(K)$ by identity theorem, implying that $\sigma|_{X\setminus h_X(K)} \equiv 0$, hence $S[\omega] = 0$ for every $\omega \in \mathcal{E}^{0,1}(X)$ with supp $\omega \Subset X \setminus h_X(K)$.

Finally, for all $f \in \mathcal{O}(Y)$, since Y is a Runge domain and $K \subset Y$, we have $h_{X(K)} \subset h_{X(Y)} = Y$. Let $h_{X(K)} \subset V \subset Y$ with V open in Y, and take $V \subset \overline{V} \subset V' \subset Y$, then by partition of unity, exists $\varphi \in \mathcal{E}(X)$ such that $\varphi|_V \equiv 1$ and supp $\varphi \subseteq V'$. Set $g = \varphi f$. Note that supp $(f - g) \subseteq Y \setminus V \subseteq Y \setminus K$, hence $T[(f - g)|_Y] = 0$. Since g is horomorphic on V, we have supp $d''g \Subset X \setminus h_X(K)$, hence by our previous claim,

$$T[f] = T[g|_Y] = S[d''g] = 0.$$

3.4. ***Runge Exhaustion***

Let X be a Riemann surface and $K \subset X$ be nonempty. Recall that $h_{X(K)}$ is the union of K and all relatively compact connected components of $X \setminus K$.

Proposition 3.15. *If K is closed, then $h_{X(K)}$ is closed.*

Proof. Let $\{C_j\}_{j\in J}$ be the connected components of $X \setminus K$, then each C_j is open, so

$$X \setminus h_{X(K)} = \bigcup_{j\in J\setminus J_0} C_j,$$

where $J_0 = \{j \in J : C_j \text{ is relatively compact}\}$, hence $X \setminus h_{X(K)}$ is open, so $h_{X(K)}$ is closed. ∎

Proposition 3.16. *If K is compact, then $h_{X(K)}$ is compact.*

Proof. Let $U \Subset X$ be a relatively compact neighborhood of K. Let $\{C_j\}_{j\in J}$ be the connected components of $X \setminus K$, then each C_j is open.

We claim that every C_j meets $\overline{U}$. Otherwise, exists some $j \in J$ such that $C_j \subset X \setminus \overline{U} \subset X \setminus U$. Since $X \setminus U$ is closed, we have $\overline{C_j} \subset X \setminus U \subset X \setminus K$. Since C_j is a connected component of $X \setminus K$, we have $\overline{C_j} = C_j$, hence C_j is closed. But C_j is also open, contradicting the connectedness of X.

Since ∂U is compact and $\{C_j\}_{j\in J}$ is a family of pairwise disjoint open subsets, there are at most finitely many C_j's meeting ∂U. Again, let

$$J_0 = \{j \in J : C_j \text{ is relatively compact}\},$$

and suppose $C_{j_1}, ..., C_{j_m}$ are those that meet ∂U, then by the previous claim all other components $C_j \subset U$, so

$$h_{X(K)} \subset U \cup \left(\bigcup_{k=1}^{m} C_{j_k}\right),$$

is relatively compact, and by the previous proposition, $h_{X(K)}$ is closed, hence $h_{X(K)}$ is compact. ∎

Corollary 3.17. *Suppose X is a non-compact Riemann surface, then there exists a sequence $(K_j)_{j=0}^{\infty}$ of compact subsets of X satisfying*

(i) *K_j is Runge in X for every j;*
(ii) *$K_{j-1} \subset K_j^{\circ}$ for every $j \geq 1$;*
(iii) *$\bigcup_{j=0}^{\infty} K_j = X$.*

Proof. Since X has countable topology, exists a sequence of compact sets $(K_j{}')_{j=0}^{\infty}$ that covers X. Let $K_0 = K_0{}'$. Choose a compact set M_0 such that $K_0 \subset M_0^{\circ}$ and set $K_1 = h(M_0)$. Choose a compact set M_1 such that $K_1 \cup K_1{}' \subset M_1^{\circ}$ and set $K_2 = h(M_1)$. Continuing this process, we can get a sequence of compact sets $(K_j)_{j=0}^{\infty}$ such that K_j is Runge in X for every j, $K_{j-1} \subset K_j^{\circ}$ for every $j \geq 1$ and $\bigcup_{j=0}^{\infty} K_j = X$.

Lemma 3.18. *For all compact $K \subset X$, there exists a Runge domain Y such that $K \subset Y \Subset X$ and Y has a regular boundary with respect to solving the Dirichlet problem.*

Proof. Choose a connected compact set $K_1 \supset K$ and a compact set K_2 with $K_1 \subset K_2^{\circ}$. Let $K_2{}' = h_X(K_2)$. For each $x \in \partial K_2{}'$, choose a chart (U, z) around x with $U \cap K_1 = \emptyset$ and $D_x := \overline{B_{r_x}(x)} \subset U$ for some $r_x > 0$. Since $\partial K_2{}'$ is compact, we can find a finite cover of $\partial K_2{}'$ by such disks $D_{x_1}, ..., D_{x_m}$. Let

$$Y_1 = K_2{}' \setminus \left(\bigcup_{k=1}^{m} D_{x_k}\right),$$

which is open and has regular boundary with respect to solving the Dirichlet problem. We claim that Y_1 is Runge in X. Let C be a connected component of $X \setminus Y_1$, then $C \cap D_{x_k} \neq \emptyset$ for some k since $C \cap \partial K_2{}' \neq \emptyset$, implying that $D_{x_k} \subset Y_1$ since D_{x_k} is connected, hence $D_{x_j} \cap C_{i'} \neq \emptyset$ for some connected component $C_{i'}$ of $X \setminus K_2{}'$, so $C_{i'} \subset C$, hence C is not compact since $C_{i'}$ is not compact. Therefore, Y_1 is Runge in X.

Let Y be the connected companennt of Y_1 containing K_1. Since Y_1 is open and X is a manifold, Y is open. Let $A := X \setminus Y_1$ and $\{A_k\}$ are connected components of A, then each A_k is closed and non-compact. If $\overline{Y} \cap A = \emptyset$, then $\overline{Y} \subset X \setminus A = Y_1$, so $Y = \overline{Y}$ is closed and open, contradiction, thus $\overline{Y} \cap A \neq \emptyset$. We claim tha each connected component C of $X \setminus Y$ meets A. Otherwise, $C \subset Y_1$, i.e. exists connected component Y' of Y_1 disjoint from Y such that $C \cap Y' \neq \emptyset$. Since C is a connected component of $X \setminus Y$ and is closed and $Y' \subset X \setminus Y$ in connected, we have $Y' \subset C$, so $\overline{Y_1} \subset C$ but $\overline{Y'} \cap A \neq \emptyset$, implying that $C \cap A \neq \emptyset$, contradiction. Therefore, each connected component of $X \setminus Y$ meets A, hence is not compact since each connected component of A is not compact, so Y is connected hence a Runge domain. Since Y_1 has regular boundary with respect to solving the Dirichlet problem, so does Y since Y is a connected component of Y_1. ∎

THEOREM 3.19. *Let X be a non-compact Riemann surface, then there exists a sequence $Y_0 \Subset Y_1 \Subset ... \Subset X$ of relatively compact Runge domains such that $\bigcup_{j=0}^{\infty} Y_j = X$ and each Y_i has a regular boundary with respect to solving the Dirichlet problem.*

Proof. Let $\left(K_j\right)_{j=0}^{\infty}$ be a sequence of compact subsets of X satisfying the conditions in Corollary 3.17. By the previous lemma, there exists a Runge domain Y_0 such that $K_0 \subset Y_0 \Subset X$ and Y_0 has a regular boundary with respect to solving the Dirichlet problem. For $j \geq 1$, suppose we already have $Y_0, ..., Y_{j-1}$, then we can find a Runge domain Y_j such that $K_j \cup \overline{Y_{j-1}} \subset Y_j \Subset X$ and Y_j has a regular boundary with respect to solving the Dirichlet problem. By construction, we have $Y_0 \Subset Y_1 \Subset ... \Subset X$, and we have

$$X = \bigcup_{j=0}^{\infty} K_j \subseteq \bigcup_{j=0}^{\infty} Y_j \subseteq X,$$

so $\bigcup_{j=0}^{\infty} Y_j = X$. ∎